

Evaluating the Quality of the LAION-2B Dataset: Insights into Large-Scale Image Dataset Assessment

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Abstract

Generative Artificial Intelligence (AI) systems have enabled the large-scale creation of images, with these technologies becoming increasingly accessible to the public. However, the development of these systems is largely dependent on the existence of massive web-scraped datasets serving as training data, such as Re-LAION-5B, which forms the focus of this research alongside the Stable Diffusion v1.5 model trained on it. Due to the nature of web-scraped training data, there has been increased concern regarding the ethical implications imposed, particularly related to bias and unintended correlations embedded within the training corpus. Specifically, non-semantic or spurious features present within the training dataset may serve as shortcuts for various downstream tasks, negatively impacting model behaviour and fairness outcomes. While these shortcuts can emerge for any task, demographic attributes represent a particularly concerning application, alongside other attributes unrelated to the inherent semantic content of the dataset.

This thesis investigates the demographic bias within the Re-LAION-5B dataset and resultant generative images produced by the Stable Diffusion v1.5 model, with a direct focus on identifying spurious features within these two components and determining whether the generative model amplifies, alters, or mitigates those spurious features identified within the respective training corpus. The study adopts a classifier-based approach derived from prior research, in which discriminative models are trained to distinguish between Re-LAION-5B and Stable Diffusion v1.5 generated images. These classifiers utilise images that have undergone specific transformations. These transformations systematically reduce the semantic information available to the model and force reliance on specific low-level image features, thereby exposing the presence of dataset specific spurious patterns that can be exploited by the model for image source identification.

Building upon this classifier based analysis, the relevant image sets used in the discrimination task are further annotated in terms of gender and skin-tone attributes. This secondary analysis investigates whether the previously identified dataset specific spurious features demonstrate correlations with these sensitive demographic attributes, allowing for the detection of harmful features that may serve as model shortcuts for deriving gender or skin-tone classifications. To further contextualise these findings and assess the real world implications of observed biases, the demographic distributions of retrieved and generated images across 200 occupation categories are compared against real world demographic statistics, revealing patterns of representation bias and potential amplification effects in the generative model.

Finally, based on the spurious features identified in the prior sections, this thesis

implements and evaluates a combined set of inference-time interventions for debiasing the Stable Diffusion v1.5 model without requiring full model retraining. These interventions demonstrably reduce spurious demographic correlations across low-level, geometric, and semantic representations, confirming that inference-time debiasing can produce measurable and substantive departures from the demographic and spurious feature profile of standard generation. By explicitly linking dataset characteristics, spurious feature learning mechanisms, and generative bias outcomes, this work contributes both methodological insights for evaluating large-scale multimodal datasets and practical interventions for the development of fairer text-to-image generation systems.

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List of Abbreviations

AUC Area Under the Curve.

BLS Bureau of Labour Statistics.

CCv2 Casual Conversations v2.

CFG Classifier-Free Guidance.

CLIP Contrastive Language-Image Pre-training.

CNN Convolutional Neural Network.

COCO Common Objects in Context.

CSAM Child Sexual Abuse Material.

FACET Fairness in Computer Vision Evaluation Benchmark.

FAISS Facebook AI Similarity Search.

Grad-CAM Gradient-weighted Class Activation Mapping.

ILOSTAT International Labour Organization Statistics.

IoU Intersection over Union.

ISCO International Standard Classification of Occupations.

ITI-GEN Inclusive Text-to-Image Generation.

LCM Latent Consistency Model.

LDA Latent Dirichlet Allocation.

LoRA Low-Rank Adaptation.

MLP Multilayer Perceptron.

MST Monk Skin Tone.

SAM Segment Anything Model.

SDv1.5 Stable Diffusion version 1.5.

VAE Variational Autoencoder.

1 Introduction

1.1 Problem Definition

The rapid advancement of text-to-image generative models has fundamentally transformed visual content creation, with diffusion-based architectures achieving unprecedented image quality and prompt controllability. Among these, Stable Diffusion [1] has emerged as particularly influential due to its open-source availability, widespread deployment, and the extensive ecosystem of tools and fine-tuned variants built upon it.

These systems rely on massive web-scraped datasets to learn the mapping between text and visual content. Re-LAION-5B, containing over 5 billion image-text pairs harvested from the internet, represents one of the largest publicly available training corpora for such models [2]. However, the scale and automated nature of web scraping introduces systematic biases that reflect and potentially amplify societal inequities present in online content [3].

Demographic bias in AI is well-documented across natural language processing [4], machine learning [5], and computer vision [Buolamwini_2018_BiasGenderClassification], yet a gap remains between detecting bias in generative outputs and understanding the specific low-level image features that cause and perpetuate it. Current research predominantly adopts an outcome-focused perspective, measuring demographic distributions in generated images against desired benchmarks [6, 7], treating models as black boxes that diagnose symptoms without exposing underlying mechanisms.

A fundamental yet underexplored dimension involves spurious features, which are low-level non-semantic image characteristics such as colour statistics, texture patterns, edge distributions, frequency domain properties, and compression artifacts [8, 9]. When these features correlate with sensitive demographic attributes, models can exploit superficial statistical regularities rather than semantic content, creating a form of bias that operates beneath standard evaluation metrics. It also remains unclear how generative training transforms these spurious patterns, making it difficult to predict how training data biases manifest in deployed systems.

Existing mitigation approaches typically require expensive interventions: curating balanced training data [10], fairness-constrained training [11], or fine-tuning on carefully constructed demographic distributions [12]. These demand substantial computational resources, creating barriers for researchers working with pre-trained models. Moreover, outcome-focused corrections may balance demographic distributions without addressing the spurious features that enable biased shortcuts,

yielding superficial fixes that fail to generalise.

This thesis addresses these gaps through a mechanistic investigation of spurious features in Re-LAION-5B and Stable Diffusion v1.5. Rather than solely measuring outcomes, it applies established methodologies to expose the low-level features that distinguish dataset sources and correlate with sensitive attributes, then evaluates targeted inference-time interventions that address root causes rather than symptoms.

1.2 Motivation

Influence and Impact: Re-LAION-5B serves as training data for multiple widely-deployed text-to-image models, while Stable Diffusion v1.5 has achieved widespread adoption across platforms such as DreamStudio and Hugging Face. Despite this pervasive influence, comprehensive analysis of spurious features in Re-LAION-5B and their propagation through Stable Diffusion v1.5 remains limited, making understanding bias mechanisms in these systems directly relevant to fairness at scale.

Computational Accessibility: Training or fine-tuning large-scale diffusion models is beyond reach for most researchers and practitioners. By focusing on inference-time methods applicable to frozen models, this research demonstrates fairness interventions accessible to smaller groups and organisations with limited computational budgets.

Reproducibility and Transparency: The open-source nature of Stable Diffusion v1.5 and the research availability of Re-LAION-5B enable findings to be independently verified, replicated, and extended, strengthening the credibility and impact of this work.

Beyond Outcome Measurement: Understanding why biases emerge, and specifically which low-level features enable biased shortcuts, supports interventions that address root causes rather than symptoms, enabling solutions that generalise rather than superficially correct specific evaluation metrics.

1.3 Aims and Objectives

The overarching aim is to apply established spurious feature detection methodologies to Re-LAION-5B and Stable Diffusion v1.5, systematically characterise the spurious features present in each, quantify their correlation with sensitive demographic attributes, and evaluate the effectiveness of existing inference-time debiasing techniques on this dataset-model pair.

Objective 1: Dataset Acquisition, Preparation, and Annotation. Retrieve images from Re-LAION-5B and generate corresponding images using Stable Diffusion v1.5 across a set of occupation categories. Annotate all images for gender and skin-tone

using established computer vision models to enable demographic analysis and spurious feature correlation assessment.

Objective 2: Spurious Feature Detection. Implement classifier-based methodologies to detect spurious features distinguishing dataset sources. Apply systematic image transformations to isolate non-semantic features and train discriminative classifiers to characterise dataset-specific spurious patterns.

Objective 3: Demographic Correlation Assessment. Analyse whether identified spurious features exhibit systematic correlations with gender and skin-tone in both datasets. Quantify the extent to which spurious features enable demographic classification shortcuts and compare distributions against real-world occupational statistics to identify representation biases.

Objective 4: Inference-Time Debiasing. Implement established inference-time debiasing techniques for Stable Diffusion v1.5 without model retraining. Evaluate their effectiveness across demographic representation and spurious feature mitigation, and assess fairness trade-offs to inform practical deployment.

Objective 5: Conclusions on Bias Mechanisms and Mitigation. Synthesise findings to characterise how spurious features manifest in Re-LAION-5B and propagate through Stable Diffusion v1.5, determining whether the generative process preserves, amplifies, or mitigates training data biases. Provide evidence-based recommendations for practitioners deploying fairer text-to-image systems.

1.4 Document Structure

The remainder of this dissertation is structured as follows. Chapter 2 provides the background and literature review, covering the Re-LAION-5B dataset and Stable Diffusion v1.5 architecture, spurious feature learning and dataset bias, demographic annotation frameworks, skin tone measurement scales, and inference-time debiasing approaches for text-to-image models. Chapter 3 describes the methodology, detailing the image retrieval and generation pipelines, demographic annotation procedures, the full image transformation pipeline for spurious feature analysis, and the inference-time debiasing configuration. Chapter 4 presents the evaluation and results, reporting the demographic distribution analysis across 200 professions, dataset classification results across all transformations, demographic attribute separability findings for both gender and skin tone, and an assessment of the debiasing pipeline against the aggregate standard Stable Diffusion v1.5 results. Chapter 5 presents the conclusions and future work, summarising the main findings and their implications for the use of Re-LAION-5B and Stable Diffusion v1.5 in downstream applications, and proposing directions for extending the demographic probing framework, improving the debiasing

pipeline, and broadening dataset assessment methodologies.

2 Background & Literature Review

2.1 Re-LAION-5B Dataset

Re-LAION-5B is a large-scale, openly released image-text dataset comprising approximately 5.85 billion image-text pairs harvested from publicly accessible web content via Common Crawl [2, 13]. It is a re-release of the original LAION-5B, motivated by the discovery of CSAM within the original corpus. The re-release applies improved safety filtering pipelines, including enhanced perceptual hashing and classifier-based detection, to remove harmful content [14].

The construction pipeline parses HTML pages indexed by Common Crawl to extract image URLs paired with alt text. These pairs are filtered using cosine similarity between image and text embeddings computed with OpenAI’s ViT-B/32 CLIP model for English pairs and a multilingual CLIP variant for all others [15], retaining pairs above 0.28 for English content and 0.26 for non-English content, a step that removes approximately 90% of the original 50 billion candidates [2]. Additional stages remove near-duplicate pairs, apply NSFW classifiers, and enforce minimum resolution and caption length requirements. The resulting dataset is stratified into three subsets summarised in Table 2.1. The LAION-1B-nolang subset frequently contains images of products, places, or content dominated by proper nouns, for which no language could be reliably detected.

Table 2.1 LAION-5B subset composition [2, 16, 17].

Subset	Approx. Size	Description
LAION-2B-en	2.32B pairs	English-language image-text pairs
LAION-2B-multi	2.26B pairs	Multilingual pairs (>100 languages)
LAION-1B-nolang	1.27B pairs	No detectable language (products, places, proper nouns)
LAION-Aesthetics v2 5+	~600M pairs	Quality-filtered subset of LAION-2B-en; aesthetic score >5.0, resolution $\geq 512 \times 512$, watermark prob. <0.5

LAION-Aesthetics v2 5+ is a quality-filtered offshoot of LAION-2B-en rather than a separate corpus, constructed using a linear classifier trained on CLIP ViT-L/14 embeddings to predict human aesthetic ratings [16]. It serves as the training corpus for Stable Diffusion v1.5.

Despite its scale, Re-LAION-5B carries significant concerns. Birhane et al. [3] audited LAION-400M, a predecessor, and identified substantial quantities of misogynistic, pornographic, and racially derogatory content, confirming that automated

filtering is insufficient at billion-scale. Beyond explicit harmful content, the web-scraped construction process introduces systematic demographic imbalances, as the statistical composition of online imagery reflects societal inequities in content production and visual representation.

A practically significant limitation of the Re-LAION-5B release is the absence of pre-computed embeddings. The original LAION-5B included pre-computed CLIP embeddings enabling efficient nearest-neighbour retrieval. In the re-release these were not made available, as the re-filtering altered the dataset composition and the embedding infrastructure was not reconstructed [14]. Any embedding-based retrieval therefore requires independently computing CLIP embeddings from raw image downloads, introducing substantial overhead. For this thesis, this absence necessitates the derivation of embeddings as part of the data acquisition pipeline.

A further consequence of Re-LAION-5B’s web-sourced construction is link attrition: the proportion of URLs that resolve successfully degrades over time as hosting services expire or content is removed [18]. Because Re-LAION-5B re-filters URLs originally crawled by Common Crawl, the gap between crawl date and download date introduces a decay component that cannot be precisely quantified without access to per-URL crawl timestamps. The surviving image pool is therefore not a random sample of the original corpus but a temporally filtered subset shaped by domain-specific hosting practices and content removal patterns. Additionally, web-scraped datasets exhibit systematic domain concentration, whereby a disproportionate share of images originates from a small number of large hosting platforms. Birhane et al. [3] demonstrate that such structural biases propagate directly into the resulting dataset, and Zeng et al. [9] similarly show that large-scale web-scraped datasets are reliably distinguishable by neural networks based on their dataset-specific content distributions.

2.2 Stable Diffusion: Architecture, Training, and Dataset Lineage

Diffusion models offer several advantages over alternative generative architectures. Unlike GANs [19], they avoid mode collapse and training instability [20], and they bypass the surrogate loss limitation of VAEs [21] that constrains reconstruction fidelity. These properties have established diffusion models as the dominant architecture for high-quality image synthesis.

Stable Diffusion is a family of open-source text-to-image models based on the Latent Diffusion Model (LDM) framework of Rombach et al. [1]. The central innovation is performing the generative process in a compressed latent space rather than at full

pixel resolution. A Variational Autoencoder [21] is trained to map high-resolution inputs into a lower-dimensional latent space; once trained, its encoder and decoder are frozen and all generative modelling is performed within this latent space. This compression dramatically reduces the computational cost relative to pixel-space diffusion models such as DALL-E [22].

The diffusion process follows the DDPM framework [23], in which a forward process corrupts a clean latent by adding Gaussian noise over T timesteps until it approximates pure Gaussian noise. A learned reverse process, implemented as a U-Net [24], iteratively denoises a noisy latent to recover a clean sample. Text conditioning is achieved via cross-attention in the U-Net’s intermediate layers, where a pretrained CLIP text encoder [15] produces embeddings injected at multiple spatial resolutions.

Classifier-Free Guidance

Classifier-Free Guidance (CFG) [25] is applied during inference to strengthen the influence of text conditioning. A single model is jointly trained as both a conditional and an unconditional model, with the text conditioning dropped with a probability of 10% during training in the case of Stable Diffusion v1.5 [17]. At inference, two forward passes are performed per denoising step and the final noise estimate is computed as:

$$\tilde{\epsilon}_{\theta}(z_t, c) = \epsilon_{\theta}(z_t, \emptyset) + s \cdot (\epsilon_{\theta}(z_t, c) - \epsilon_{\theta}(z_t, \emptyset)) \quad (2.1)$$

where z_t is the noisy latent at timestep t , c is the text conditioning, $\emptyset$ denotes the null unconditional token, and $s \geq 1$ is the guidance scale. Higher guidance scales produce outputs more strongly adherent to the prompt at the cost of stochastic diversity. Critically for bias analysis, higher CFG scales push the model more strongly toward its learned conditional distribution, amplifying demographic stereotypes and spurious correlations present in the training data.

Training Lineage

Stable Diffusion v1.5 was released by RunwayML as a continuation of training from v1.2, itself initialised from v1.1 weights produced by CompVis [17]. Training was conducted on LAION-Aesthetics v2 5+ using the AdamW optimiser with a batch size of 2048 at 512×512 pixel resolution. The training corpus is a direct derivative of LAION-2B-en, inheriting its web-scraped composition, demographic imbalances, and the spurious statistical regularities documented in Section 2.1. The open-source availability of both model weights and documented training data lineage makes Stable Diffusion v1.5 particularly well-suited for mechanistic bias analysis, as findings can be independently verified and directly linked to known properties of the training corpus.

Later models in the Stable Diffusion family, including SDXL [26] and Stable Diffusion 3 [27], rely on proprietary training corpora whose composition has not been disclosed, substantially limiting the capacity for independent bias auditing and reinforcing the methodological value of focusing on v1.5.

2.2.1 Low-Rank Adaptation of Diffusion Models

Low-Rank Adaptation (LoRA) [28] is a parameter-efficient fine-tuning technique in which the weight update ΔW for a matrix $W \in \mathbb{R}^{d \times k}$ is decomposed into two low-rank matrices $A \in \mathbb{R}^{d \times r}$ and $B \in \mathbb{R}^{r \times k}$, where $r \ll \min(d, k)$:

$$W' = W + \Delta W = W + AB$$

Because only A and B are trained while W is frozen, the number of trainable parameters is reduced from $d \times k$ to $r(d + k)$. LoRA adapters can be fused directly into base weights at inference time, eliminating runtime overhead.

Latent Consistency Models (LCMs) [29] introduce a distillation-based objective enabling high-quality generation in two to four denoising steps rather than the twenty to fifty required by standard DDPM samplers. LCM-LoRA [30] applies this distillation as a LoRA adapter rather than requiring full model retraining, implementing a consistency function that maps predictions at different noise levels to the same clean image, allowing the sampling trajectory to skip intermediate timesteps.

Critically, LCM-LoRA’s training objective is purely geometric: the adapter is optimised to reproduce the output distribution of the base model more efficiently, with no training signal derived from demographic content, aesthetic labels, or semantic categories. The LCM distillation loss operates on the consistency of latent-space trajectories across noise levels, ensuring that predictions at different timesteps map to the same clean image. The adapter weights therefore encode information about the structure of the denoising process rather than the content of the training images. This distinguishes speed-based LoRAs categorically from concept LoRAs or style LoRAs, which are explicitly trained to shift the model’s semantic prior toward a specific concept or appearance, and which consequently alter what the model generates. Consequently, applying LCM-LoRA to Stable Diffusion v1.5 does not introduce additional demographic signal and is therefore unlikely to confound measurements of the base model’s demographic prior, making it suitable for use in the generation pipeline of this thesis.

2.2.2 Aesthetic Quality Assessment in Large-Scale Image Datasets

Aesthetic quality in web-scraped datasets varies enormously. To address this, Schuhmann [16] trained a linear predictor on CLIP ViT-L/14 embeddings using human preference ratings from the Simulacra Aesthetic Captions (SAC) dataset [31], producing a lightweight scalar aesthetic score. This predictor was used to construct the LAION-Aesthetics v2 subsets, with the 5+ threshold retaining images above a score of 5.0 out of 10.

In the context of evaluating generative outputs, aesthetic scoring serves a different purpose. Diffusion models operating at reduced step counts can produce structurally coherent images that nonetheless exhibit quality failures such as posterisation or large undifferentiated regions from incomplete denoising, failures not captured by detection-based validation. A conservative threshold of 3.0 is adopted in this work as a minimum viability gate, positioned well below the training cutoff to avoid introducing systematic upward quality bias into the generated subset while still rejecting genuinely corrupted outputs.

2.2.3 Face Detection Models and Evaluation Benchmarks

Face detection produces axis-aligned bounding boxes with confidence scores for human faces within an image. Modern detectors are predominantly single-stage architectures that perform classification and regression jointly, offering substantially higher throughput than two-stage approaches [32]. The WIDER Face benchmark [33] is the standard evaluation dataset, comprising 32,203 images with 393,703 annotated faces across three difficulty tiers based on face size and occlusion. For applications requiring detection of clearly visible, unoccluded faces in generative or retrieval settings, Easy and Medium Average Precision are the more relevant indicators, as the Hard partition targets conditions unlikely to arise in such contexts.

RetinaFace [32] established a strong baseline through joint detection and five-point landmark regression within a Feature Pyramid Network. SCRFD [34] introduced sample and computation redistribution to produce a family of efficient detectors at reduced FLOPs. YOLOv8-face adapts the YOLOv8 architecture [35] to face detection, offering competitive accuracy on Easy and Medium partitions with substantially higher throughput, making it well-suited for large-scale filtering pipelines where speed is a primary constraint.

For this work, the relevant properties are single-face precision and inference throughput, since structural filtering is applied to hundreds of thousands of candidates. Small-face recall is a secondary concern given that the structural validation criteria require face regions of sufficient size to support downstream FaceMesh landmark

extraction.

2.2.4 Facial Landmark Estimation: MediaPipe FaceMesh

Facial landmark estimation recovers the positions of anatomically defined keypoints within a detected face region. MediaPipe FaceMesh [36] predicts 468 three-dimensional face landmarks from a single RGB image using a lightweight convolutional network, producing dense coverage across the full facial surface including eye contours, eyebrows, lips, nose, and outer face boundary.

FaceMesh serves a functionally distinct role from face detection in structural validation. Detection confirms the presence and approximate location of a face; landmark estimation confirms that the detected region contains a geometrically coherent face with sufficient surface detail for downstream processing. A detected bounding box may enclose a face-like region that is too blurred, too occluded, or too poorly rendered to yield reliable landmarks, and FaceMesh failure on such regions provides a principled rejection criterion. The convex hull of the predicted landmarks also provides a precise polygonal mask of the facial skin region, enabling tightly cropped face extractions that exclude surrounding hair, clothing, or background, a property that is directly relevant when the extracted region is used for skin tone classification where non-skin pixels would introduce spurious colour signal.

This functional complementarity is the basis for the two-stage face processing pipeline adopted in this work: YOLO-based detection identifies candidate face regions efficiently at scale, and FaceMesh subsequently validates geometric coherence and produces the segmented face crop used in downstream demographic annotation.

2.3 Similarity Search in High-Dimensional Embedding Spaces

CLIP embeddings [15] are typically 512- or 768-dimensional floating-point vectors, and retrieval is framed as a nearest-neighbour search in this shared space. When similarity is measured by cosine similarity, vectors are L2-normalised prior to indexing and inner product search is performed, since for unit vectors the two are equivalent.

Naive exact search scales as $O(Nd)$, requiring $N \times d$ operations per query, which becomes computationally intractable for databases containing hundreds of millions to billions of entries. Efficient similarity search therefore requires indexing structures that reduce the number of comparisons per query.

The FAISS Library

The Facebook AI Similarity Search (FAISS) library [37] is a widely adopted open-source library providing indexing structures and algorithms for similarity search and clustering of dense vectors. It supports both CPU and GPU implementations and spans a spectrum from exact to approximate nearest-neighbour search.

Exact Search: IndexFlatIP

The simplest index, IndexFlatIP, performs exact inner product search with $O(Nd)$ complexity, exhaustively computing the inner product between a query and every indexed vector. No approximation is applied, guaranteeing that true nearest neighbours are returned for every query. Despite being exhaustive, it substantially outperforms unoptimised linear scan by exploiting BLAS-level matrix operations and CPU SIMD acceleration.

Approximate Search Methods

For databases where exact search is too slow, FAISS provides approximate nearest-neighbour index types that trade retrieval exactness for substantially reduced query times. IndexIVFFlat partitions the embedding space into k Voronoi cells using k -means clustering and at query time searches only the n_{probe} nearest cells, reducing complexity to approximately $O(n_{\text{probe}} \cdot N/k)$ [38]. IndexHNSWFlat organises indexed vectors into a Hierarchical Navigable Small World graph [39], yielding empirically logarithmic query complexity $O(\log N)$ with strong recall and low memory overhead. IndexIVFPQ combines inverted file partitioning with Product Quantisation [38], compressing vectors into quantised subvectors to enable billion-scale retrieval on commodity hardware at the cost of quantisation error, which corrupts similarity scores relative to true distances.

Approximate search methods additionally introduce search error, the failure to visit index regions containing true nearest neighbours during traversal. For applications where retrieval correctness is critical, such as when retrieved items form the empirical basis of a downstream analysis, exact search with IndexFlatIP eliminates both error sources by guaranteeing complete traversal with no vector compression, and is therefore adopted in this work to ensure that retrieval-induced approximation error does not confound downstream bias measurements.

2.4 Spurious Features in Machine Learning

Defining Spurious Features and Spurious Correlations

A key challenge in deploying machine learning models reliably is their tendency to learn features that are predictive within the training data but carry no causal or semantic relationship to the target variable. Such features are termed spurious features, and the statistical associations they form with target labels are known as spurious correlations [40, 41]. Formally, a spurious correlation exists when a feature f co-occurs with a label y in the training distribution but this co-occurrence does not persist under distribution shift, i.e., $p_{\text{train}}(f | y) \neq p_{\text{test}}(f | y)$, or when f carries no causal information about y despite being statistically correlated with it. A model that exploits such a feature in lieu of causally relevant ones is said to learn a shortcut [42].

Geirhos et al. [42] characterise a shortcut as any decision rule that performs well on the training distribution by exploiting a feature correlated with the correct label but which does not reflect the underlying task concept. Critically, they argue that shortcut learning is not a pathological edge case but a rational consequence of empirical risk minimisation: gradient-based optimisation will consistently favour whichever feature reduces training loss most efficiently.

Taxonomies of Spurious Features in Computer Vision

Spurious features in visual data can be broadly categorised by the level of the image hierarchy at which they reside and whether they reflect properties of the subject, the context, or the acquisition process.

At the pixel and statistical level, low-level image attributes including mean colour channel values, brightness distributions, contrast statistics, and jpeg compression artefacts can serve as reliable proxies for class labels when the dataset collection process introduces systematic correlations between acquisition conditions and class membership. Jo and Bengio [43] demonstrate that CNNs trained on benchmark datasets are inclined to learn surface statistical regularities rather than high-level abstractions, achieving high accuracy even when these regularities bear no semantic relationship to the class concept. Geirhos et al. [44] further demonstrate that ImageNet-trained CNNs exhibit a strong texture bias, relying on texture for object classification rather than shape.

At the frequency domain level, Wang et al. [45] demonstrate that CNNs can exploit high-frequency features imperceptible to the human eye as spurious shortcuts, and that adversarially introduced high-frequency perturbations can substantially alter model decisions without any visible change to image content. This sensitivity is

particularly concerning in web-sourced datasets where different image origins carry systematically different frequency signatures arising from variation in camera hardware, processing pipelines, and compression settings.

At the contextual and background level, spurious features can arise from statistical regularities in scene composition rather than in the subject itself. Beery et al. [46] show that classifiers trained on images where animals consistently appear against certain background types fail severely when the same animal appears against an unfamiliar background. Xiao et al. [47] demonstrate that models achieve non-trivial classification accuracy when trained on background-only images with foreground objects masked, establishing that contextual scene statistics constitute a pervasive spurious signal. Torralba and Efros [48] show that models trained on one visual dataset and evaluated on another exhibit sharp performance drops, demonstrating that much of what models learn reflects dataset-specific regularities rather than generalisable visual concepts.

At the semantic and co-occurrence level, spurious correlations can be encoded in higher-level statistical regularities such as the tendency for certain object categories to co-occur with certain scenes, or for certain demographic attributes to be associated with particular activities. Zhao et al. [49] demonstrate that gender-stereotypical object co-occurrences in captioning datasets cause models to amplify those associations in their outputs. Singh et al. [50] further show that performance degrades substantially when objects appear without their typical co-occurring context, confirming that semantic co-occurrence regularities function as learned shortcuts.

At the structural and geometric level, classical edge detection algorithms such as the Canny edge detector [51] reduce an image to a binary boundary map capturing object contours while discarding colour and texture. Class-agnostic segmentation tools such as the Segment Anything Model [52] extend this to higher-quality contour representations. Monocular depth estimation encodes the relative spatial geometry of a scene without reference to object semantics. Since these structural properties are systematically shaped by the contexts in which images are captured, they can carry dataset-specific or demographic-specific spurious signals invisible to pixel-level analysis. Brendel and Bethge [53] demonstrate that models operating solely on small independent image patches achieve surprisingly high accuracy on ImageNet, confirming that local structural features within small receptive fields are sufficient to support classification without any holistic understanding of image content.

At the person level, a person’s appearance, shape, spatial location, and pose may each independently correlate with demographic group membership when the collection conditions of a dataset introduce systematic differences in how different groups are photographed. This has direct consequences for fairness-through-blindness approaches, which attempt to remove protected attribute information by occluding the

person in an image [54, 55]. Because person-level spurious features are intertwined with background signals and encoded across multiple representational layers simultaneously, such occlusion-based interventions may leave substantial demographic information intact, a limitation examined in detail in Section 2.4.

At the semantic representation level, richer image representations that preserve object identity while discarding pixel-level detail can themselves carry dataset-specific spurious signals. Semantic segmentation maps retain the spatial distribution of scene categories while removing texture and colour. Object detection representations preserve object co-occurrence statistics while discarding internal pixel content. Image captions discard all visual information in favour of natural language but still encode semantic regularities reflecting the statistical composition of the underlying image distribution. VAE reconstructions, as implemented in latent diffusion architectures [1], provide a further diagnostic: if reconstructed images remain discriminable after compression, the bias resides primarily in semantic content rather than low-level acquisition artefacts. Taken together, these representation types demonstrate that spurious features are not confined to the pixel domain but permeate the full hierarchy of visual abstraction.

Spurious Features and Demographic Bias

When a spurious feature correlates with a sensitive attribute such as gender, race, or skin tone, a model that exploits the shortcut will systematically produce disparate outcomes across demographic groups. Buolamwini and Gebru [56] provide a foundational empirical demonstration: commercial gender classifiers trained on datasets overrepresenting lighter-skinned individuals exhibit substantially higher error rates for darker-skinned individuals, attributable in part to the model relying on skin-tone correlates as classification shortcuts. Mehrabi et al. [5] catalogue the mechanisms by which demographic shortcuts enter models, including representation bias (underrepresentation of certain groups in training data), measurement bias (features measured differently across groups), and aggregation bias (heterogeneous sub-populations pooled under a single model).

Critically, features that appear neutral, including colour statistics, pose distributions, and background scenes, can function as proxies for demographic attributes when dataset composition creates systematic correlations between low-level properties and group membership. Meister et al. [8] term these gender artifacts and demonstrate that they are effectively impossible to fully remove from standard visual datasets without also removing information necessary for downstream tasks. This proxy mechanism means that debiasing interventions which remove explicit demographic signals may leave intact the spurious low-level features that encode

equivalent information, producing only superficial corrections.

Spurious Features in Text-to-Image Generative Systems

The concern about spurious features takes on a distinct character in text-to-image generative models. Whereas discriminative models learn to exploit spurious correlations to make predictions, generative models trained on biased data learn to reproduce those correlations in their outputs: biased statistical regularities present in the training corpus become embedded in the model’s learned prior over images, causing the model to synthesise spurious feature patterns even when the conditioning prompt contains no information that should elicit them.

The training pipeline of latent diffusion models such as Stable Diffusion [1] illustrates this concretely. The model learns a joint distribution over image latents conditioned on text embeddings derived from CLIP [15]. Because CLIP is itself trained on web-scale image-text data, it encodes the same compositional and societal biases present in internet imagery [2, 3], and these biases are subsequently inherited by diffusion models conditioned on its embeddings. Spurious correlations in the training data between text tokens and low-level image features are absorbed into the diffusion model’s score function, causing those features to be hallucinated at generation time even in the absence of explicit demographic prompting.

Bianchi et al. [7] demonstrate this empirically, showing that text-to-image models not only reflect demographic under-representation present in their training data but actively amplify stereotypical associations: occupation prompts produce generated images with demographic distributions more extreme than those in real-world labour statistics. Luccioni et al. [6] extend this finding across a broader set of models, confirming that the amplification effect is systemic. Critically, neither study addresses which specific low-level image features encode these biases, treating the model as a black box and measuring only demographic outcomes.

Understanding Bias in Large-Scale Visual Datasets

The taxonomy above provides a conceptual decomposition of spurious features across different levels of image representation, but does not quantify their relative contribution to dataset-level bias. Zeng et al. [9] operationalise this through a transformation-based framework, aiming to identify the concrete forms of bias that make large-scale pre-training datasets distinguishable by neural networks. Their work builds on the dataset classification task introduced by Torralba and Efros [48] and revisited by Liu and He [57], who found that three widely-used datasets, YFCC100M [58], CC12M [59], and DataComp-1B [60], collectively referred to as YCD, can be

classified with 84.7% accuracy. While this confirms the datasets are biased, it does not explain which image properties are responsible. Zeng et al. address this gap through a transformation-based framework.

Transformation-Based Bias Quantification

The central contribution is a transformation-based framework: apply a transformation that preserves one type of visual information while removing all others, train a dataset classifier on the transformed images, and use the resulting accuracy to measure how much bias that type of information carries. For each transformation, 1M images per dataset are used for training and 10K for validation using a ConvNeXt-Tiny model trained for 30 epochs. On unmodified images, a reference accuracy of 82.0% is achieved.

At the semantic level, segmentation using ViT-Adapter-Large with the ADE20K vocabulary achieves 69.8%; object detection using ViTDet-Huge with LVIS achieves 61.9%; and image captioning with LLaVA 1.5 achieves 63.8% for short captions and 66.1% for long captions. These results show that semantic content, specifically which categories of objects appear in each dataset, is a primary source of dataset bias.

The structural transformations produce a striking result: Canny edge detection achieves 71.0%, SAM-based contour extraction reaches 73.2%, and monocular depth estimation with Depth-Anything-V2 yields 73.1%. These representations contain no object labels, yet they outperform most semantic transformations, showing that the shapes and spatial arrangements of objects differ significantly across datasets and carry as much discriminative information as object category.

Pixel and patch shuffling experiments separate local structure from global spatial layout. Randomly shuffling individual pixels removes all spatial structure, reducing the classifier to colour statistics alone; accuracy falls to 52.2%, just above the 33.3% chance level. Shuffling 16×16 patches instead recovers nearly all discriminability at 81.2%. This shows that local spatial structure within small image regions is sufficient to distinguish datasets, and that global composition matters much less than local texture and shape. A mean-RGB experiment compressing each image to a single average colour per channel achieves 48.5%, roughly 15 percentage points above chance, suggesting that even the global colour palette of images differs systematically across datasets. Frequency domain analysis shows high-pass filters yield 79.2% and low-pass filters yield 70.4%, indicating that spurious signals exist across the full frequency spectrum and cannot be eliminated by targeting a single frequency range.

A VAE experiment provides further context: passing images through the bottleneck of a pretrained Stable Diffusion VAE reduces accuracy from 82.0% to 77.4%, showing that low-level properties such as jpeg compression artefacts

contribute relatively little to the overall bias compared to semantic and structural content. Finally, combining two transformations together increases accuracy beyond what either achieves alone, for example combining pixel shuffling with SAM contours reaches 74.2% compared to 73.1% for SAM contours alone, demonstrating that different types of spurious signals carry partly independent information. This finding is directly relevant to the multi-transformation approach used in this thesis.

Propagation of Bias into Generative Models

A central question for this thesis is whether biases in training data carry through into generative model outputs. Zeng et al. test this by training separate unconditional Diffusion Transformer (DiT) models on each of the YCD datasets and classifying the resulting synthetic images. These generated images can be classified with 77.6% accuracy, nearly matching the 82.0% reference on real images, demonstrating that generative models reproduce the distributional biases of their training data to a degree largely preserved in their outputs. A second experiment using a single pretrained SDXL-Turbo model conditioned on LLaVA-generated captions achieves 58.1%, confirming that semantic bias is partly transmitted even when the generative model is not dataset-specific. Together, these results show a clear link between training data composition and generative output, directly supporting the central hypothesis of this thesis.

This distinction establishes a two-stage bias pipeline that is central to the present work: biases originate in the training dataset and are subsequently encoded into the generative model, where they may be preserved or amplified during synthesis. Re-LAION-5B represents the source stage of this pipeline, and Stable Diffusion v1.5 represents the generative stage, making the relationship between their respective spurious feature distributions a primary object of analysis.

Semantic Explanation of Dataset Differences

To explain why the datasets are distinguishable, Zeng et al. apply Grad-CAM to the reference classifier, showing that the model focuses on specific objects to identify dataset origin. A logistic regression model trained on binary object-presence vectors achieves 52.0–52.4% accuracy depending on vocabulary, well above chance but well below the 82.0% of the neural classifier, showing that object identity alone does not fully explain discriminability and that lower-level structural signals also play a role. Open-ended language analysis using LDA topic modelling characterises YFCC as outdoor and human-centred, DataComp as product-focused with clean backgrounds, and CC as a mix of the two.

These findings directly motivate the probe design adopted in this thesis, where transformations are selected to isolate semantic, structural, and low-level statistical components of the image. By evaluating classification performance under each transformation, the contribution of each feature type to dataset-specific bias can be quantified independently. When these spurious features correlate with demographic attributes, they provide a mechanism through which demographic information can be inferred from non-semantic image properties. The probe-based framework therefore enables not only the detection of dataset bias, but the identification of feature types that act as proxies for demographic attributes.

Evaluation of Image Captioning Quality

The transformation-based framework of Zeng et al. requires that generated captions faithfully reflect image content, as a captioning model producing degenerate descriptions would corrupt the semantic signal the transformation is designed to preserve. Caption-based representations discard all pixel-level information, isolating semantic content entirely. If dataset classification remains above chance under this transformation, the evidence strongly suggests that the discriminative signal resides in systematic differences in object co-occurrence, scene composition, or semantic structure rather than in low-level image statistics. Selecting an appropriate captioning model therefore requires a principled evaluation protocol. Reference-based n-gram metrics such as BLEU [61], METEOR [62], and CIDEr [63] assess surface-level lexical overlap with human-authored references, while diversity metrics [64] detect template collapse invisible to precision measures. At the semantic level, Sentence-BERT [65] provides a paraphrase-robust measure of meaning preservation, and CLIPScore [66] offers reference-free visual grounding by computing cosine similarity between CLIP image and text embeddings. RefCLIP-S extends this as the harmonic mean of CLIPScore and maximum reference similarity, achieving the highest correlation with human judgements of any single metric. No single metric is sufficient, and the failure modes of each motivate the multi-metric evaluation protocol employed in this thesis.

Gender Artifacts in Visual Datasets

Meister et al. [8] systematically investigate the nature and prevalence of gender artifacts in large-scale computer vision datasets. A gender artifact is defined as a visual cue that is both learnable by a modern image classifier and interpretable, meaning it has a recognisable human corollary such as colour, shape, or scene composition. The work is motivated by a gap in the fairness literature: fairness-through-blindness techniques assume that removing a person’s visible appearance from an image is sufficient to

eliminate gendered signal. Meister et al. challenge this by asking to what extent gendered information can truly be removed from a visual dataset. The gender artifact model is treated as a diagnostic tool for understanding dataset correlations, not a system for predicting gender identity; the gender labels used are derived from external annotator perception and are used solely to reveal distributional differences in the data.

Datasets and Model

The analysis focuses on MS-COCO [67] and OpenImages [68]. Gender labels for COCO are inferred from captions via keyword matching [49]; for OpenImages they come from the MIAP subset [69]. Both datasets are skewed male at 69.2% and 62.9% respectively. The gender artifact model is a ResNet-50 [70] pretrained on ImageNet, with performance reported as AUC where 50% represents chance-level discrimination.

Resolution and Colour Artifacts

Images downsampled to 7×7 pixels, at which almost all semantic content is destroyed, yield an AUC of 59.2% on COCO and 63.9% on OpenImages in colour. Repeating the experiment in greyscale, the AUC falls to near chance at 28×28 , showing that colour is the primary carrier of gender-correlated signal at low resolutions. A logistic regression classifier trained on just the three mean RGB channel values achieves 58.0% and 59.1% on COCO and OpenImages respectively, likely due to a compositional confound: male-labelled images are disproportionately set in outdoor sports contexts dominated by green tones, while female-labelled images more often feature close-up portraits with lighter average pixel values.

Person and Background Artifacts

A systematic occlusion protocol separates gender artifacts arising from the person from those arising from the background. With all person pixels visible, the model achieves 93.4% AUC on COCO. Removing the background reduces this only slightly to 92.7%, confirming person appearance as the dominant signal under normal conditions.

When the person’s appearance is replaced by a white segmentation mask but the background remains visible, the model still achieves 79.6% AUC, showing that the silhouette alone carries substantial gendered information. Removing the background as well reduces performance to 74.8%, confirming that silhouette shape and position are meaningful gender artifacts independent of scene context. Training on the 17 COCO keypoint locations alone yields 64.8%, and a normalised pose-only representation still achieves 58.0%, establishing body pose as an independent gender artifact.

Under the MaskRect condition, which matches the occlusion strategy used by several prominent mitigation methods [54, 55], the AUC remains at 70.8% on COCO and 63.1% on OpenImages, directly undermining the core assumption of fairness-through-blindness approaches. Class Activation Map analysis explains why: when the person is occluded, the model shifts attention onto background objects, associating indoor objects with female labels and outdoor objects with male labels.

Contextual Object Artifacts

A logistic regression classifier trained on a binary object-presence vector achieves 75.4% AUC on COCO, with objects most predictive of female-labelled images including hair dryers, handbags, and teddy bears, and those most predictive of male-labelled images including skateboards, ties, and snowboards. Critically, gendered information is broadly distributed across the object vocabulary rather than concentrated in a few obviously gendered objects: 30 randomly selected objects from the 80 annotated COCO categories still yield 66.8% AUC, confirming that no small subset of objects could be removed to meaningfully eliminate object-level artifacts.

Analysis of Fairness-Through-Blindness Methods

Applying the adversarial debiasing approach of Wang et al. [55] to COCO images reveals that 76.8% of removed pixels come from the background rather than the person, with outdoor scene objects such as giraffes, trains, and buses most frequently affected. Because gendered signals are spread throughout the image rather than localised to the person, the method suppresses scene-level information necessary for downstream tasks rather than the person-centric gendered information it was designed to target.

Discussion and Broader Implications

The authors draw three main conclusions. First, gender artifacts are so widely distributed across image structure that removing them without also removing task-relevant information is not feasible. Second, if a model can infer gender labels from mean pixel colour or the presence of a skateboard, then what it has learned reflects the compositional biases of the dataset rather than gender expression in any meaningful sense. Third, the authors recommend moving away from fairness-through-blindness approaches towards fairness-aware methods that explicitly account for distributional differences across demographic groups.

2.5 Demographic Annotation in Large-Scale Image Datasets

Studying demographic bias in large-scale visual datasets requires annotating images with sensitive attributes such as gender and skin tone. Unlike object category labels where a clear ground truth can in principle be defined, demographic annotations are contested, sitting at the intersection of biological, social, perceptual, and cultural factors. This section surveys annotation practices, examines sources of measurement bias and label uncertainty, reviews skin tone annotation frameworks, and outlines the core limitations of visual demographic inference.

2.5.1 Human and Automated Annotation Practices

Demographic attributes in visual datasets have traditionally been labelled by human annotators. Early benchmarks such as Labelled Faces in the Wild [71] exhibited variation in gender and race but were not designed for fairness analysis. MS-COCO [67] includes the category person without structured demographic annotations, while OpenImages [68] adds subclasses such as man and woman without a fairness-driven protocol. The More Inclusive Annotations for People (MIAP) dataset [69] introduces non-binary gender presentation labels under a formal multi-annotator framework. These efforts demonstrate that human demographic annotation is feasible at scale but is methodologically complex, with disagreement and label uncertainty intrinsic to the task.

Inter-annotator agreement for demographic labels is typically lower than for object categories, reflecting the interpretive ambiguity of socially constructed attributes. Aroyo and Welty [72] argue that annotator disagreement often encodes genuine differences in perception and interpretation, and that collapsing annotations to a single majority label can obscure meaningful variation.

At the scale of web-scraped datasets, manual annotation is not practically feasible. Large-scale studies therefore rely on automated demographic inference using pre-trained classifiers. FairFace [73] is commonly used due to its improved demographic coverage relative to earlier benchmarks.

A purpose-built benchmark for evaluating automated demographic classifiers is the Casual Conversations v2 (CCv2) dataset [74], comprising over 26,000 video clips from more than 5,000 consenting participants recorded across diverse lighting and environmental conditions. Demographic labels in CCv2 were provided directly by participants through self-reporting, while trained annotators labelled apparent skin tone using both the Fitzpatrick Skin Type and Monk Skin Tone scales. This dual-annotation design separates self-identified from perceived attributes, providing a principled basis for evaluating whether automated classifiers capture perceived

appearance rather than identity.

A complementary benchmark is the Fairness in Computer Vision Evaluation (FACET) dataset [75], providing annotations for approximately 32,000 images drawn from the Segment Anything 1 Billion (SA-1B) dataset, covering perceived gender presentation, perceived skin tone using the Monk Skin Tone scale, perceived age group, and additional fairness-relevant attributes. Annotations were produced by trained human raters under a structured multi-stage pipeline and support systematic evaluation across demographic attributes and their intersections.

However, automated annotation introduces a structural circularity. Demographic classifiers are themselves trained on human-annotated datasets that encode the same perceptual subjectivities and representational imbalances described above. As demonstrated by Buolamwini and Gebru [56], performance disparities in commercial gender classifiers reflect underlying dataset imbalances. When such models are used to annotate new datasets, they propagate the statistical structure and potential biases of their training data into downstream analyses, transforming annotation bias from an observable human disagreement into a systematic model-mediated measurement artefact that is no longer directly observable or correctable.

2.5.2 Measurement Bias and Label Uncertainty

Measurement bias arises when the label assignment process systematically distorts the metric it intends to measure [5]. Unlike object categories, demographic attributes lack stable visual ground truth and are inferred from perceptual cues, making measurement error often structured rather than random. Buolamwini and Gebru [56] show that darker-skinned females experience significantly higher misclassification rates than lighter-skinned males. When automated classifiers trained on such data annotate large-scale web datasets, these subgroup-specific error patterns become embedded in the generated labels.

These issues are particularly consequential when demographic labels function as covariates against which feature distributions are measured. Structured annotation error can produce apparent correlations between visual artefacts and demographic attributes that reflect classifier error patterns rather than properties of the underlying data. Without ground truth labels at scale, disentangling genuine distributional structure from measurement artefact becomes statistically intractable. Careful selection and documentation of annotation models is therefore essential when demographic attributes are used to audit spurious correlations in large-scale visual datasets.

2.5.3 Skin Tone Annotation Frameworks

The most widely used skin tone framework in computer vision is the Fitzpatrick Skin Type (FST) scale, originally developed in dermatology to classify sun-reactive skin response to ultraviolet exposure [76]. Its categories are anchored in self-reported photoreactivity rather than directly observable colour properties, creating a construct mismatch when applied to image-based annotation. Subsequent work has documented the compression of variation at the darker end of the scale [77], reducing sensitivity precisely where subgroup disparities are often most pronounced.

The Monk Skin Tone (MST) scale was developed to address these limitations as a ten-point ordinal framework explicitly designed for technology and machine learning applications [78]. Unlike Fitzpatrick, the MST scale is anchored to visual reference swatches intended for photographic assessment rather than sun-reactive history. A large-scale survey comparing Fitzpatrick, MST, and a 40-shade commercial palette found that Fitzpatrick was perceived as significantly less representative, particularly among darker-skinned participants, while the ten-point MST scale was perceived as comparably inclusive to the much finer-grained commercial palette [79].

Empirical work evaluating MST estimation under diverse lighting demonstrates substantial degradation in fine-grained ten-class performance, particularly in intermediate tone ranges [80]. When tones are grouped into broader light, medium, and dark categories, estimation stability improves markedly across lighting conditions. For large-scale web-scraped imagery where illumination is uncontrolled, grouped treatment provides greater robustness and reduces boundary-level annotation noise. Grouping in this context is not a simplification of the construct but a strategy to reduce measurement variance that could otherwise obscure or artificially induce spurious correlations.

A further methodological risk is shortcut learning during skin tone estimation. Experimental evidence shows that full-image CNN models exploit background, clothing, or contextual artefacts correlated with skin tone labels rather than pigmentation itself, as revealed by Grad-CAM analyses, whereas models trained on segmented facial skin regions exhibit activations concentrated on skin pixels [81]. For this reason, skin tone classification in this thesis is applied to detected facial skin regions rather than full images.

Schumann et al. [78] introduced the Monk Skin Tone Examples (MST-E) dataset comprising 1,515 images and 31 videos of 19 subjects spanning the full ten-point MST scale, photographed across a range of poses, expressions, and lighting conditions. Final annotations were provided by the creator of the MST scale, Dr Ellis Monk. The dataset was designed to support annotator training and to investigate annotation consistency under varying environmental conditions, and also enables evaluation of automated skin

tone estimation methods across specific capture conditions rather than relying solely on aggregate accuracy metrics.

2.5.4 Limitations of Visual Demographic Inference

A fundamental limitation of demographic annotation from visual data is the distinction between perceived attributes and self-identified attributes. Systems that infer gender from images predict whether a subject’s appearance conforms to culturally dominant norms of gendered presentation, not gender identity. Scheuerman et al. [82] demonstrate that commercial gender classification systems operationalise gender as a binary perceptual category and exhibit systematic failures on individuals whose presentation diverges from normative expectations. Consequently, when generative outputs are compared to demographic statistics such as labour force participation rates, the annotation variable and the statistical benchmark may refer to different constructs.

A second limitation concerns the instability of photometric inference under varying acquisition conditions. Skin tone classifiers are sensitive to illumination colour temperature, exposure, and post-processing. In uncontrolled web-scale datasets such as Re-LAION-5B, images are captured under heterogeneous conditions, creating a confound between acquisition artefacts and demographic inference quality. This issue is further compounded in synthetic imagery, where lighting is a learned prior embedded in the generative model rather than an external environmental factor.

A related conceptual distinction must be drawn between race and skin tone. Race is a socially constructed category shaped by historical, political, and cultural processes that cannot be reliably inferred from visual appearance alone. Skin tone is a photometric property of visible surface pigmentation. While correlated with racial categorisation in many societies, the two constructs are not equivalent, and fairness audits that operationalise race through perceived visual categorisation introduce additional layers of annotator subjectivity [83]. Skin tone, though still perceptually mediated, is more directly grounded in observable colour properties and more closely aligned with the low-level image features examined in spurious feature analysis.

Taken together, these limitations highlight that demographic labels derived from visual inference function as measurement instruments rather than ground truth variables. Recognising this constraint is essential when interpreting spurious feature distributions in both web-scraped datasets and generative model outputs.

2.6 Inference-Time Debiasing of Text-to-Image Diffusion Models

Approaches to mitigating demographic bias in text-to-image diffusion models can be categorised by the stage of the pipeline at which they intervene. Training-time methods address bias through modifications to the training data or objective function [10, 11]. Fine-tuning methods adjust pre-trained weights on demographic reference corpora [12]. Both families require substantial computational resources and high-quality annotated data, placing them beyond reach for practitioners working with pre-trained deployed systems [84].

Inference-time methods operate on a frozen model during generation, requiring no retraining and no access to weights beyond those publicly available. This family is particularly relevant for widely deployed open-source systems such as Stable Diffusion v1.5, where the only available intervention points are the conditioning signal, the sampling trajectory, and the model’s internal attention mechanisms. The methods surveyed below address demographic bias through three distinct mechanisms: editing internal model representations via closed-form weight modifications, manipulating the text conditioning signal through learned inclusive embeddings, and guiding the denoising trajectory through external signals targeting spatial structure or colour distribution.

Weight Editing: Unified Concept Editing

Unified Concept Editing (UCE) [85] directly edits the cross-attention projections of a diffusion model through a closed-form solution, without gradient-based training. The value projection matrices W_v of the U-Net’s cross-attention layers encode associations between text token embeddings and the visual features attended to during denoising. UCE frames debiasing as a constrained least-squares problem over W_v at each attention layer: given target concept embeddings representing biased associations to be corrected and anchor embeddings representing unrelated content to be preserved, UCE computes a closed-form update ΔW_v that minimises change to anchor representations while steering target representations toward a more balanced demographic prior. This closed-form solution is derived from a linearisation of the attention layer’s behaviour around the target token embeddings, requiring no forward passes beyond computing a small number of text embeddings. Because the edit is applied layer-wise and accumulated across attention layers, each layer contributes independently to the overall concept redirection, and the closed-form solution at each layer does not account for the downstream effects of edits applied to other layers.

A key strength is scalability to concurrent edits across multiple concepts,

demonstrated across up to 35 occupational prompts alongside style erasure and content moderation within a single edited model. However, because the edit applies uniformly across all spatial positions and all prompts that activate the targeted concept representation, UCE cannot selectively debias outputs in specific image regions or under specific contextual conditions. The linear approximation underlying the closed-form solution also constrains the method’s ability to correct strongly non-linear biases embedded simultaneously across multiple layers. It also provides no mechanism for specifying a precise target demographic distribution.

Inclusive Prompt Embeddings: ITI-GEN

ITI-GEN [86] operates at the level of the text conditioning signal rather than the model weights. It begins from the observation that demographic attributes such as skin tone are difficult to express unambiguously through natural language, and resolves this by replacing linguistic specification with image-grounded specification. The method learns a set of inclusive tokens, which are short learnable embedding vectors inserted into the CLIP text encoder’s input space, optimised to produce generated images uniformly distributed across the categories of a target demographic attribute.

For each attribute category, a set of reference images is selected, and the corresponding token embedding is trained via two losses through backpropagation through the frozen diffusion model. A direction alignment loss encourages the learned token to move the text embedding toward the CLIP image embedding of the reference images for that category. A semantic consistency loss penalises divergence from the original prompt’s semantic content, ensuring the inclusive token supplements rather than overwrites the user’s intended generation context. For A attributes each with K categories, one requires $A \times K$ reference image sets rather than K^A joint combinations, and the resulting tokens are composable at inference time by concatenation. ITI-GEN does not modify model weights and is directly compatible with ControlNet [87] and other conditioning frameworks.

A reproducibility study by Gallo Fernandez et al. [88] identifies two important limitations. First, ITI-GEN is susceptible to learning spurious proxy features from its reference images: if reference images for a demographic category are systematically correlated with background, clothing, or lighting, the learned token may encode these spurious cues rather than the target attribute itself. Second, the exponential growth of the joint attribute combination space with increasing numbers of attributes makes inclusive generation for all joint combinations increasingly difficult.

Self-Discovering Latent Directions: FairGen

FairGen [84] addresses a structural limitation shared by both UCE and ITI-GEN: their dependence on externally curated reference datasets or annotation models to define target demographic directions. FairGen instead discovers demographic latent directions directly from the model through a self-supervised procedure, attaching lightweight adapters to the cross-attention layers of the frozen U-Net and optimising them to learn the direction in the model’s latent feature space along which a target attribute varies. This is achieved by manipulating the noise prediction function through a noise composition objective that exploits the structure of the DDPM training process, allowing the latent direction for a given demographic attribute to be extracted without requiring any labelled demographic data.

Orthogonal regularisation between adapters for different attributes is a critical design choice: without it, adapters trained on correlated attributes such as gender and age, which co-vary in training data, tend to encode overlapping directions that interfere destructively when activated together, degrading both fairness and image quality. At inference time, a distribution indicator tracks the demographic composition of images generated so far in a batch and steers subsequent generation to correct any accumulated imbalance, enforcing fairness globally across a batch rather than independently per image. The primary limitation is that the discovered directions are model-specific and non-transferable, requiring the self-discovering training process to be repeated for each base model architecture.

Spatial Conditioning: ControlNet

ControlNet [87] adds fine-grained spatial conditioning to a pre-trained diffusion model, enabling the structure of generated images to be constrained by auxiliary signals such as pose skeletons, depth maps, Canny edge maps, or semantic segmentation masks. The central architectural challenge is to introduce spatial conditioning without destabilising the large pre-trained backbone. ControlNet resolves this through a trainable copy of the frozen U-Net encoder and middle block that receives the additional conditioning signal and processes it through the same encoder structure. The outputs are injected back into the frozen backbone via residual connections at each encoder level, allowing the decoder to integrate spatial information during the upsampling pass where image structure is synthesised.

To prevent the randomly initialised conditioning branch from corrupting the frozen backbone during early training, all residual connections are implemented as zero-initialised convolution layers. At training onset these produce zero output, so the backbone behaves identically to the unmodified model. As training proceeds, the zero convolution weights grow from zero in proportion to the gradient signal, introducing

conditioning information progressively. Multiple ControlNet signals can be composed at inference time by summing their residual contributions, enabling simultaneous constraint of multiple spatial properties without retraining.

In the context of debiasing, ControlNet is relevant because many demographic spurious features operate at the structural level. As documented by Meister et al. [8], body pose distributions, person-to-frame scale, and spatial framing patterns constitute significant gender artifacts not removed by purely semantic or colouristic debiasing interventions. By providing an explicit structural conditioning signal such as a neutral canonical pose skeleton, ControlNet can constrain the spatial properties of generated images independently of the semantic and demographic properties governed by the text prompt and inclusive token embeddings, enabling both forms of control to be composed without mutual interference.

Colour Distribution Guidance: SW-Guidance and Chamfer-Based Alignment

A further class of inference-time methods targets the colour statistics of generated images directly, motivated by the observation that text-to-image models trained on aesthetically filtered web-scraped corpora learn strong colour priors that may correlate with demographic attributes. Consistent with the spurious feature analysis of Meister et al. [8], who demonstrate that mean RGB channel values are sufficient to discriminate gender-labelled images above chance, colour distributions in training data can function as low-level proxies for demographic group membership.

SW-Guidance [89] is a training-free method that introduces colour conditioning into the diffusion sampling process by modifying the noise prediction at each denoising step. At each step the gradient of the Sliced 1-Wasserstein distance between the colour distribution of the current partially denoised image and a reference palette is computed and added to the denoising network’s noise prediction, functioning as an additional guidance term analogous to CFG [25]. A notable practical limitation is that computing the guidance gradient requires backpropagation through the entire denoising U-Net at every timestep, introducing substantial overhead that makes SW-Guidance impractical for large-scale image generation.

Shum et al. [90] propose an alternative colour alignment approach that avoids backpropagation through the denoising network. Their method projects diffusion terms, either the partially denoised latent representation or the predicted clean image $\hat{x}_0$, directly into a conditional colour manifold defined by a reference palette, using the Chamfer distance to establish correspondences between the colour point clouds of the generated and reference images:

$$d_{\text{Chamfer}}(X, Y) = \sum_{x \in X} \min_{y \in Y} \|x - y\|_2^2 + \sum_{y \in Y} \min_{x \in X} \|x - y\|_2^2 \quad (2.2)$$

At each denoising step where colour alignment is active, an iterative matching procedure computes the Chamfer-optimal correspondence and takes a gradient step to project the predicted $\hat{x}_0$ toward the reference distribution, which is then used in place of the original predicted clean image in the standard DDPM recomposition formula. Alignment is applied only during early denoising timesteps, reflecting the empirical observation that global colour structure is established early in the denoising trajectory while fine structural and semantic details emerge during later steps. Below a configurable threshold, alignment is disabled to avoid corrupting the fine-grained content forming during late denoising.

2.7 Occupations as a Probe for Demographic Bias in Text-to-Image Generation

Having established how bias manifests in datasets and propagates through generative models, an appropriate probing framework is required to measure these effects empirically. Occupations represent one of the most well-validated analytical lenses for this purpose. Job titles carry historically accumulated societal associations between professional roles and demographic attributes, extensively documented in real-world labour statistics and deeply reflected in the large-scale text and image corpora on which modern models are trained. Neutral occupation prompts therefore provide a transparent and interpretable window onto a model’s learned demographic priors, making them a standard experimental instrument across both natural language processing and text-to-image generation research.

2.7.1 Occupational Stereotyping in Language and Vision Systems

Evidence that AI systems acquire and reproduce occupational stereotypes is well established across a range of model families. Nadeem et al. [91] demonstrate through their large-scale StereoSet benchmark, covering 120 profession-related target terms, that all major pretrained language models including BERT, GPT-2, RoBERTa, and XLNet exhibit significant stereotypical preferences in profession-related contexts. Notably, they find a consistent positive relationship between a model’s language modelling ability and the strength of its stereotypical biases, suggesting that stereotype acquisition is a systematic consequence of learning from real-world text distributions. This is directly relevant to the present work because the CLIP text encoders that condition text-to-image models are pretrained on precisely such distributions, providing a theoretically grounded pathway through which occupational stereotypes propagate into generated imagery.

In the vision-language domain, Harrison et al. [92] demonstrate that human annotators systematically apply role-based names more frequently to male athletes than to female athletes performing identical activities, with an average of 46% sport-related names for male subjects versus 35% for female subjects. Because captioning datasets of this kind form part of the training corpora for large vision-language models, such annotation asymmetries constitute a direct mechanism through which occupational concepts become demographically skewed in model representations.

2.7.2 Occupational Bias in Text-to-Image Models

Within text-to-image generation, Naik and Nushi [93] provide a comprehensive audit confirming that occupation prompts reliably surface severe representational biases in both DALL-E v2 and Stable Diffusion v1. Generating images with neutral prompts such as a photo of a doctor and annotating outputs for perceived gender, race, and age, they find that model outputs deviate dramatically from real-world occupational demographics as reported by the US Bureau of Labor Statistics. Occupations such as chief executive officer and computer programmer yield images that are almost exclusively male and white, while nurse and housekeeper yield almost exclusively female images. Chuang et al. [94] further demonstrate that these biases are embedded in the structure of the CLIP text embedding space itself, with demographic directions geometrically separable from occupational content, confirming that occupation prompts are a particularly reliable and interpretable probe for the demographic biases arising from web-scale training data.

2.7.3 Occupation Classification: The ISCO-08 Framework

For occupation-based probing to yield generalisable results, the prompt set must be drawn from a principled taxonomy covering the full breadth of occupational space rather than over-representing stereotypically gendered or racialised roles selected by convenience. The International Labour Organization’s International Standard Classification of Occupations (ISCO-08) [95] provides this foundation, organising occupations into a four-level hierarchy: ten major groups, 43 sub-major groups, 130 minor groups, and 436 unit groups, each assigned a numeric code and skill-level designation ranging from elementary occupations at level 1 through highly qualified professionals at level 4.

In this thesis, ISCO-08 serves as the organising framework for the occupation prompt set. By sampling job titles systematically across all ten major groups, the resulting set spans a representative range of social prestige, skill requirements, and

cultural connotation, ensuring that bias measurements reflect the model's general behaviour across occupational space rather than being driven by a small subset of roles with strong stereotypical associations. The use of an internationally standardised classification also facilitates comparison with prior work that has likewise adopted ISCO-08 as a reference point [93].

3 Methodology

3.1 Image Retrieval

3.1.1 Data Acquisition and Subsetting

Re-LAION-2B-en-research-safe is distributed as 128 shards of approximately 3.62 GB each. Due to storage and computational constraints, processing was restricted to four shards selected uniformly at random, yielding an initial subset of 65,552,809 URL-caption pairs. Because LAION shards are generated during distributed web crawling without semantic ordering, shard index does not correspond to content category, domain, or temporal structure; random selection therefore provides a principled basis for treating the subset as representative of the broader corpus. Under a binomial sampling model, a worst-case 95% margin of error of approximately $\pm 0.04\%$ is achieved for proportion estimates at this scale. Nonetheless, unknown crawl-time biases such as domain saturation effects cannot be excluded without access to LAION crawl metadata, and all findings are reported as representative estimates rather than exact population parameters.

3.1.2 Image Source Distribution

Analysis of valid image origins revealed that 18.79% of successfully retrieved images originated from five domains: `cdn.shopify.com` (3,053,782 images), `i.pinimg.com` (2,585,519), `i.ebayimg.com` (966,820), `images-na.ssl-images-amazon.com` (886,054), and `thumbs.dreamstime.com` (678,516). This concentration indicates strong domain centralisation, with large commercial platforms contributing disproportionately to the retrieved image pool. Product imagery and stock photography may therefore be overrepresented relative to content from smaller independent domains.

3.1.3 Image Validity Criteria

Since Re-LAION-5B distributes URL-caption pairs rather than image binaries, each URL was resolved and the corresponding image downloaded. An image was deemed valid only if it satisfied all of the following conditions: a GET request completed successfully with an HTTP 2xx status code within a 10-second timeout; the response body was successfully decoded and converted to RGB; and embedding computation succeeded. Where embedding computation failed for a batch, recursive binary splitting was applied to isolate and exclude individual corrupted images without falling back to serial processing.

3.1.4 Decay Rate

From 65,552,809 attempted URLs, 43,491,938 images were successfully retrieved and validated, yielding a decay rate of:

$$\text{Decay Rate} = 1 - \frac{43,491,938}{65,552,809} \approx 0.337$$

This level of link attrition is consistent with prior observations of web-scale dataset degradation [2, 18]. Link decay is not necessarily uniform: if certain domains or content types become unavailable at disproportionate rates, the surviving image pool may be systematically skewed relative to the original dataset composition, introducing survivorship bias that cannot be quantified post-hoc.

3.1.5 Image Preprocessing and Embedding Computation

All images were preprocessed using the canonical pipeline for `openai/clip-vit-large-patch14`, including resizing to 224×224 pixels with aspect ratio preservation, centre cropping, and channel-wise normalisation using CLIP-specific mean and standard deviation values. CLIP image embeddings were extracted using the same model (embedding dimensionality: 768; patch resolution: 14×14) under `torch.no_grad()` in evaluation mode with GPU acceleration.

Each embedding vector $\mathbf{e}$ was L2-normalised:

$$\mathbf{e}_{\text{norm}} = \frac{\mathbf{e}}{\|\mathbf{e}\|_2}$$

This enforces unit-length representations and enables cosine similarity retrieval via inner-product equivalence, as described in Section 3.2.2. Results were accumulated and flushed to Parquet files incrementally, with checkpointing to support fault tolerance and resumability across the multi-month processing window.

While the original LAION-5B filtering pipeline employed CLIP ViT-B/32, this work uses the higher-capacity ViT-L/14 variant, which offers 768-dimensional embeddings, finer spatial granularity, and stronger semantic alignment in retrieval benchmarks. Since the objective is bias analysis rather than replication of LAION filtering thresholds, the use of a higher-capacity backbone is methodologically appropriate. The pipeline was executed on an NVIDIA RTX 3060 Ti GPU with 64 GB of system RAM over approximately two to three months.

Images were accessed exclusively via publicly available URLs for academic research purposes only. No authentication mechanisms were bypassed, no image binaries were redistributed, and no attempt was made to identify individuals. Analysis was conducted at aggregate statistical levels, with the research objective being to

analyse bias rather than amplify or deploy it.

The primary limitations of the retrieval process are as follows. Only 4 of 128 shards were processed, so findings apply strictly to the processed subset. Unknown crawl-time biases cannot be excluded without access to LAION crawl metadata. Approximately 33.7% of URLs were unavailable, introducing potential survivorship bias. Domain concentration from five platforms accounts for 18.79% of valid images. Embeddings reflect biases inherent to the CLIP pretraining corpus, and results are encoder-dependent.

3.2 Profession-Conditioned Image Retrieval

3.2.1 Profession-Conditioned Subset Construction

Following large-scale embedding computation, profession-conditioned imagery was selected as the primary analytical lens because occupational prompts provide a transparent and interpretable probe for a model’s learned demographic priors, as established in Section 2.7. A set of 200 diverse professions was constructed and aligned with ISCO-08 occupational categories to ensure systematic coverage across skill levels and domain areas; the full list is provided in the Appendix.

An image is considered structurally valid if it contains exactly one clearly detectable face associated with an identifiable person bounding box and satisfies the spatial and anatomical consistency criteria defined in Section 3.3.4. This definition is applied consistently across both the retrieval and generation pipelines.

3.2.2 FAISS Index Construction

To enable efficient similarity-based retrieval over the 43 million L2-normalised embeddings, a dedicated FAISS IndexFlatIP index was constructed per shard, accompanied by a position-to-path mapping. Since embeddings were L2-normalised during extraction, inner product search over IndexFlatIP is equivalent to cosine similarity retrieval. Index construction processed each Parquet file row-group by row-group, with periodic atomic checkpointing to ensure crash recovery. At inference time, shard indexes were streamed sequentially, with per-shard results aggregated and globally re-ranked by similarity score.

3.2.3 Prompt Design

The retrieval template used for each profession was:

"A photo of a/an {profession}"

The grammatical article was assigned programmatically based on the initial character of the profession name. Gender-neutral phrasing was used throughout to avoid reinforcing gender priors at the query stage.

3.2.4 Negative Prompt Dominance Filtering

A negative prompt dominance filter was applied to suppress retrieval of semantically irrelevant content. For each candidate image, its CLIP embedding was compared against embeddings derived from the following negative prompts: Cartoon, NSFW, Sex, Naked, Clothing, Object, Sign, Logo, Document, Paper, Page, Animal. A candidate was rejected if the maximum cosine similarity to any negative concept satisfied:

$$\text{sim}_{\text{neg}} \geq \text{sim}_{\text{pos}} - \delta$$

where sim_{pos} is the cosine similarity to the target profession prompt and $\delta = 0.01$ is a dominance margin. This enforces strict positive dominance: an image is retained only when the target profession prompt is unambiguously more similar to the image than any negative concept by a margin exceeding the threshold.

3.2.5 Embedding-Level Deduplication

To prevent the same image from appearing multiple times within a given profession's result set, each candidate embedding was hashed via its byte representation, and duplicates were discarded prior to result accumulation. Cross-profession deduplication was not enforced, as the same image may legitimately be semantically relevant to multiple professions.

3.2.6 Retrieval Parameters

For each of the 200 professions, the top 125,000 candidates were retrieved per shard and aggregated globally. A cosine similarity threshold of 0.15 was applied to remove near-random matches while maintaining a sufficiently large candidate pool for downstream structural filtering. Following structural validation, the top 1,000 highest-scoring structurally valid images were retained per profession. This two-stage design decouples semantic relevance from structural suitability.

3.2.7 Face Detection Model Selection

Face detection is a critical gating step in both the retrieval and generation pipelines: images without a clearly detectable face are rejected prior to downstream demographic annotation, so systematic failure on particular face scales or poses would introduce filtering bias. Four candidate models were evaluated on the WIDER Face validation set [33], reporting AP at IoU threshold 0.5 (AP@0.5) overall and stratified by face size, as well as COCO-style mAP. Table 3.1 summarises all results and Figure 3.1 shows the corresponding precision-recall curves.

Table 3.1 Face detection model evaluation on the WIDER Face validation set. AP@0.5 is reported overall and stratified by face size. mAP@0.50:0.95 is the COCO-style average across IoU thresholds. [†] Evaluated using the `batch_face` library, which uses a different model checkpoint and preprocessing pipeline from the original RetinaFace implementation; results reflect this deployment configuration rather than the model’s published capability.

Model	AP@0.5	mAP@0.5:0.95	Small	Medium	Large
YOLOv12l-Face	0.905	0.556	0.783	0.915	0.948
SCRFD	0.844	0.478	0.614	0.884	0.954
RetinaFace [†]	0.319	0.176	0.011	0.376	0.712
MediaPipe	0.117	0.040	0.000	0.015	0.497

MediaPipe [96] achieved only 0.117 AP@0.5 with zero recall on small faces, exhibiting a recall ceiling consistent with its design as a conservative single-face detector optimised for mobile inference. RetinaFace [32], evaluated via the `batch_face` library, achieved 0.319 AP@0.5 with near-zero small-face recall; the precision-recall curve reveals systematic detection suppression rather than a precision-recall trade-off, and the library-level deviation from the original implementation limits interpretability. SCRFD [34] achieved 0.844 AP@0.5 overall but its small-face AP of 0.614 falls substantially below YOLOv12l-Face. YOLOv12l-Face achieved the highest overall AP@0.5 at 0.905 and the highest mAP@0.50:0.95 at 0.556, with superior small-face detection at 0.783. Its precision-recall curve sustains near-perfect precision to approximately 0.8 recall, a substantially wider high-precision operating range than SCRFD, which begins degrading at around 0.6 recall. YOLOv12l-Face was therefore selected as the face detection model for all filtering and annotation pipelines.

3.2.8 Human-Centric Structural Filtering

CLIP-based retrieval does not guarantee that retrieved images contain clearly identifiable individuals. A multi-stage structural validation pipeline was applied to enforce human-centric image suitability. Images were loaded in batches using

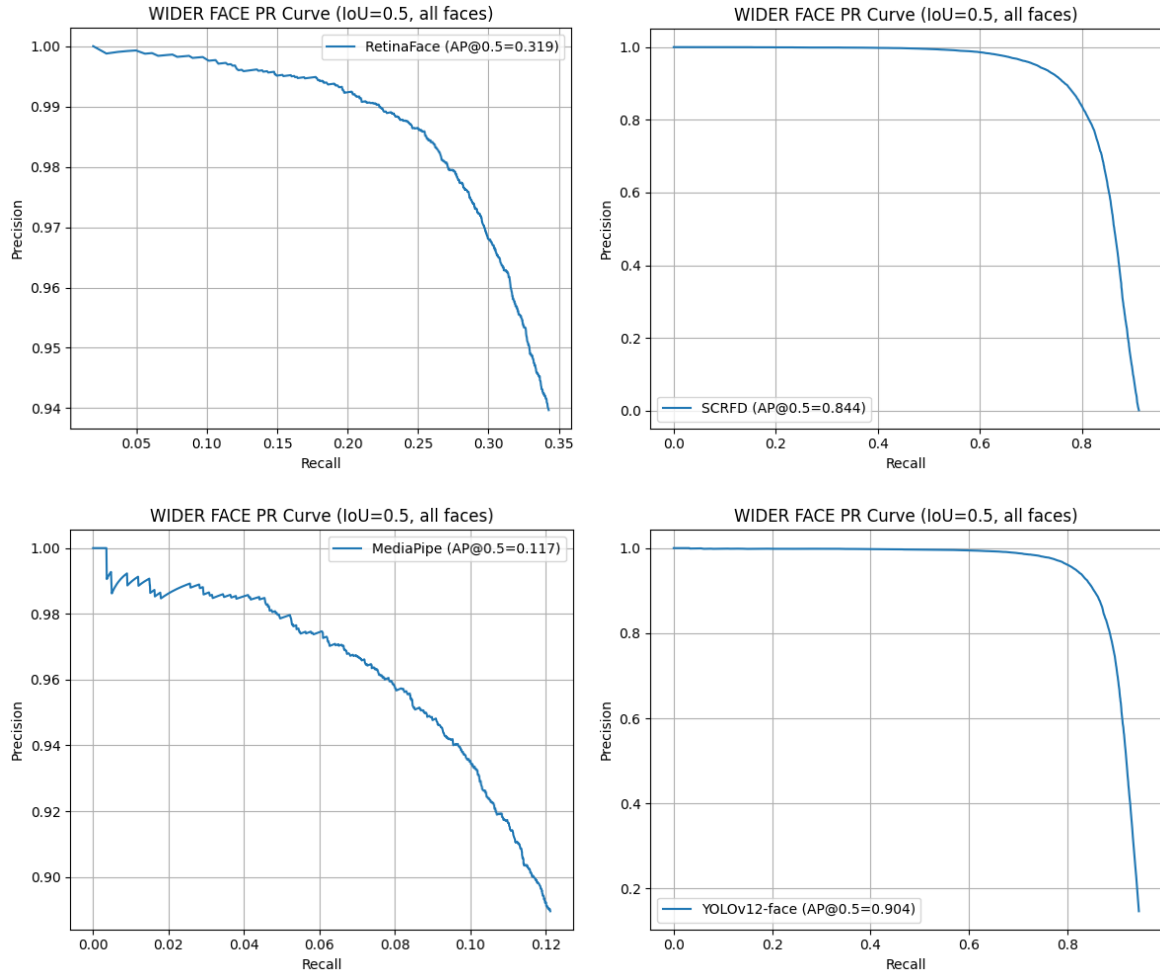


Figure 3.1 Precision-recall curves at IoU= 0.5 on the WIDER Face validation set for all four evaluated face detection models. RetinaFace and MediaPipe both exhibit a recall ceiling below 0.35 at near-perfect precision, indicating systematic detection suppression. SCRFD maintains high precision to approximately 0.6 recall before degrading. YOLOv12l-Face achieves the widest high-precision operating range, sustaining near-perfect precision to approximately 0.8 recall, consistent with its superior AP@0.5 and small-face detection performance.

TurboJPEG for JPEG inputs with OpenCV as fallback. Face detection required exactly one detected face with confidence ≥ 0.6 ; person detection required at least one detected person with confidence ≥ 0.5 . Spatial consistency was verified by requiring the face centroid to lie within a person bounding box, or IoU between the face and person bounding boxes to be ≥ 0.3 , with a face-to-person area ratio exceeding 0.02. Images satisfying face but not person detection criteria were admitted provided MediaPipe FaceMesh segmentation succeeded on the face region, avoiding unnecessary rejection of images where the face is clearly detectable but the person bounding box is slightly misaligned.

Following successful detection, MediaPipe FaceMesh was applied to the face region of interest. Landmarks were used to compute a convex hull mask from which a

tightly cropped face ROI was extracted with a 10-pixel padding. Images for which FaceMesh failed to return landmarks were discarded.

3.2.9 Filtering Outcomes and Acceptance Rate Distribution

Across all 200 professions, 25,125,679 candidate images were submitted to structural validation, of which 8,420,684 satisfied all criteria, yielding an overall acceptance rate of 33.51% (mean 33.51%, median 33.67%, standard deviation 10.71%). The minimum observed acceptance rate was 9.55% and the maximum 59.21%.

Figure 3.2 illustrates the distribution of per-profession acceptance rates. The distribution is approximately unimodal with a visible left tail, consistent with the expectation that CLIP retrieval quality is profession-dependent: occupations associated with clear single-person visual conventions yield more structurally valid results than those whose online representation is dominated by equipment, environments, or group scenes. The overall rejection rate of approximately 66% confirms that CLIP-based retrieval alone is insufficient for demographic analysis.

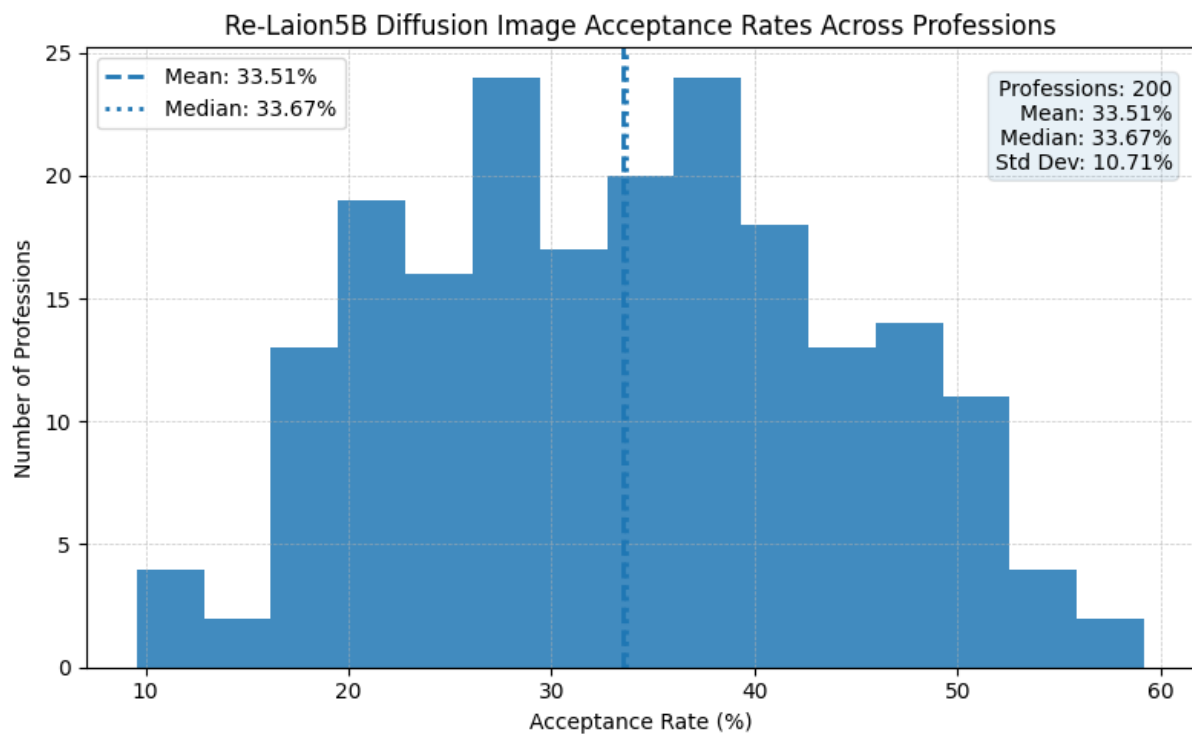


Figure 3.2 Distribution of structural acceptance rates across 200 professions following CLIP retrieval and YOLO-based structural filtering of Re-LAION-5B images. The distribution is approximately unimodal with a mean of 33.51% and median of 33.67%, with a visible left tail reflecting professions whose online visual representation is dominated by non-person content.

3.2.10 Subset Design and Statistical Justification

For each of the 200 professions, 1,000 images were retained following structural validation, yielding a balanced profession-conditioned subset of 200,000 images. Under a binomial model for binary feature presence, a per-profession sample of $n = 1,000$ yields a worst-case 95% margin of error of:

$$\text{MOE} = z_{0.975} \cdot \sqrt{\frac{p(1-p)}{n}} = 1.96 \cdot \sqrt{\frac{0.25}{1000}} \approx \pm 3.1\%$$

This margin is sufficiently small to detect meaningful cross-profession differences in spurious feature prevalence. When aggregated across all 200 professions, the full 200,000-image subset yields highly precise overall estimates (worst-case $\pm 0.22\%$ at 95% confidence). Equal sampling across professions produces a balanced evaluation set that prevents high-frequency professions from disproportionately influencing aggregate statistics, though it implies that aggregate prevalence estimates reflect an equal-weighted average over professions rather than the natural profession frequency distribution within Re-LAION-5B.

3.3 Image Generation Pipeline

3.3.1 Model Selection

A corresponding set of synthetic images was required to enable direct comparison between training data and generative model outputs. Model selection was governed by two criteria: first, training data traceability to a documented LAION-derived corpus; and second, inference throughput sufficient to generate 200,000 images within available computational constraints.

Applying the traceability criterion first, candidate models were restricted to those with a publicly documented training lineage derived from Re-LAION-5B. Three configurations satisfied this requirement: Stable Diffusion v1.5 [17], SDv1.5 with LCM LoRA [17, 97], and Stable Diffusion v1.2 [98], all trained exclusively on LAION-Aesthetics v2 5+ [16]. Subsequent iterations were excluded: Stable Diffusion 2.0 and 2.1 employed a different text encoder architecture [99] introducing confounds; SDXL, Stable Diffusion 3, and later releases were trained on proprietary datasets of undisclosed composition [26, 27], severing the documentable link to Re-LAION-5B that this analysis requires.¹

Among the three qualifying configurations, model selection was based on

¹Additional architectures screened during preliminary evaluation — SSD-1B [100], PixArt-Alpha [101], and Segmind Vega [102] — were excluded solely on the basis of training data provenance.

inference throughput. Per-image inference times are reported in Table 3.2. SDv1.5 with LCM LoRA achieved 1.5 seconds per image, compared to 5.6 seconds for standard SDv1.5 and 9.1 seconds for SDv1.2. At the scale of 200,000 images, this corresponds to a projected generation time of approximately 83 hours versus 311 hours, making the LCM-accelerated configuration the only practically feasible option within the available computational window.

Table 3.2 Per-image inference times for LAION-derived model configurations at 512×512 resolution. Times are the mean over four generation runs on an NVIDIA RTX 3060 Ti. [†]LCM LoRA acceleration applied.

Model	Training Corpus	Time (s/image)
SDv1.5 LCM LoRA [†] [17, 97]	LAION-Aesthetics v2 5+	1.5
SDv1.5 [17]	LAION-Aesthetics v2 5+	5.6
SDv1.2 [98]	LAION-Aesthetics v2 5+	9.1

Sample outputs from each configuration are shown in Figure 3.3. SDv1.5 and SDv1.5 LCM both produce consistently full-body, anatomically plausible compositions, with SDv1.5 LCM retaining structural reliability at substantially reduced inference cost and only a marginal reduction in fine texture detail. SDv1.2 produces outputs of comparable quality but at 9.1 seconds per image, offering no throughput advantage.

SDv1.5 was selected as the base model, loaded with the `latent-consistency/lcm-lora-sdv1-5` adapter [97] and fused for fast inference. The default DDPM scheduler was replaced with `LCMScheduler`, enabling high-quality generation through consistency distillation [30]. A four-step inference schedule was used throughout; preliminary testing confirmed that fewer than four steps introduces visible distortion while additional steps yield no measurable improvement. Generation was performed in half-precision (FP16) on GPU.

3.3.2 Prompt Design and Justification

The prompt template used for all profession-conditioned generation was:

```
full-body realistic photo of a {profession}, standing,
centered, in a professional setting appropriate for their
occupation, sharp focus, 35mm lens, natural professional
lighting
```

Full-body framing was specified explicitly to override Stable Diffusion’s strong prior toward portrait-style close-ups [1] and to ensure compatibility with downstream structural validation, which requires a full detected person bounding box [103].



Figure 3.3 Sample outputs from the ten model configurations screened during preliminary evaluation, generated using a fixed Doctor prompt at 512×512 resolution. Four samples are shown per configuration. Of these, only SDv1.5, SDv1.5 LCM, and SDv1.2 satisfy the LAION training lineage criterion and are formally evaluated for model selection.

Standing and centred positioning reduces compositional ambiguity from edge-cropping artefacts and multiple subjects [104]. An abstract contextual phrase rather than a specific environment was used, as explicit environment tokens are known to amplify demographic stereotypes [6, 12]. Sharp focus improves edge fidelity for downstream face and person detection under low-step sampling [30], a 35mm focal length token promotes naturalistic full-body framing [105], and neutral professional lighting suppresses noise artefacts and supports aesthetic quality scores above the downstream threshold [12].

3.3.3 Guidance Scale and Its Methodological Implications

The classifier-free guidance scale was fixed at $s = 1.0$ across all validated generation runs. At $s = 1$, no guidance amplification is applied and the model produces outputs conditioned purely on its learned prior. Standard Stable Diffusion deployment uses guidance scales of $s = 7-7.5$; this choice was deliberately avoided because, as described in Section 2.2, elevated guidance amplifies any demographic or compositional bias encoded in the learned prior. By fixing $s = 1$, any demographic bias identified represents a conservative lower bound: at guidance scales used in production deployment the same biases would be further amplified. If bias is detectable at its minimum, it follows that the bias is structural and attributable to model weights and training data rather than inference-time configuration.

3.3.4 Structural Validation Criteria and Unfiltered Control Subset

Unlike retrieval from Re-LAION-5B, generative models produce images without guarantees of compositional suitability. Common failure modes include multiple subjects, truncated bodies, disconnected or misaligned faces, and low-frequency blurred renderings from low-step inference. Structural validation is therefore a necessary preprocessing stage. Generation proceeded asynchronously: a ValidationWorker pool decoupled CPU-bound validation from GPU generation, allowing the main generation loop to remain active while workers processed previously generated images in parallel. Per-profession resumability was implemented so that professions with sufficient existing valid images were skipped without re-running generation.

An image was accepted only if it satisfied all of the following conditions. The YOLO person detection model must detect at least one bounding box with confidence ≥ 0.4 . The YOLO face detection model must detect exactly one face with confidence ≥ 0.4 . MediaPipe FaceMesh must successfully return at least 10 landmarks for the face region, from which a convex hull mask and tightly cropped face ROI with 10-pixel

padding were derived. For spatial and anatomical consistency, a composite score was computed per detected person bounding box:

$$\text{score} = 10 \cdot \mathbf{1}[\text{centroid_inside}] + 2 \cdot \mathbf{1}[\text{upper_half}] + \min\left(\frac{A_{\text{face}}}{A_{\text{person}}}, 0.05\right) \cdot 20$$

where $\mathbf{1}[\text{centroid_inside}]$ indicates that the face centroid lies within the person bounding box, $\mathbf{1}[\text{upper_half}]$ indicates the face centroid lies in the upper half of the person bounding box, and $A_{\text{face}}/A_{\text{person}}$ is the face-to-person area ratio. Images for which no valid spatial association could be established were rejected, as were those with a face-to-person area ratio below 0.002. Finally, each accepted image was scored using a CLIP-based aesthetic regressor (ViT-L-14 backbone, laion2b_s32b_b82k pretraining) and images with an aesthetic score below 3.0 were rejected, as described in Section 2.2.2.

3.3.5 Structural Acceptance Rates for Generated Images

Across all 200 professions, the mean structural acceptance rate for generated images was 66.38% (median 71.00%, standard deviation 16.60%), compared to 33.51% for Re-LAION-5B retrieval. The substantially higher acceptance rate is consistent with the effect of full-body prompt engineering, which biases the diffusion model toward compositions satisfying single-person detection and face-person spatial association requirements. Nonetheless, per-profession variability remains substantial, indicating that certain occupations activate compositional priors that systematically resist full-body single-subject framing. The systematic difference in mean acceptance rate between retrieved and generated subsets must be considered when comparing demographic or spurious feature prevalence across the two data sources.

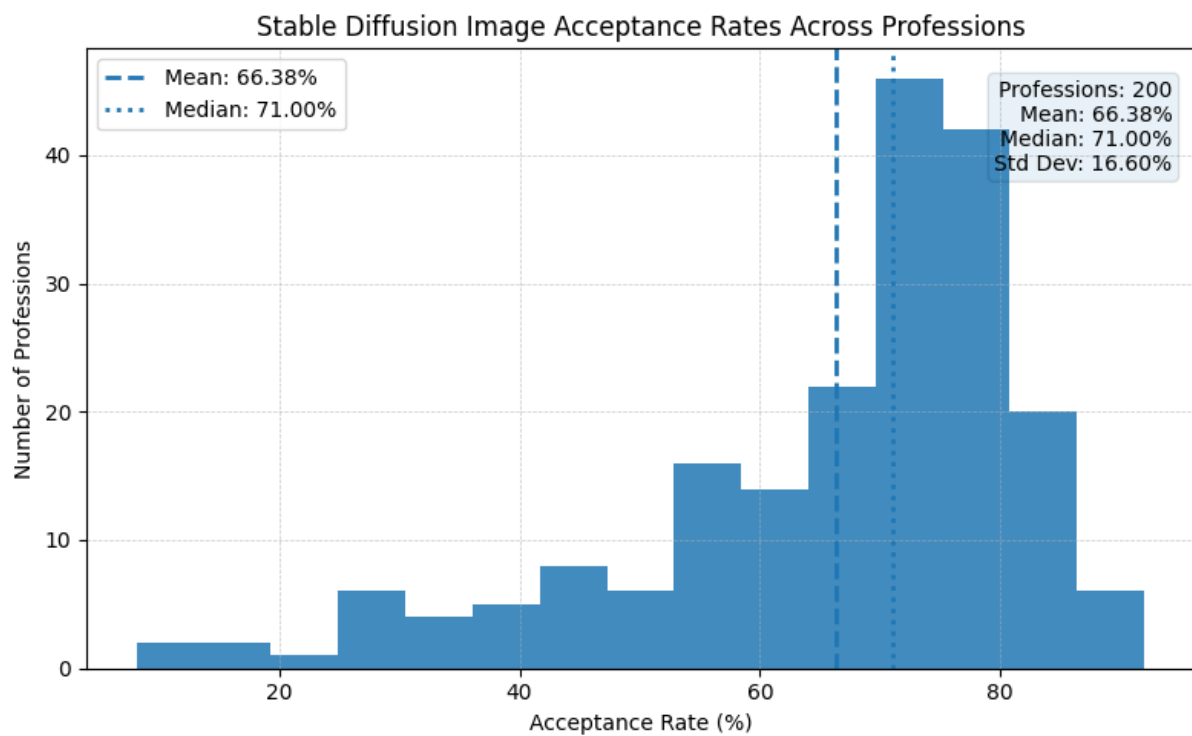


Figure 3.4 Distribution of structural acceptance rates across 200 professions for Stable Diffusion v1.5 generated images. The distribution is left-skewed with a mean of 66.38% and median of 71.00%, substantially higher than the Re-LAION-5B retrieval acceptance rate of 33.51%.

3.4 Demographic Annotation

To enable bias analysis across gender and skin tone dimensions, every face-containing image in both the Re-LAION-5B retrieval set and the Stable Diffusion v1.5 generation set required demographic labels. This section describes the common pre-processing applied to all annotation tasks, the candidate model evaluation used to select a gender classifier, and the equivalent evaluation used to select a skin tone estimator. All demographic labels are treated as model-derived proxies of perceived attributes rather than ground truth, and are used exclusively to analyse distributional correlations; the limitations of this approach are discussed in Section 2.5.4.

3.4.1 Pre-processing

All images were passed through a consistent face segmentation step using MediaPipe FaceMesh [36]. FaceMesh fits a dense 468-landmark mesh to the detected face region; a convex hull was computed from the landmarks, and pixels outside the hull were replaced with a uniform black fill, yielding a tightly cropped, background-removed face image. This is the same FaceMesh configuration employed during structural validation, and the same face crop produced there was reused here. Applying segmented face crops rather than full bounding-box crops prevents models from exploiting background cues that may correlate incidentally with demographic attributes [81], and ensures that accuracy differences across candidate models reflect genuine differences in representational capacity rather than the image region each model receives.

For the FACET benchmark [75], an agreement-based filter was applied to the skin tone annotations prior to model evaluation. The agreement ratio for each sample was computed as the proportion of total annotator votes assigned to the majority skin tone label; only samples with an agreement ratio of at least 0.2 were retained. This threshold was selected as the minimum viable filter to suppress low-agreement samples that would otherwise inflate per-model error rates, while avoiding the substantial reduction in available sample count that a higher threshold would produce.

3.4.2 Gender Annotation

Candidate Models

Five off-the-shelf gender classifiers were evaluated on segmented face crops: the HuggingFace Gender Classifier [106], a ViT-B/16 model fine-tuned for binary gender classification; the Realistic Gender Classifier [107], a SigLIP-backbone classifier trained on high-quality portrait imagery; DeepFace [108], a multi-attribute facial analysis

library; InsightFace [109], a face analysis toolkit with an integrated demographic attribute head; and FairFace [73], a classifier trained on a demographically balanced dataset.

Evaluation Protocol

Models were evaluated on Casual Conversations v2 (CCv2) [74] and FACET [75], both pre-processed as described above. DeepFace was evaluated exclusively on FACET due to the absence of a batch inference interface. Two CCv2 conditions were evaluated: full images and face-only crops. Accuracy was computed as the proportion of correct binary predictions and decomposed into male and female subgroup rates.

Results

Table 3.3 summarises overall and per-class accuracy across models on the CCv2 face-only and FACET conditions.

Table 3.3 Gender classification accuracy on CCv2 face-only segmented crops and FACET. Overall, male (M), and female (F) accuracies are reported. DeepFace was not evaluated on CCv2 face-only due to the absence of a batch inference interface. The Realistic Gender Classifier achieves the highest overall accuracy on CCv2 face-only and the best balance between male and female subgroup accuracy.

Model	CCv2 Face-Only			FACET		
	Overall	M	F	Overall	M	F
HF Gender Classifier	0.715	0.963	0.561	0.789	0.877	0.569
Realistic Gender Clf.	0.886	0.924	0.862	0.823	0.875	0.695
InsightFace	0.771	0.784	0.763	0.772	0.821	0.643
FairFace	0.801	0.983	0.687	0.834	0.906	0.653
DeepFace	—	—	—	0.780	0.938	0.384

The Realistic Gender Classifier achieved an overall accuracy of 88.6% on CCv2 face-only, with a male–female gap of 6.2 percentage points. This gap is substantially smaller than those of the HuggingFace classifier (40.2 pp), FairFace (29.6 pp), and DeepFace on FACET (55.4 pp), all of which exhibit a systematic tendency to classify ambiguous or darker-toned faces as male. InsightFace achieved the smallest male–female gap (2.1 pp on CCv2 face-only) but at the cost of an 11.5 percentage point reduction in overall accuracy, indicating that its more balanced per-class rates arise from lower male recall rather than improved female recall.

Per-Skin-Tone Performance of the Selected Model

Table 3.4 disaggregates the Realistic Gender Classifier's CCv2 face-only accuracy by MST level, quantifying the degree to which annotation quality varies across the demographic dimension the thesis analyses.

Table 3.4 Per-MST-level accuracy of the Realistic Gender Classifier on CCv2 face-only segmented crops. Overall, male (M), and female (F) sub-group accuracies and sample counts (N) are reported. Accuracy is highest for mid-range tones (MST 4–6) and degrades at both extremes, with female accuracy declining more sharply at darker tones.

MST	N	Overall	M Acc.	F Acc.
1	1,190	0.756	0.977	0.681
2	13,337	0.870	0.968	0.833
3	33,732	0.893	0.946	0.870
4	38,273	0.893	0.939	0.871
5	55,797	0.906	0.918	0.895
6	23,074	0.879	0.884	0.875
7	6,438	0.826	0.906	0.761
8	4,183	0.803	0.925	0.680
9	1,671	0.662	0.948	0.488
10	168	0.726	1.000	0.570

Overall accuracy peaks at MST 5 (90.6%) and declines toward both the lightest and darkest tones. Male accuracy remains consistently high across all MST levels while female accuracy degrades sharply at darker tones, falling to 48.8% at MST 9, consistent with the intersectional accuracy disparities documented in prior work [56]. These limitations must be accounted for when interpreting gender-disaggregated results at the darker end of the skin tone range. Despite these disparities, the Realistic Gender Classifier offered the best combination of overall accuracy, male–female balance, and inference throughput and was therefore selected for final annotation.

3.4.3 Skin Tone Annotation

Scale Selection

The Monk Skin Tone (MST) scale [78] was adopted over the Fitzpatrick scale [76]. The Fitzpatrick scale was developed to predict ultraviolet radiation sensitivity and has been criticised for conflating clinical and perceptual dimensions and for providing insufficient resolution at dark skin tones [77, 78]. A large-scale survey by Heldreth et al. [79] found that Fitzpatrick was perceived as significantly less inclusive by darker-skinned respondents, while MST was rated comparably to a 40-shade commercial palette despite its coarser granularity.

Skin tone predictions were produced as raw ten-point MST scores and additionally collapsed into a three-bin grouping following Matias and Neto [81]: Light (MST 1–3), Mid (MST 4–7), and Dark (MST 8–10). The three-bin representation is used as the primary grouping for aggregate bias analyses, as the coarser granularity is more robust to annotation noise under variable imaging conditions, though full ten-point MST distributions are additionally reported where finer-grained inspection is warranted.

Candidate Models

Four skin tone estimation approaches were evaluated: the SkinTone Classifier Library [110], an off-the-shelf tool deriving MST estimates from colour histograms; a RandomForest histogram-based regressor trained on CCv2 following Matias and Neto [81]; DenseNet-121 [81] trained for MST classification in LAB colour space; and VGG-16 [80] fine-tuned for MST estimation and subsequently ablated across four design axes. All trained models were evaluated on a held-out CCv2 validation split, the Monk Skin Tone Estimation (MSTE) benchmark, and FACET, using exact accuracy, within-one accuracy, and three-bin accuracy.

Baseline Results

Table 3.5 reports global accuracy for the baseline models. The SkinTone Library assigns near-zero probability to MST 1–3 across all datasets, rendering it unsuitable for tasks requiring reliable coverage of light skin tones. The RandomForest generalises poorly beyond its training distribution, concentrating predictions in MST 4–6 with near-zero accuracy at the extremes. DenseNet-121 generalises more evenly across the tone range. VGG-16 exceeded DenseNet-121 on all CCv2 metrics and was selected for further optimisation via the ablation study below.

Table 3.5 Global accuracy of baseline skin tone models. E = exact, ± 1 = within-one, B = three-bin. All trainable models were trained on CCv2.

Model	MSTE			CCv2			FACET		
	E	± 1	B	E	± 1	B	E	± 1	B
SkinTone Library	0.112	0.281	0.425	0.084	0.298	0.603	0.073	0.239	0.421
RandomForest	0.132	0.335	0.410	0.346	0.756	0.700	0.176	0.550	0.484
DenseNet-121	0.210	0.517	0.564	0.303	0.692	0.697	0.210	0.529	0.576

VGG-16 Ablation Study

A factorial ablation was conducted across four binary design dimensions, yielding sixteen VGG-16 variants evaluated under two label spaces (MST10 and MST3). The four axes were output formulation (classification versus regression), colour space (RGB versus CIE L*a*b*), label granularity (MST10 versus MST3), and background treatment (black background versus average skin-tone background fill). Tables 3.6 and 3.7 report results for all variants.

Table 3.6 VGG-16 MST10 variant accuracy. E = exact, ± 1 = within-one, B = three-bin. BB = black background, AB = average skin-tone background. Bold indicates the best value in each column.

Mode	Space / BG	MSTE			CCv2			FACET		
		E	± 1	B	E	± 1	B	E	± 1	B
Class.	RGB / BB	0.161	0.423	0.577	0.336	0.717	0.735	0.172	0.465	0.587
Class.	RGB / AB	0.217	0.445	0.529	0.341	0.636	0.717	0.119	0.384	0.547
Class.	LAB / BB	0.228	0.493	0.568	0.346	0.697	0.723	0.145	0.437	0.544
Class.	LAB / AB	0.162	0.445	0.492	0.283	0.669	0.674	0.148	0.442	0.583
Regr.	RGB / BB	0.159	0.424	0.566	0.375	0.804	0.778	0.165	0.471	0.593
Regr.	RGB / AB	0.132	0.367	0.528	0.374	0.784	0.783	0.173	0.499	0.572
Regr.	LAB / BB	0.146	0.423	0.549	0.350	0.776	0.734	0.185	0.532	0.587
Regr.	LAB / AB	0.163	0.494	0.556	0.370	0.778	0.775	0.207	0.567	0.596

Table 3.7 VGG-16 MST3 variant accuracy. Since the label space contains only three classes, exact accuracy and three-bin accuracy are identical. Bold indicates the best value in each column.

Mode	Space / BG	MSTE		CCv2		FACET	
		Exact	± 1	Exact	± 1	Exact	± 1
Class.	RGB / BB	0.557	0.939	0.771	0.994	0.583	0.983
Class.	RGB / AB	0.545	0.945	0.765	0.993	0.581	0.972
Class.	LAB / BB	0.478	0.883	0.723	0.957	0.534	0.944
Class.	LAB / AB	0.540	0.901	0.727	0.990	0.502	0.908
Regr.	RGB / BB	0.627	0.992	0.801	1.000	0.640	0.996
Regr.	RGB / AB	0.584	0.978	0.799	1.000	0.619	0.997
Regr.	LAB / BB	0.578	0.996	0.749	1.000	0.604	0.997
Regr.	LAB / AB	0.526	0.995	0.777	0.999	0.598	0.998

Model Selection

Three conclusions emerge from the ablation. First, the regression formulation outperforms classification on the metrics most relevant to downstream analysis: in the

MST10 setting, RGB regression achieves CCv2 within-one accuracy of 80.4% and three-bin accuracy of 77.8%, compared to 71.7% and 73.5% for the classification variant. Regression produces a continuous output rounded to the nearest integer at inference time, distributing prediction probability smoothly across adjacent MST classes rather than committing fully to a single discrete class, which is methodologically preferable when the downstream analysis relies on three-bin groupings. Second, RGB outperforms LAB as the input colour space, suggesting that the CCv2 training distribution already provides sufficient implicit lighting variation to support luminance-robust representations. Third, black background consistently matches or outperforms the average skin-tone background across regression variants, suggesting that the additional chrominance introduced by the mean face fill provides no representational benefit and may partially confound the skin tone signal.

The final selected model is the VGG-16 MST10 Regression RGB Black-Background variant, achieving the performance summarised in Table 3.8.

Table 3.8 Final selected model performance: VGG-16 MST10 Regression RGB Black-Background.

Dataset	Exact Acc.	Within-One Acc.	Three-Bin Acc.
CCv2 (held-out)	0.375	0.804	0.778
MSTE	0.159	0.424	0.566
FACET	0.165	0.471	0.593

The lower MSTE and FACET exact accuracies relative to CCv2 reflect two factors: distributional shift from the CCv2 training distribution, and the MSTE benchmark’s design of assigning identical ground truth labels across illumination conditions while the model was trained without explicit lighting augmentation. Three-bin accuracy is more robust to both effects, as the broader grouping boundaries absorb within-class prediction variance.

Final Annotation Pipeline

The selected gender classifier and skin tone model were integrated into a unified annotation pipeline. For each input image the pipeline executes in sequence: FaceMesh segmentation produces a black-background face crop; the Realistic Gender Classifier yields a binary gender label and confidence score; the VGG-16 MST10 Regression model yields a continuous MST score rounded to the nearest integer and mapped to a Light/Mid/Dark bin. The resulting labels are written to a per-image JSONL manifest. The pipeline supports GPU batch inference and is fully resumable via the manifest file.

3.4.4 Real-World Occupational Demographic Reference Data

To contextualise observed demographic distributions against ground-truth labour force composition, two complementary reference datasets were derived. The US-specific baseline was sourced from the Bureau of Labor Statistics (BLS) Current Population Survey Table 11 (CPSAAT11), reporting employed civilian workers by detailed occupation, sex, race, and Hispanic or Latino ethnicity for the 2024 reference year. The global baseline was sourced from ILOSTAT, providing employment data disaggregated by sex and two-digit ISCO-08 occupational group across reporting countries. The US baseline serves as the primary reference due to the Western-web-dominated composition of large-scale web-scraped datasets such as Re-LAION-5B, as reflected in prior work on text-to-image models [6, 7].

The 200 professions were mapped to BLS labels via a hierarchical pipeline: first a manually curated authoritative mapping covering ambiguous cases; then automated matching via case-insensitive substring containment, keyword overlap scoring, and fuzzy string matching. Of the 200 professions, 197 were successfully matched and 3 were intentionally excluded. Professions matched to BLS categories with fewer than 50,000 employed workers were retained for gender analysis but excluded from race-stratified comparisons.

For the global baseline, professions were mapped to ISCO-08 four-digit codes via the WageIndicator WISCO database [111], truncated to two-digit sub-major group codes for joining against ILOSTAT data. Because ILOSTAT reports at the ISCO-2 level, all professions sharing the same two-digit group receive identical global female percentages. This coarser granularity means the global baseline should be interpreted as a broad occupational group-level reference rather than a per-profession ground truth.

3.4.5 Race-to-Skin-Tone Distribution Mapping

Since the BLS reference data reports demographic composition in terms of racial categories while the annotation pipeline produces MST skin tone labels, a principled mapping was required. This mapping was derived empirically using FairFace [73] as a calibration corpus. The VGG-16 MST regression model was applied to FairFace face crops to predict a discrete MST label per image, and for each of the seven FairFace race categories a conditional distribution $P(\text{MST} \mid \text{race})$ was computed. The expected MST distribution for a given profession p was then computed as:

$$P(\text{MST} = m \mid p) = \sum_r P(\text{MST} = m \mid \text{race} = r) \cdot w_{r,p} \quad (3.1)$$

where $w_{r,p}$ is the BLS-reported racial composition fraction for race group r in profession p . This provides a profession-specific expected skin tone reference distribution against which observed MST distributions can be compared.

This approach carries several limitations. The FairFace racial categories do not align perfectly with BLS categories, requiring approximate mappings for Asian and Hispanic subgroups. The mapping inherits systematic errors from the VGG-16 model and reflects the skin tone distribution of FairFace’s image collection rather than the true population distribution. Inspection of the resulting conditional distributions reveals that all racial groups map predominantly to the Mid bin, reflecting the tonal range of the FairFace corpus rather than the full MST scale; the utility of these distributions for real-world skin tone comparison is therefore limited, as discussed further in Section 4.5.

3.5 Image Transformation Pipeline for Spurious Feature Analysis

Having established demographic annotation labels for both image sets, the analysis turns to characterising the spurious features present in Re-LAION-5B and Stable Diffusion v1.5 outputs. Unless otherwise stated, each sample corresponds to a single image-derived representation (for example, a transformed image, caption, or segmented face region) with demographic and dataset labels inherited from the source image. This design isolates representation-level signal by pairing fixed labels with transformed inputs, ensuring that any predictive performance arises from retained feature information rather than changes in annotation. Within each task (gender or skin tone), the underlying image set, label assignments, and data splits are held constant across all transformations, ensuring that observed differences arise solely from the representation. Data splits differ between the gender and skin tone tasks due to stratified sampling requirements, but remain fixed within each task. All probes within a given task are trained and evaluated under identical data splits and training protocols, with fixed random seeds to eliminate training stochasticity as a source of variation. Each transformation-specific probe is trained independently to isolate feature-specific signal without cross-representation leakage. All comparisons between datasets are conducted at the level of probe outputs on transformed representations, and are interpreted within-task, focusing on relative differences between transformations rather than absolute performance across demographic attributes.

The transformations applied serve two distinct probing objectives. The first, following Zeng et al. [9], applies image modifications to isolate specific visual feature categories as probes for dataset-discriminating signal, testing whether a classifier can distinguish Re-LAION-5B, Stable Diffusion v1.5 outputs, and COCO images from each transformation alone. The second, following Meister et al. [8], applies a subset of these representations as probes for demographic signal, testing whether gender and skin tone can be predicted from low-level visual features independently of semantic content. Where a transformation serves both objectives it is described once and cross-referenced in the demographic probing context.

The dataset classification task is performed across three classes: Re-LAION-5B images, Stable Diffusion v1.5 generated images, and COCO images. The inclusion of COCO as a structurally independent reference enables per-class accuracy attribution from the confusion matrix. Under a binary classification over Re-LAION-5B and Stable Diffusion v1.5, a high overall accuracy cannot distinguish whether the discriminating signal originates entirely from one dataset or from both simultaneously. By extending to a three-class problem, per-class accuracies (A_L, A_S, A_C) become individually

interpretable: if $A_S \gg A_L \approx A_C \approx 1/3$, the signal is localised to the generative outputs; if $A_L \gg A_S \approx A_C \approx 1/3$, the signal is localised to Re-LAION-5B, pointing to properties of its web-scraping and filtering pipeline. The chance level $1/3$ serves as the lower bound for each class, replacing the binary chance level of $1/2$.

Not all transformations applied in the dataset classification setting are extended to the demographic probing setting. Transformations that present the full image to the probe model were deliberately excluded, as these representations retain both person and scene content without isolating either region. Any demographic signal recovered from such inputs cannot be attributed unambiguously to person appearance or to scene-level contextual correlates, undermining the interpretive goal of localising signal within specific, separable information channels. Where a full-image representation is nonetheless relevant to demographic probing, a modified representation is derived that discards scene-level content while retaining the target channel; this is described for each affected transformation below.

Where the same transformation appears in both settings, the probe model is matched to the structure and dimensionality of the representation and kept deliberately simple, following Meister et al. [8]. If a linear or shallow model recovers demographic signal from a compressed representation, that is strong evidence the signal is directly encoded there rather than being an artefact of a powerful classifier exploiting non-linear interactions. Accordingly, mean RGB is probed with logistic regression over the three-dimensional colour vector rather than a ConvNeXt-Tiny operating on a uniform-colour canvas, and object detection is probed via logistic regression over detection feature vectors rather than a convolutional classifier over the rendered white-canvas image, avoiding the reintroduction of spatial layout as a confound. Where a spatial input is unavoidable a ResNet-50 model is used, consistent with Meister et al. and providing a conservative upper bound relative to the ConvNeXt-Tiny used in dataset classification.

Monocular Depth Estimation

The depth transformation isolates the spatial geometry of each image by suppressing colour and texture cues while retaining only per-pixel relative depth values encoding scene structure, object positioning, and subject-to-background distance. Zeng et al. report that depth-transformed images achieve 73.1% classification accuracy across the YCD datasets, confirming that spatial geometry constitutes a meaningful axis of dataset-level bias. Depth maps were produced using Depth-Anything-V2 [112], resizing each image to 384×384 pixels prior to inference. The predicted depth tensor was upsampled to original image dimensions via bicubic interpolation, normalised to $[0, 255]$, and saved as a three-channel PNG by replicating the grayscale channel across

all colour channels for compatibility with the ConvNeXt-Tiny classifier. A final resize to 224×224 pixels was applied after depth computation. The ViT-S variant of Depth-Anything-V2 was used in place of the larger variant employed by Zeng et al.; their ablation demonstrates that pre-trained model size has minimal impact on downstream classification accuracy for this transformation [9, Table 3]. Representative depth maps are shown in Figure 3.9.

For the demographic probing experiments, the dense depth map risks reintroducing person silhouette information that could correlate with gender independently of depth geometry. A novel extension therefore samples the depth map only at the seventeen COCO keypoint locations identified by the pose estimator, producing a sparse seventeen-dimensional depth vector per image. Because only scalar depth values at discrete joint locations are retained, the representation exposes no silhouette geometry while still capturing the relative depth profile of the body. Depth values at visible keypoints were sampled at the nearest integer pixel coordinate, with joints below confidence 0.5 assigned a depth value of zero. The raw depth vector was per-person z-score normalised across visible joints to remove absolute subject-to-camera distance, with both raw and normalised variants saved to JSONL.

Edge Detection

Edge detection transforms an image into a sparse boundary map capturing object contours, silhouettes, and spatial intensity discontinuities while suppressing colour, texture, and semantic content. Zeng et al. report that Canny edge detection achieves 71.0% classification accuracy and SAM-based contour extraction reaches 73.2% across the YCD datasets; the 2.2 percentage point difference is within the range attributable to dataset composition differences rather than methodological superiority, and the computational overhead of SAM’s automatic mask generator across the full three-dataset corpus is disproportionate to the interpretive gain. Canny edge detection was therefore adopted as the sole edge representation.

Edge maps were derived using the classical Canny algorithm [51] as implemented in OpenCV. Each image was resized to 224×224 pixels, converted to greyscale, and smoothed with a 3×3 Gaussian kernel prior to gradient computation. Resizing before edge extraction avoids the edge thinning and thickening artefacts that arise when downsampling a precomputed edge map. Hysteresis thresholds of $t_1 \approx 85$ and $t_2 = 255$ were applied, retaining weak edges only when connected to strong edges, matching the implementation used by Zeng et al. Example edge maps are shown alongside depth and pose outputs in Figure 3.9.

Frequency Domain Filtering

Frequency domain filtering decomposes an image into low-frequency content (smooth colour gradients, global illumination, coarse structure) and high-frequency content (fine textures, sharp edges, compression artefacts). Zeng et al. report classification accuracies of 70.4% for low-pass and 79.2% for high-pass filtered images, both close to the 82.0% reference, confirming that spurious signals exist simultaneously across the full frequency spectrum and that mitigation targeting a single frequency range is insufficient.

Frequency filtering was applied using an ideal circular mask with a hard cutoff radius of $r = 40$ pixels, matching the primary configuration of Zeng et al. Each image was resized to 224×224 pixels, converted to greyscale, and the two-dimensional discrete Fourier transform computed via `numpy.fft.fft2` with frequency shift to centre the zero-frequency component. The low-pass mask retains all components within the circular region; the high-pass mask is its complement. Each masked spectrum was inverse-transformed, with the real component extracted and normalised to $[0, 255]$ via per-image min-max scaling before saving as a greyscale PNG. Both low-pass and high-pass representations were derived from a single FFT pass per image. Representative outputs are shown in Figure 3.5.

Pixel Shuffling, Patch Shuffling, and Mean RGB

Three related transformations progressively reduce the spatial information available to a classifier, from complete spatial destruction through controlled local structure preservation to global colour summarisation, isolating the contribution of global composition, local structure, and colour statistics independently.

Pixel shuffling randomly permutes the spatial position of every pixel while preserving the overall colour distribution exactly, reducing the available signal to global colour statistics alone. Zeng et al. report that this reduces classification accuracy from 82.0% to 52.2%, just above the 33.3% chance level, confirming that colour statistics carry only modest discriminative signal and that spatial structure is the dominant axis of dataset bias. Each image was resized to 224×224 pixels, and pixels were randomly permuted with a per-image deterministic seed derived from the sum of the RGB channel values of the top-left pixel.

Patch shuffling generalises this by permuting fixed-size non-overlapping patches rather than individual pixels, preserving local structure within each patch while randomising global arrangement. Zeng et al. demonstrate that 16×16 patch shuffling recovers nearly all discriminability at 81.2%, establishing that local spatial structure within small image regions is sufficient to distinguish datasets. Patch shuffling was applied at four scales (2×2 , 4×4 , 8×8 , and 16×16 pixels) to characterise the accuracy



Figure 3.5 Frequency domain transformation outputs applied to three profession-conditioned images. Low-Pass Filter: reconstruction from frequency components within the circular cutoff radius $r = 40$, retaining smooth global luminance and colour gradients while suppressing fine-grained detail. High-Pass Filter: the complementary residual, isolating fine textures, sharp edges, and compression artefacts. Both representations are derived from a single FFT pass per image and saved as greyscale PNGs.

profile across patch sizes rather than relying on a single scale. Each patch size constitutes an independent classification experiment.

Mean RGB compresses each image to a single average colour per channel, filling the entire canvas with that uniform colour and leaving only the three-dimensional mean colour vector as the available signal. Zeng et al. report that even this extreme reduction achieves 48.5% accuracy, approximately 15 percentage points above chance, confirming that global colour distributions carry a non-trivial spurious signal. The raw mean RGB vector and a unit-range normalised equivalent were additionally recorded to a JSONL statistics file to support auxiliary linear probe experiments. Mean RGB also features in the demographic feature isolation experiments following Meister et al., where the raw three-dimensional vector is supplied directly to a logistic regression classifier rather than the filled uniform-colour image being passed to the ConvNeXt-Tiny network. Figure 3.6 illustrates the three spatial decomposition transformations.

Variational Autoencoder Reconstruction

The VAE reconstruction transformation passes each image through the encoder and decoder of a pretrained Variational Autoencoder, attenuating fine-grained low-level properties such as JPEG compression artefacts and sensor noise while preserving semantically meaningful content. As a spurious feature probe, it isolates whether dataset-discriminating signal survives latent compression. Zeng et al. report that VAE-reconstructed images achieve 77.4% classification accuracy compared to the 82.0% reference, confirming that low-level acquisition properties contribute only marginally to the overall discriminative signal. Reconstructions were produced using the KL-regularised autoencoder (`autoencoder_kl_64x64x3`) from the Stable Diffusion v1.5 architecture [1], with images resized to 256×256 pixels prior to encoding and the posterior mode used for deterministic reconstructions. The decoded output was clamped to $[0, 1]$ and resized to 224×224 pixels for classifier compatibility.

A methodological consideration is that the VAE used for reconstruction is the same architecture and checkpoint embedded within Stable Diffusion v1.5. Generated images are therefore passed through the same decoder that produced them, meaning any latent-space artefacts introduced during generation are preserved with higher fidelity than for Re-LAION-5B or COCO images. This asymmetry should be considered when interpreting per-class accuracy on VAE-reconstructed images: higher separability for Stable Diffusion outputs may partially reflect architectural proximity rather than solely the presence of spurious semantic features. Example VAE reconstructions are shown in Figure 3.8.

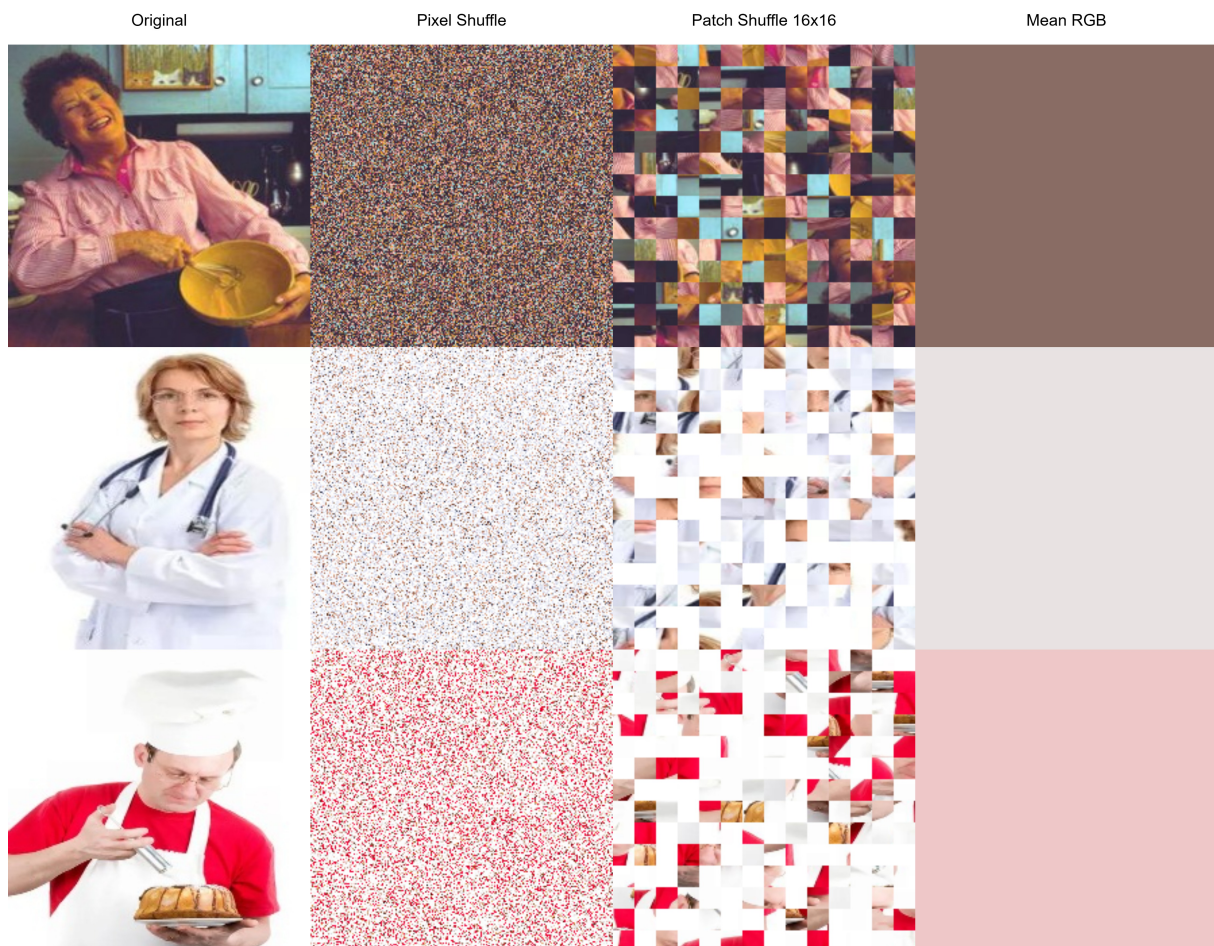


Figure 3.6 Spatial decomposition transformations applied to three profession-conditioned images. Pixel Shuffle: all pixels randomly permuted, preserving the global colour distribution while destroying all spatial structure. Patch Shuffle 16×16 : non-overlapping patches randomly permuted, preserving local texture within each patch while disrupting global composition. Mean RGB: the entire image replaced by a uniform fill of the per-channel spatial mean, retaining only the three-dimensional global colour vector.

Semantic Segmentation

Semantic segmentation assigns a class label to every pixel using a fixed category vocabulary, retaining the spatial distribution of object categories while discarding all texture, colour, and appearance information. Zeng et al. report that segmentation using the ADE20K vocabulary achieves 69.8% classification accuracy, lower than purely structural transformations such as depth and edge detection, suggesting that object identity alone does not fully account for the discriminative signal captured by those representations.

Segmentation maps were produced using Mask2Former [113] with a Swin-Large backbone pretrained on ADE20K. Each image was resized to 512×512 pixels prior to inference. The predicted per-pixel class index map was resized to 224×224 pixels via nearest-neighbour interpolation before palette application, ensuring that interpolation operates directly on discrete class labels. This work uses the Swin-Large variant rather than the BEiT_{v2}-ViT-Adapter-Large backbone used by Zeng et al., which was excluded on computational grounds. Both models use the same ADE20K vocabulary and produce semantically equivalent output representations; since the downstream classification task operates on the category distribution rather than pixel-level precision, interpretive conclusions remain valid, though absolute accuracy may differ slightly from the 69.8% reported by Zeng et al. Representative segmentation maps are shown in Figure 3.7.

Object Detection

Object detection reduces each image to a set of labelled bounding boxes, retaining object identity, approximate spatial location, and co-occurrence statistics while discarding all pixel-level appearance information. Zeng et al. report that object detection achieves 61.9% classification accuracy across the YCD datasets, lower than segmentation and structural transformations, suggesting that object identity and co-occurrence alone account for a smaller share of dataset-level bias than spatial geometry.

Four detection vocabularies were considered. COCO (80 classes) was excluded as its vocabulary is too coarse to capture the range of occupation-relevant objects. LVIS (1,203 classes) was excluded as its long tail of rare categories adds detection noise without improving interpretability. Object365 (365 classes) was excluded as it lacks the breadth of occupation-relevant clothing and implement classes available in OpenImages V7. OpenImages V7 (601 classes) was selected for its broad and balanced range of categories directly relevant to professional contexts.

Eight detection models were evaluated on a 10,000-image subset for throughput viability, summarised in Table 3.9. ViTDet-Huge [114] and GroundingDINO

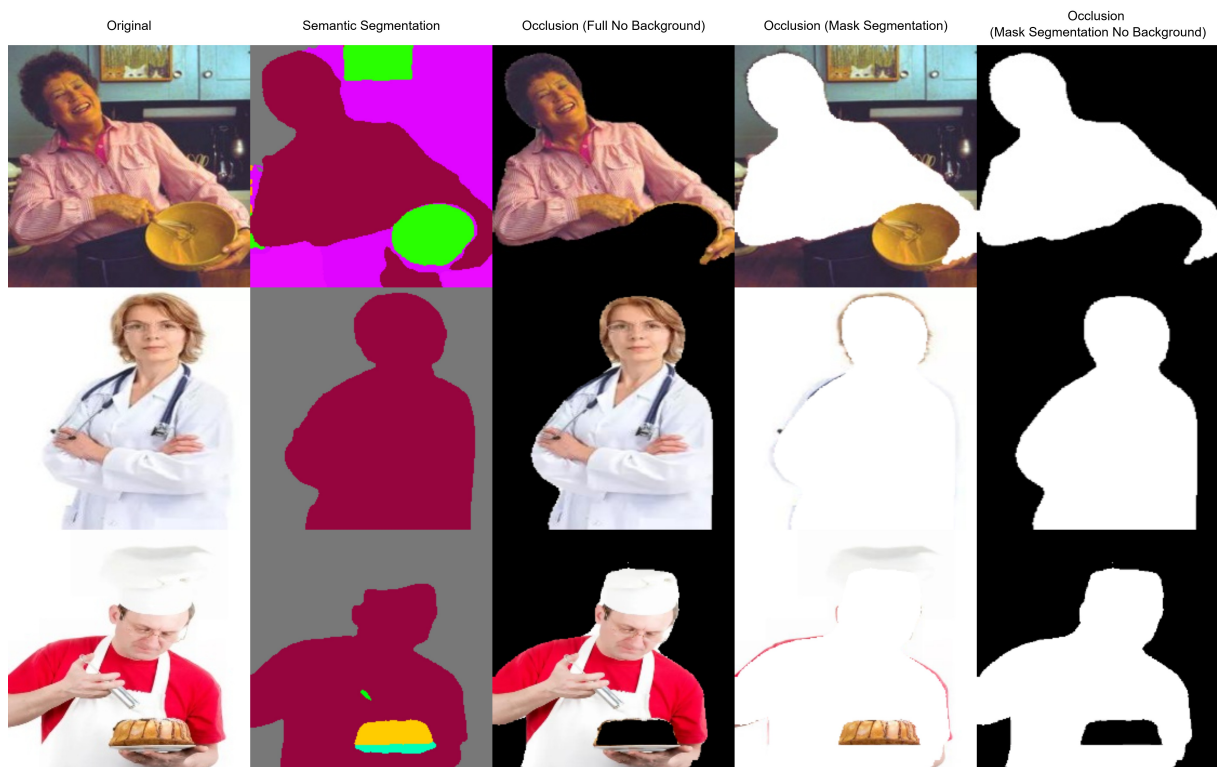


Figure 3.7 Semantic segmentation and segmentation-derived occlusion representations applied to three profession-conditioned images. Semantic Segmentation: ADE20K colour-coded pixel-label map produced by Mask2Former with a Swin-Large backbone. Occlusion (Full No Background): the subject isolated from the scene with all background pixels zeroed. Occlusion (Mask Segmentation): the person region replaced by a uniform white fill, removing the subject while preserving background context. Occlusion (Mask Segmentation No Background): the person mask only, rendered as a white silhouette on a black background.

[115] both exceeded one hour of inference time and were excluded as impractical. RT-DETR-X [116] was excluded as its available pretrained weights target COCO and Object365 rather than OpenImages V7. The YOLOv8x-World variants achieved substantially degraded performance on OpenImages V7 when prompted with the class list rather than natively trained, as shown in Table 3.10. YOLOv8x with OpenImages V7 pretrained weights was selected, achieving mAP@0.5:0.95 of 0.3614 with practical throughput of approximately 26 images per second.

Table 3.9 Object detection model inference time on a 10,000-image subset of Re-LAION-5B at 640×640 pixel input resolution.

Model	Time (10k images)	Architecture
YOLOv8x (OIV7)	~7 min 06 sec	CNN, closed-vocabulary
YOLOv8m (OIV7)	~5 min 47 sec	CNN, closed-vocabulary
YOLOv8x-World	~8 min 11 sec	CNN, open-vocabulary
YOLOv8x-Worldv2	~7 min 23 sec	CNN, open-vocabulary
YOLOv8x + World (Ensemble)	~12 min 44 sec	Ensemble
RT-DETR-X	~6 min 24 sec	Transformer, end-to-end
GroundingDINO	>60 min	Vision-language model
ViTDet-Huge	>60 min	Transformer backbone

Table 3.10 Detection performance on the OpenImages V7 validation set.

Model	mAP@0.5	mAP@0.5:0.95	Precision	Recall
YOLOv8x (OIV7)	0.4345	0.3614	0.4786	0.4374
YOLOv8m (OIV7)	0.4094	0.3336	0.4791	0.4007
YOLOv8x-World	0.1370	0.1140	0.2044	0.2568
YOLOv8x-Worldv2	0.1403	0.1160	0.1847	0.2600

Detection was performed at 640×640 pixel resolution with a confidence threshold of 0.1 and an IoU threshold of 0.5. Twelve human body-part classes present in the OpenImages V7 vocabulary were excluded from all outputs, as these carry person-level appearance information rather than object co-occurrence signal. Detected bounding boxes were rendered as overlays on a uniform white canvas, removing all pixel-level appearance information and retaining only object identity, spatial position, and bounding box geometry, with outputs resized to 224×224 pixels for classifier compatibility. Representative detection outputs are shown in Figure 3.8.

A label remapping strategy was applied to normalise class name variants and reduce gender-correlated label noise. The standard remapping consolidates gendered person labels (Man, Woman, Boy, Girl) into a single Person class. The restricted remapping additionally consolidates clothing and accessory labels carrying gender-correlated associations (Dress, Brassiere, High heels, Lipstick) into generic counterparts (Clothing, Footwear, Cosmetics, Fashion accessory), enabling a cleaner

test of whether dataset-discriminating signal persists once gender-correlated object vocabulary is suppressed. Table 3.11 summarises both configurations.

Table 3.11 Label remapping configurations applied during object detection.

Original Label	Standard Remap	Restricted Remap	Category
Man, Woman, Boy, Girl	Person	Person	Person
Dress, Skirt, Miniskirt, Brassiere	—	Clothing	Clothing
Suit, Coat, Jacket, Jeans, Shirt	—	Clothing	Clothing
Shorts, Trousers, Tie, Scarf, Belt	—	Clothing	Clothing
Hat, Sun hat, Cowboy hat, Fedora, Sombrero	—	Clothing	Clothing
High heels, Sandal, Boot, Sock	—	Footwear	Footwear
Lipstick, Face powder	—	Cosmetics	Cosmetics
Perfume, Hair spray	—	Personal care	Personal care
Earrings, Necklace, Tiara, Handbag	—	Fashion accessory	Fashion accessory

Image Captioning

Image captioning converts each image into a natural language description, reducing the visual representation to semantic content while discarding the underlying pixel representation. As a spurious feature probe, caption-based classification tests whether dataset-discriminating signal survives this extreme semantic compression: if a classifier operating on caption embeddings reliably distinguishes dataset origin, the bias must reside in systematic differences in scene content, object co-occurrence, or compositional conventions. Zeng et al. report classification accuracies of 63.8% for short captions and 66.1% for long captions using LLaVA 1.5 7B [117].

Ten models were assessed for throughput viability. LLaVA 1.5 7B, TinyLLaVA, and LLaVA-OneVision were excluded due to prohibitive inference times from autoregressive generation requirements limiting effective batch sizes to 2. BLIP2-Flan-T5-XL was excluded as 8-bit quantisation introduced an unnecessary constraint with no speed advantage. Moondream2 does not support batched inference. The five viable models — BLIP Large [118], Florence 2 Base [119], Kosmos 2 [120], ViT-GPT2, and Tiny — were advanced to quality evaluation. Table 3.12 summarises throughput results.

The five viable models were benchmarked on the Karpathy test splits of COCO 2014 [67] and Flickr30k [122] using BLEU@1–4 [61], METEOR [62], CIDEr [63], 4-gram diversity, SBERT [65], CLIP-S, and RefCLIP-S [66]. Results are summarised in Tables 3.13 and 3.14.

BLIP Large achieved the highest scores across the most diagnostic metrics on both evaluation datasets. Its CLIP-S scores of 0.6646 on COCO and 0.6550 on Flickr30k indicate strong visual grounding, and its RefCLIP-S scores of 0.7279 and

Table 3.12 Image captioning model throughput assessment on a single GPU.

Model	Batch Size	Time (25 images)	Viable
LLaVA 1.5 7B [117]	2	~27.0s	No
TinyLLaVA-Phi-2-SigLIP-3.1B	2	~180.0s	No
LLaVA-OneVision-Qwen2-0.5B	2	~180.0s	No
BLIP2-Flan-T5-XL [121]	25	~2.3s	No
Moondream2	1	N/A	No
BLIP Large [118]	25	~2.3s	Yes
Florence 2 Base [119]	25	~2.6s	Yes
Kosmos 2 [120]	25	~1.9s	Yes
ViT-GPT2	25	~2.1s	Yes
Tiny	25	~2.2s	Yes

Table 3.13 Classical caption quality metrics on the Karpathy test splits of COCO 2014 and Flickr30k. Low 4-gram diversity (underlined) indicates degenerate template-based generation.

Model	Dataset	BLEU@4	METEOR	CIDEr	4-gram Div.	SBERT
BLIP Large	COCO	0.1278	0.2096	0.0325	0.3406	0.1158
ViT-GPT2	COCO	0.1290	0.1918	0.0307	0.3246	0.0904
Tiny	COCO	0.1270	0.1964	0.0367	<u>0.0547</u>	0.0562
Kosmos 2	COCO	0.0947	0.2014	0.0050	0.3334	0.1449
Florence 2	COCO	0.0842	0.1754	0.0013	0.2978	0.1373
BLIP Large	Flickr30k	0.1254	0.1934	0.0268	0.4554	0.0866
ViT-GPT2	Flickr30k	0.1306	0.1890	0.0306	0.4935	0.0682
Tiny	Flickr30k	0.1386	0.1915	0.0375	<u>0.1499</u>	0.0479
Kosmos 2	Flickr30k	0.0992	0.2044	0.0051	0.4492	0.1079
Florence 2	Flickr30k	0.0855	0.1834	0.0011	0.4504	0.1194

0.6868 are the highest of all five candidates, confirming that its captions are simultaneously well-aligned with image content and semantically coherent relative to human references. A critical failure mode was identified in the Tiny model, which achieved a 4-gram diversity of only 0.0547 on COCO, indicating that fewer than 6% of all 4-grams across generated captions were unique — a hallmark of degenerate template-based generation that demonstrates why traditional n-gram metrics alone are insufficient. BLIP Large was therefore selected as the captioning model.

A word-level gender-neutral remapping was applied to all generated captions prior to saving, replacing gendered person references (woman, man) with person, gendered child references (boy, girl) with child, and gendered pronouns (he, she, his, her) with their gender-neutral equivalents (they, their, themselves). This ensures that downstream classifiers cannot exploit gendered language introduced by the captioning model as a proxy for dataset origin. Captions were generated using the prompt "An image of", following the captioning protocol of Peng et al. [120], with a maximum output length of 30 tokens and greedy decoding. Representative captions are shown

Table 3.14 CLIP-based caption quality metrics on the Karpathy test splits of COCO 2014 and Flickr30k. CLIP-S measures image-caption alignment scaled to $[0, 2.5]$; RefCLIP-S is the harmonic mean of CLIP-S and maximum reference similarity.

Model	Dataset	CLIP-S	RefSim	RefCLIP-S
BLIP Large	COCO	0.6646	0.8148	0.7279
ViT-GPT2	COCO	0.6040	0.8289	0.6951
Tiny	COCO	0.4110	0.6087	0.4853
Kosmos 2	COCO	0.5568	0.6314	0.5861
Florence 2	COCO	0.5927	0.6536	0.6163
BLIP Large	Flickr30k	0.6550	0.7330	0.6868
ViT-GPT2	Flickr30k	0.5290	0.6713	0.5867
Tiny	Flickr30k	0.4777	0.6371	0.5409
Kosmos 2	Flickr30k	0.5600	0.5766	0.5619
Florence 2	Flickr30k	0.6059	0.5759	0.5853

alongside object detection and VAE outputs in Figure 3.8.

Pose Estimation

Pose estimation reduces each image to a set of COCO-17 body joint coordinates, retaining only skeletal geometry while discarding all pixel-level appearance information. Following Meister et al. [8], pose representations isolate whether demographic attributes can be inferred from body geometry alone. Unlike preceding transformations that produce rendered image outputs consumed by the ConvNeXt-Tiny classifier, the pose pipeline produces 34-dimensional normalised keypoint vectors as direct input to a low-capacity MLP probe. This transformation is applied exclusively to Re-LAION-5B and Stable Diffusion v1.5 image sets; COCO images are excluded as they are not guaranteed to contain people in the occupational contexts under analysis, and pose estimation on images without clearly visible subjects produces uninformative zero vectors.

Seventeen model configurations across five architectures were evaluated on all person-annotated images in the COCO 2017 validation set to identify the best trade-off between accuracy and throughput, with results summarised in Table 3.15. MoveNet and MediaPipe were excluded on accuracy grounds, achieving AP below 0.55 even in Oracle mode. HRNet variants achieved the highest AP values but operate at under 9 FPS end-to-end, rendering full-corpus extraction prohibitively slow. YOLOv8l-Pose was selected for its superior throughput of 39.7 FPS combined with an AP of 0.659, with the practical advantage of requiring no external person detector. The accuracy gap relative to YOLOv8x-Pose (0.015 AP) is small and does not affect the downstream representational objective, since the pipeline discards all appearance information and retains only normalised joint geometry.

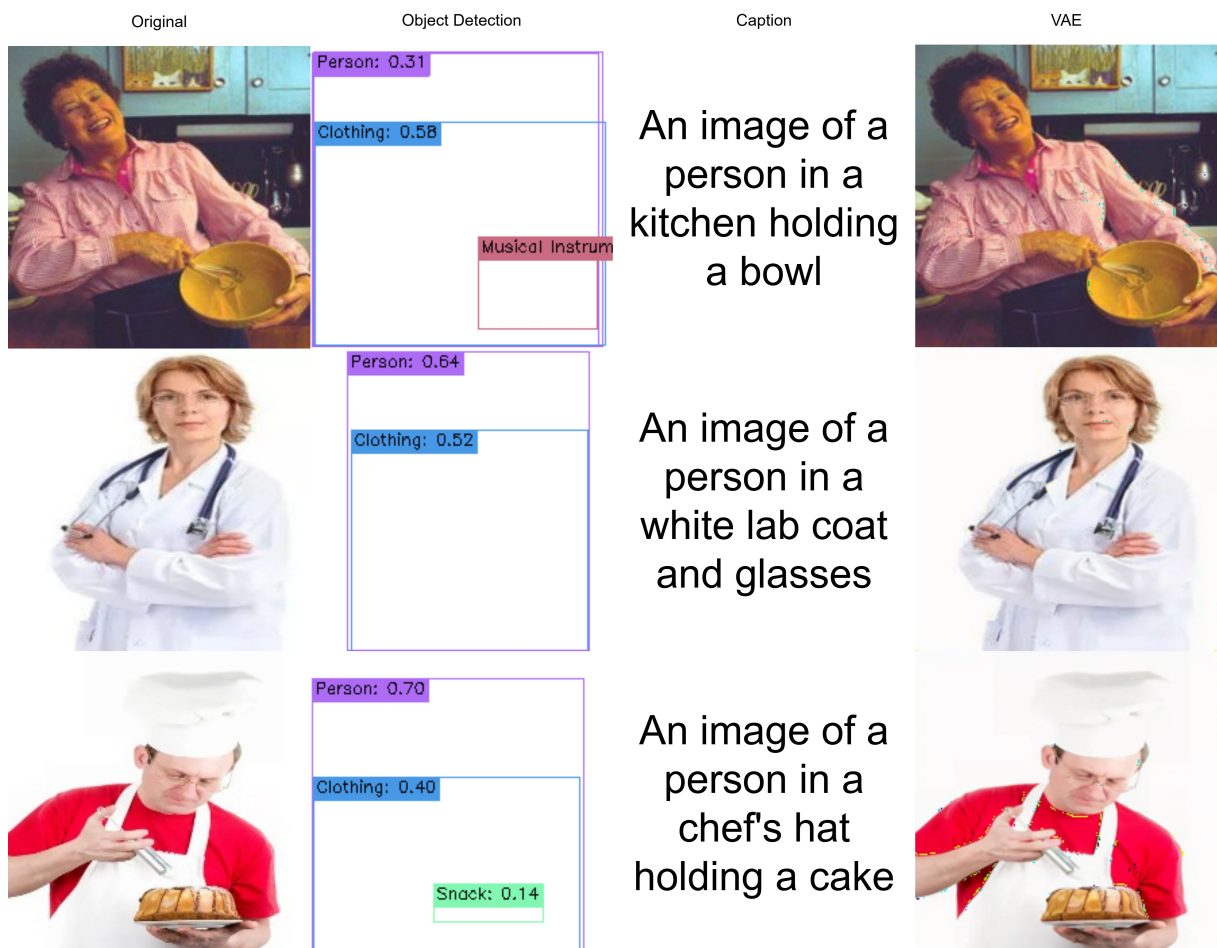


Figure 3.8 Semantic and latent-space transformation outputs applied to three profession-conditioned images. Object Detection: bounding boxes rendered on a white canvas using YOLOv8x with OpenImages V7 weights. Caption: short natural language descriptions generated by BLIP Large using the "An image of" prefix, following gender-neutral remapping. VAE: reconstructions produced by the KL-regularised autoencoder of Stable Diffusion v1.5.

Table 3.15 Pose estimation model evaluation on COCO 2017 validation set. Standard mode uses the model’s integrated person detector. Oracle mode provides ground-truth bounding boxes. FPS is measured end-to-end on a single GPU.

Model	AP	AP ₅₀	AP ₇₅	AR	FPS
<i>YOLOv8 (integrated detection + keypoints)</i>					
YOLOv8n-Pose	0.479	0.744	0.520	0.538	27.0
YOLOv8s-Pose	0.581	0.830	0.642	0.638	37.1
YOLOv8m-Pose	0.633	0.860	0.704	0.691	33.0
YOLOv8l-Pose	0.659	0.871	0.735	0.717	39.7
YOLOv8x-Pose	0.674	0.871	0.747	0.730	27.9
<i>MoveNet (requires external person detector)</i>					
MoveNet-Lightning (Standard)	0.419	0.621	0.456	0.450	9.9
MoveNet-Thunder (Oracle)	0.548	0.779	0.594	0.584	4.1
<i>MediaPipe (requires external person detector)</i>					
MediaPipe-C2 (Standard)	0.243	0.405	0.255	0.333	3.6
<i>HRNet (requires external person detector)</i>					
HRNet-W32 (Standard)	0.714	0.879	0.789	0.769	8.7
HRNet-W32-384 (Oracle)	0.776	0.933	0.853	0.808	8.5
HRNet-W48 (Standard)	0.723	0.880	0.793	0.778	7.9

For each image, YOLOv8l-Pose was applied at native image resolution with a person detection confidence threshold of 0.25. When multiple detections were returned, the detection with the highest box confidence score was retained. Joints with YOLO keypoint confidence at or above 0.5 were assigned visibility $v = 2$ and retained; joints below this threshold were zeroed and assigned $v = 0$. Pose geometry was normalised to remove absolute scale and image position: visible joints were centred by subtracting the mean joint position, and the bounding box area of visible joints was scaled to a fixed target area of 4,000 pixels, following the normalisation of Meister et al. The 34-dimensional normalised keypoint vector serves as direct input to the downstream MLP probe. Figure 3.9 shows skeleton overlays for visual reference; the downstream MLP probe receives the 34-dimensional normalised keypoint vector directly, with no rendered image produced by the pose pipeline.

Occlusion

Occlusion transformations selectively remove or isolate spatial regions of each image by replacing either the person or the background with a uniform fill, producing a controlled series of representations that expose different sources of demographic signal. Following Meister et al. [8], five occlusion variants are produced per image, applied exclusively to Re-LAION-5B and Stable Diffusion v1.5 images as demographic probes. COCO images are excluded because they are not conditioned on occupational queries and are not guaranteed to depict people in the professional contexts required

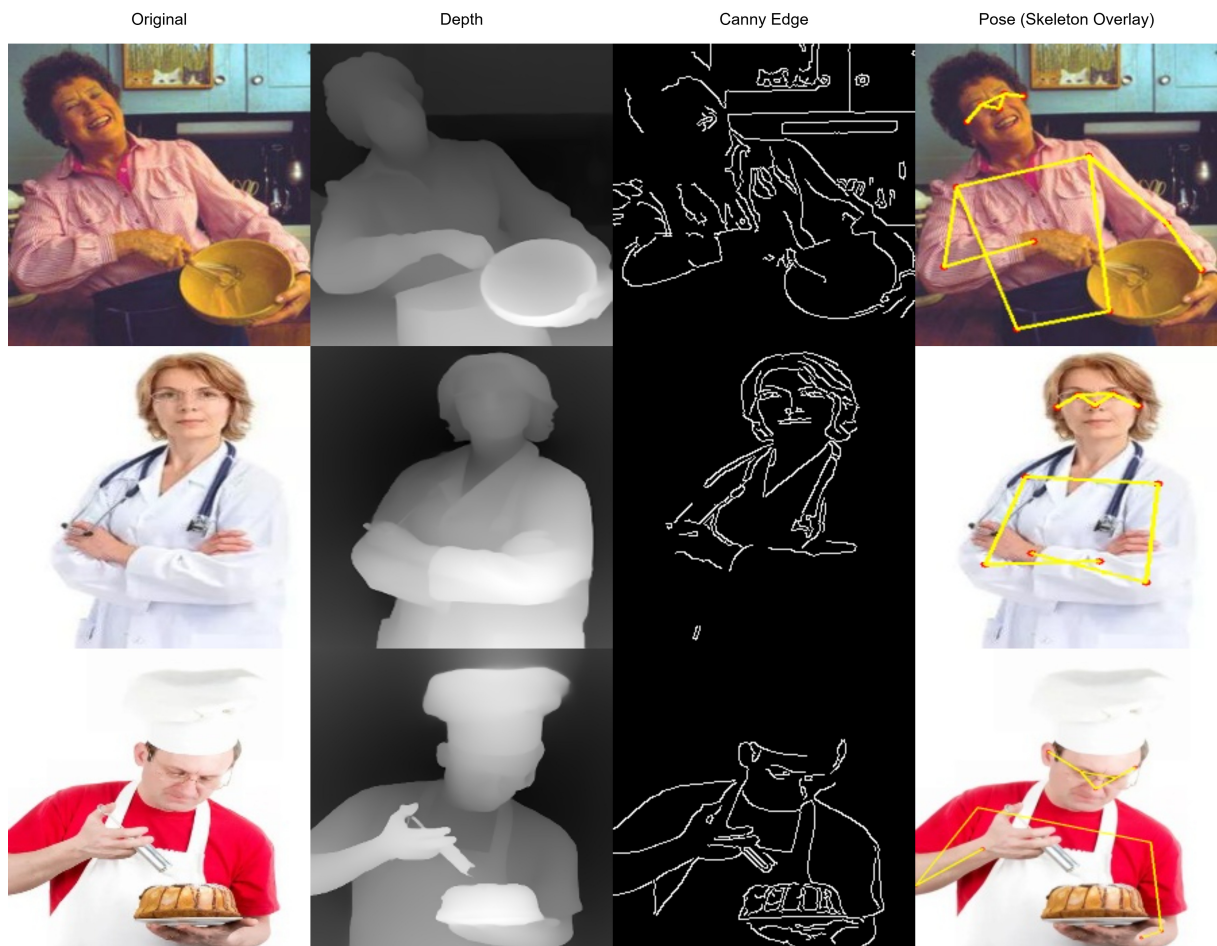


Figure 3.9 Representative outputs of three structural transformation pipelines. Depth: dense relative depth map produced by Depth-Anything-V2 Small. Canny Edge: sparse boundary map produced by classical Canny edge detection. Pose (Skeleton Overlay): COCO-17 keypoint skeleton estimated by YOLOv8l-Pose, overlaid on the original image for visual reference; the actual classifier input is the 34-dimensional normalised keypoint vector.

for gender and skin tone probing. This work replicates the occlusion experiments of Meister et al. in the context of profession-conditioned imagery and extends them to skin tone prediction in addition to gender.

The five variants are: Full NoBg, which retains person pixels within the segmentation mask and replaces all background pixels with black; MaskSegm, which replaces all pixels within the person segmentation mask with white while retaining the original background; MaskSegm NoBg, which renders the person mask as a white silhouette on a black background; MaskRect, which replaces the axis-aligned bounding box of the person with white while retaining the background; and MaskRect NoBg, which renders a white rectangle corresponding to the person bounding box against a black background. The naming convention follows Meister et al. directly.

A key methodological difference from Meister et al. is that their analysis operates on COCO images with ground-truth person segmentation masks. In this work, person masks must be predicted by a segmentation model. Five instance segmentation models were evaluated on single-person images from the COCO 2017 validation set for model selection, with results in Table 3.16. YOLACT was excluded at 79.5% detection rate, missing approximately one in five persons. Lang-SAM achieved the highest IoU (0.846) but at 0.35 iterations per second is approximately 3.7 times slower than Mask2Former COCO, rendering it impractical at full dataset scale. Mask2Former ADE20K performs semantic rather than instance segmentation and was excluded as it conflates all person pixels in multi-person images. Mask2Former COCO was selected for its detection rate of 96.5% evaluated model, combined with an IoU of 0.829 and precision of 0.899.

Table 3.16 Segmentation model evaluation on single-person images from the COCO 2017 validation set, used for model selection only.

Model	Det. Rate	Success Rate	IoU	Prec.	Throughput
Mask2Former COCO	0.965	0.698	0.829	0.899	1.28 it/s
Mask R-CNN	0.941	0.707	0.813	—	1.63 it/s
Lang-SAM	0.891	0.637	0.846	0.931	0.35 it/s
Mask2Former ADE20K	0.946	0.676	0.785	—	1.86 it/s
YOLACT	0.795	0.613	0.789	—	3.00 it/s

For each image, Mask2Former COCO was applied in half-precision under `torch.no_grad()` to produce a binary person mask, upsampled to original image dimensions via nearest-neighbour interpolation. The five occlusion variants were derived as element-wise operations on the original RGB image. All occlusion fills use a uniform white value (RGB = [255, 255, 255]) for masked regions and black (RGB = [0, 0, 0]) for background suppression, consistent with Meister et al., who demonstrate that results are robust to the choice of fill colour. Final output images were resized to

224×224 pixels for classifier compatibility. Representative outputs are shown in Figures 3.7 and 3.10.

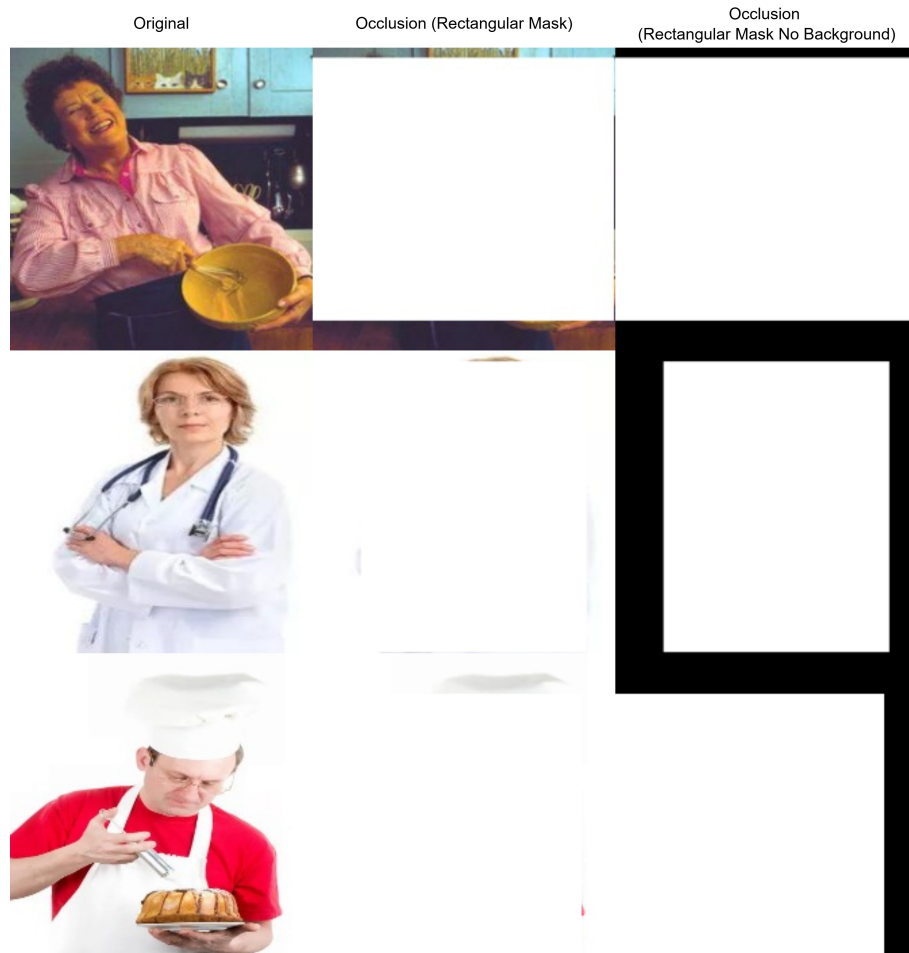


Figure 3.10 Rectangular occlusion transformations applied to three profession-conditioned images. Occlusion (Rectangular Mask): the subject region replaced by a uniform white rectangle, removing the person while preserving background context. Occlusion (Rectangular Mask No Background): the inverse mask applied with a black background, retaining only the occluded region boundary.

3.6 Inference-Time Debiasing of Stable Diffusion v1.5

The preceding analyses characterise the spurious features and demographic biases present in Re-LAION-5B and in the standard Stable Diffusion v1.5 generation pipeline. A complementary question is whether inference-time debiasing interventions applied to the same base model can produce measurable shifts in its demographic output distribution. This section describes the debiased generation pipeline used to produce a 10,000-image corpus against which the effectiveness of these interventions is evaluated in Section 4.4. Given the computational constraints of this work and the requirement that the base model remain Stable Diffusion v1.5, the analysis is restricted

to inference-time approaches; retraining and weight editing methods are acknowledged as potentially more thorough interventions but are outside the scope of the present evaluation.

3.6.1 Experimental Design

The debiased corpus consists of 10,000 images generated for a single profession, doctor, rather than distributed across the full 200-profession list. Concentrating 10,000 images on a single profession yields sufficient statistical power to detect demographic shifts attributable to the interventions themselves, whereas spreading them across 200 professions would yield only 50 images per profession per demographic condition, too few for reliable distributional comparison. The doctor profession was selected as it exhibits substantial demographic skew in the unbiased generation pipeline and has a well-documented real-world demographic baseline.

3.6.2 ITI-GEN Demographic Conditioning

ITI-GEN [86] learns a set of inclusive token embeddings that steer the diffusion model toward a more balanced demographic distribution without modifying model weights. Token embeddings were trained on CCv2 [74] across two attribute axes: gender (female, male) and skin tone (spanning the ten-point MST scale), producing learned perturbation embeddings indexed over all gender-skin tone combinations. The training prompt was "an image of a person", following the ITI-GEN convention of using a neutral descriptive prompt for embedding training. Embeddings were loaded from a checkpoint at epoch 19 and applied to the generation prompt "A picture of a doctor" via the `prompt_prepend` mechanism, concatenating inclusive embeddings with the CLIP text encoding before passing the combined representation to the UNet cross-attention layers.

3.6.3 Pose Conditioning via ControlNet

A fixed canonical pose was enforced across all debiased generated images using ControlNet [87] with an OpenPose conditioning signal. The conditioning image was a single reference pose skeleton depicting a neutral standing figure, applied at a ControlNet scale of 0.6. Without a fixed structural constraint, the model's learned pose prior may itself be demographically skewed, causing variations in body geometry across generated images to introduce a confound into the demographic annotation pipeline. Enforcing a canonical pose ensures that any observed demographic variation

in the debiased corpus reflects differences in appearance-level attributes rather than pose-correlated compositional differences.

3.6.4 Colour Alignment

ITI-GEN operates at the level of semantic text conditioning and does not directly constrain the colour distribution of generated images. A complementary inference-time colour alignment strategy was therefore added to steer the denoising trajectory toward the colour distribution of a skin-tone-diverse reference image. Two approaches were implemented and compared.

SW-Guidance [89] applies Sliced Wasserstein Distance guidance at each denoising step by decoding the current latent to pixel space and computing a gradient signal minimising the SWD between the decoded image’s colour distribution and that of a reference. Despite producing visually favourable colour distributions in preliminary qualitative evaluation, SW-Guidance was excluded on throughput grounds: the repeated decode–gradient–encode operations at each denoising step result in inference times incompatible with generating 10,000 images within the available computational window.

The Chamfer colour alignment approach of Shum et al. [90] performs colour steering directly in latent space without decoding at each step. A reference image is blurred and encoded to a latent representation at the start of generation. At each DDPM backward step, the predicted clean sample $\hat{x}_0$ is projected toward the reference latent distribution using an iterative Chamfer distance matching procedure: nearest-neighbour correspondences between spatial positions of the current and reference latents are established, a displacement field is computed from matched correspondences, and the projected latent is blended back into the denoising step via a learning-rate-scaled update. Because the Chamfer projection operates entirely within the latent space, it imposes no additional decode–encode overhead per step and achieves throughput comparable to standard generation. Chamfer colour alignment was accordingly selected for the full-scale debiased generation run.

3.6.5 Generation Configuration

The debiased corpus was generated using Stable Diffusion v1.5 (runwayml/stable-diffusion-v1-5) with a fine-tuned VAE (stabilityai/sd-vae-ft-mse), producing 512×512 pixel images. The generation prompt was "A picture of a doctor", with classifier-free guidance at a scale of 7.0. Unlike the standard generation pipeline, which used $s = 1.0$ to minimise inference-time amplification of learned priors, the debiased pipeline uses the standard deployment

scale to ensure that the model’s demographic priors are fully active, providing a more demanding test of whether the debiasing interventions can override them. The denoising schedule used 50 inference steps over the canonical 1,000-step scaled-linear noise schedule ($\beta_{\text{start}} = 0.00085$, $\beta_{\text{end}} = 0.0120$), following the standard Stable Diffusion v1.5 configuration. A fixed negative prompt was applied throughout: "longbody, lowres, bad anatomy, bad hands, missing fingers, extra digit, fewer digits, cropped, worst quality, low quality".

The full generation space comprised twenty demographic combinations, spanning two gender categories (female, male) and ten skin tone levels (MST 1–10) from the CCv2 attribute set. Five hundred images were generated per combination, yielding a total corpus of 10,000 images. A base random seed of 42 was used, with per-image seeds derived deterministically as $\text{seed} + n$ for iteration index n , ensuring full reproducibility across all combinations.

ITI-GEN inclusive token embeddings were loaded from the epoch 19 checkpoint trained on CCv2 [74] under the training prompt "an image of a person", and applied to the generation prompt via the `prompt_prepend` mechanism as described in Section 3.6.2. ControlNet OpenPose conditioning

(`lllyasviel/control_v11p_sd15_openpose`) was applied at a conditioning scale of 0.6 using a single fixed reference pose skeleton (`basic_pose.jpg`), enforcing a canonical standing posture across all generated images as described in Section 3.6.3. LCM-LoRA acceleration was not employed; preliminary evaluation found that its reduced step count, combined with the per-step Chamfer projection modifications applied at each denoising iteration, produced images of insufficient visual coherence for downstream annotation and probing.

The Chamfer colour alignment projection was applied to the predicted clean sample $\hat{x}_0$ at every denoising step where the DDPM timestep $t > 600$, corresponding to approximately the first twenty of the fifty inference steps. Restricting projection to early high-noise timesteps follows the placement strategy of Shum et al. [90], where colour steering applied at low-noise timesteps risks corrupting fine-grained semantic detail. The projection used a learning rate of 0.05 and a blur factor of 5, which downsamples the reference image to one-fifth of its spatial resolution before re-upsampling prior to VAE encoding, suppressing high-frequency texture in the reference signal and retaining only the broad colour distribution. The colour reference image (`doctor_mean_rgb.png`) was a uniform-fill image derived from the mean RGB of the standard SDv1.5 doctor generation corpus, providing a skin-tone-neutral colour prior without encoding any scene-level content.

4 Evaluation

This chapter presents the empirical results of the analyses described in the methodology. The evaluation proceeds in four stages. First, the demographic distributions of the annotated Re-LAION-5B and Stable Diffusion v1.5 corpora are characterised at both aggregate and profession level. Second, the dataset classification results from the spurious feature transformation pipeline are reported and interpreted, with per-class accuracy attribution used to localise the source of discriminative signal within each transformation. Third, the demographic probing results are presented, assessing whether gender and skin tone can be inferred from low-level image representations independently of semantic content. Fourth, the debiased generation corpus is evaluated against the aggregate Stable Diffusion v1.5 results to assess whether the inference-time interventions produce measurable demographic shifts. The chapter concludes with a unified discussion of findings, their implications, and the limitations of the evaluation.

4.1 Demographic Distribution Analysis

4.1.1 Aggregate Gender and Skin Tone Distributions

Figures 4.1 and 4.2 show the aggregate gender and skin tone distributions across all 200 professions for both Re-LAION-5B and Stable Diffusion v1.5. Figure 4.3 shows the full MST 1–10 skin tone distribution for both datasets.

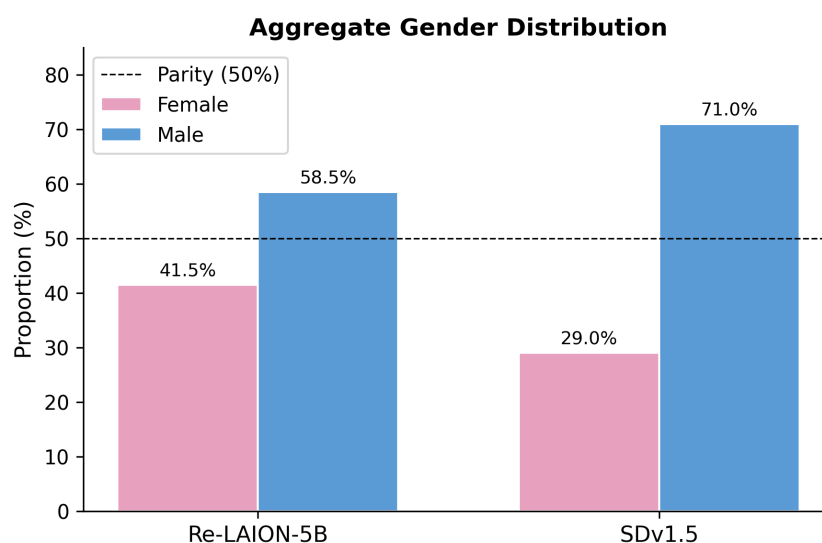


Figure 4.1 Aggregate gender distribution across all 200 professions for Re-LAION-5B and Stable Diffusion v1.5. The dashed line indicates demographic parity at 50%.

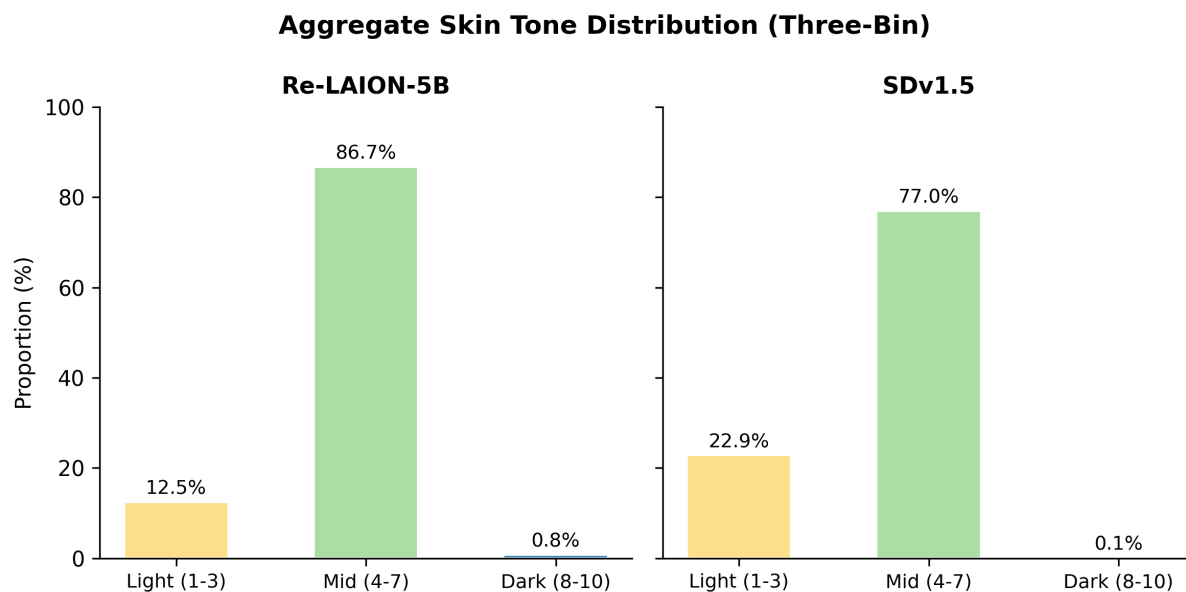


Figure 4.2 Aggregate skin tone distribution using the Light (MST 1–3), Mid (MST 4–7), and Dark (MST 8–10) three-bin grouping for Re-LAION-5B and Stable Diffusion v1.5.

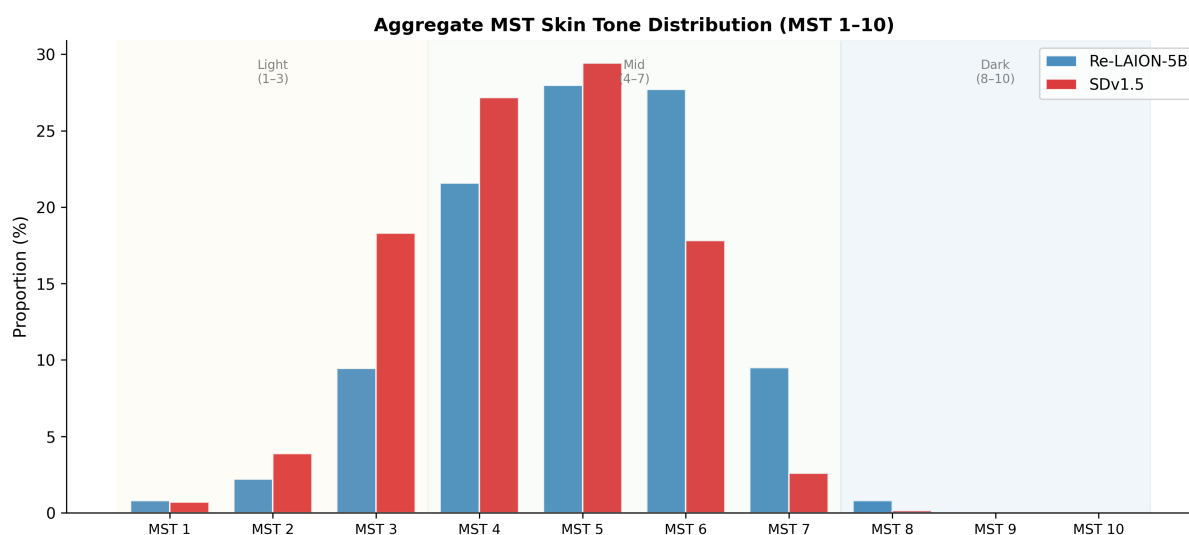


Figure 4.3 Aggregate MST 1–10 skin tone distribution for Re-LAION-5B and Stable Diffusion v1.5. Shaded regions indicate the Light (MST 1–3), Mid (MST 4–7), and Dark (MST 8–10) bin boundaries.

Both datasets exhibit a pronounced male skew across the full 200-profession corpus, but the imbalance is substantially more severe in Stable Diffusion v1.5 than in Re-LAION-5B. Re-LAION-5B contains 83,003 female-labelled images (41.5%) against 116,997 male-labelled images (58.5%), a gap of 17 percentage points relative to parity. Stable Diffusion v1.5 outputs widen this gap to 42 percentage points, with only 58,037 female-labelled images (29.02%) against 141,963 male-labelled images (70.98%). The generative model therefore does not merely reflect the gender distribution of its training corpus but amplifies it substantially, producing outputs that are considerably more male-skewed than the web-scraped imagery from which it learned.

The skin tone distributions reveal a structurally different pattern. Both datasets are overwhelmingly dominated by Mid-bin predictions (MST 4–7), with Re-LAION-5B at 86.74% and Stable Diffusion v1.5 at 76.99%. The MST 1–10 distribution in Figure 4.3 reveals that within the Mid bin, the two datasets differ in their modal value: Re-LAION-5B peaks at MST 5 and MST 6 with approximately 28% each, while Stable Diffusion v1.5 peaks at MST 4 and MST 5, representing a shift toward lighter tones within the Mid range. The more pronounced divergence lies at the extremes: Re-LAION-5B has a Light-bin proportion of 12.45% and a Dark-bin proportion of 0.81%, indicating that non-Mid predictions are distributed, albeit very unevenly, across both ends of the tone scale. Stable Diffusion v1.5 shows a substantially higher Light-bin proportion of 22.87% but a Dark-bin proportion of just 0.14% — a reduction of 83% relative to Re-LAION-5B. MST values 9 and 10 are effectively absent in both datasets, with Stable Diffusion v1.5 generating near-zero proportions even at MST 8. This near-complete collapse of dark skin tone representation in the generative outputs, combined with the shift toward lighter tones within the Mid bin, suggests that the model has absorbed a strong light skin tone prior from its training corpus and concentrates its outputs toward the lighter half of the tonal range.

4.1.2 Per-Profession and Per-ISCO-Group Analysis

Figures 4.4, 4.5, and 4.6 break down the gender and skin tone distributions by ISCO-08 major occupational group. Tables 4.1 and 4.2 list the ten professions with the strongest observed gender and light skin tone skew in each dataset respectively.

The ISCO-08 group breakdown in Figure 4.4 reveals that gender skew is pervasive across all occupational categories in both datasets, but its magnitude and pattern differ markedly. In Re-LAION-5B, Clerical Support is the only major group to exceed demographic parity at 58.26% female, consistent with the historically female-dominated composition of administrative and secretarial roles. All remaining groups fall below parity, with Plant & Machine Operators showing the strongest male skew at 77.85% male, followed by Armed Forces at 76.5% and Managers at 64.53%. In

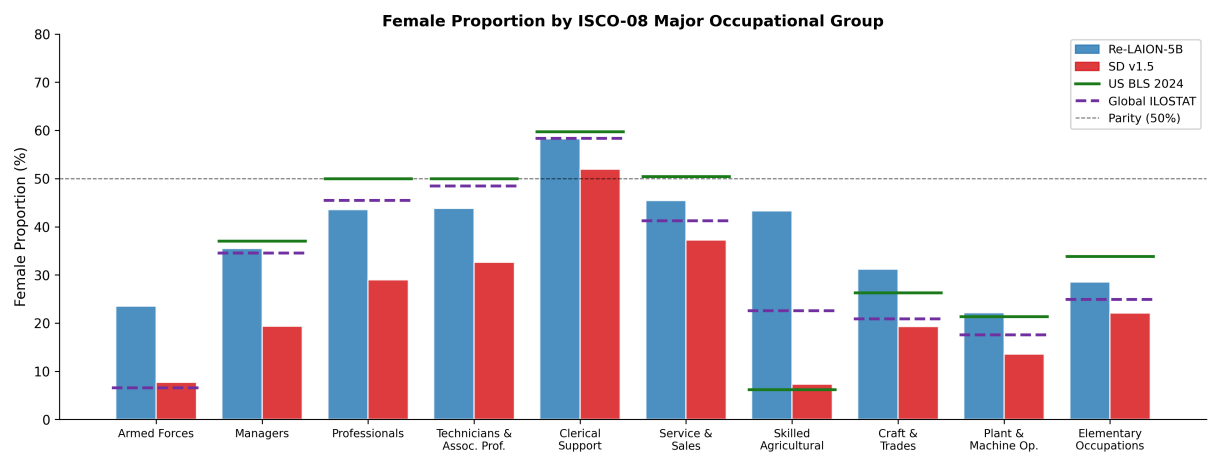


Figure 4.4 Female proportion by ISCO-08 major occupational group for Re-LAION-5B and Stable Diffusion v1.5. The dashed line indicates demographic parity at 50%.

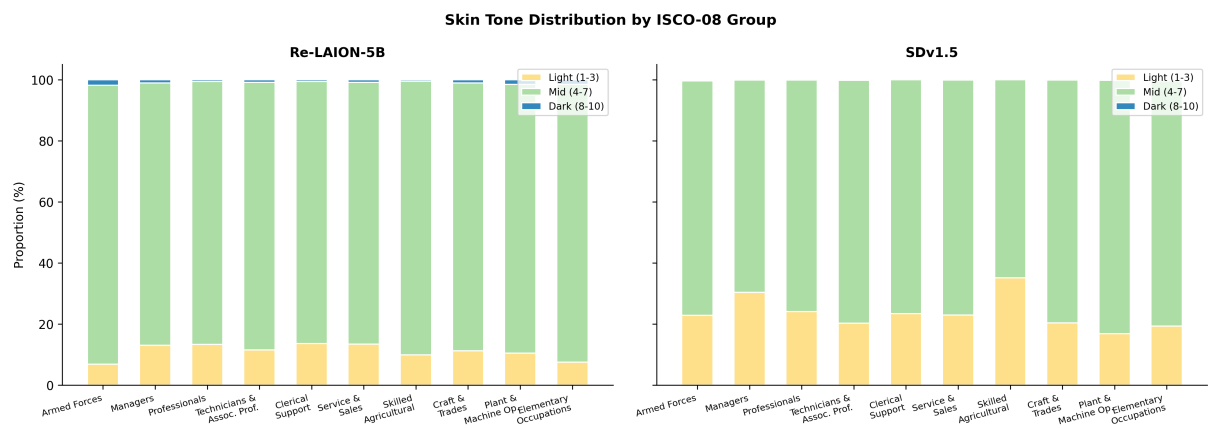


Figure 4.5 Skin tone distribution (Light, Mid, Dark bins) by ISCO-08 major occupational group for Re-LAION-5B and Stable Diffusion v1.5.

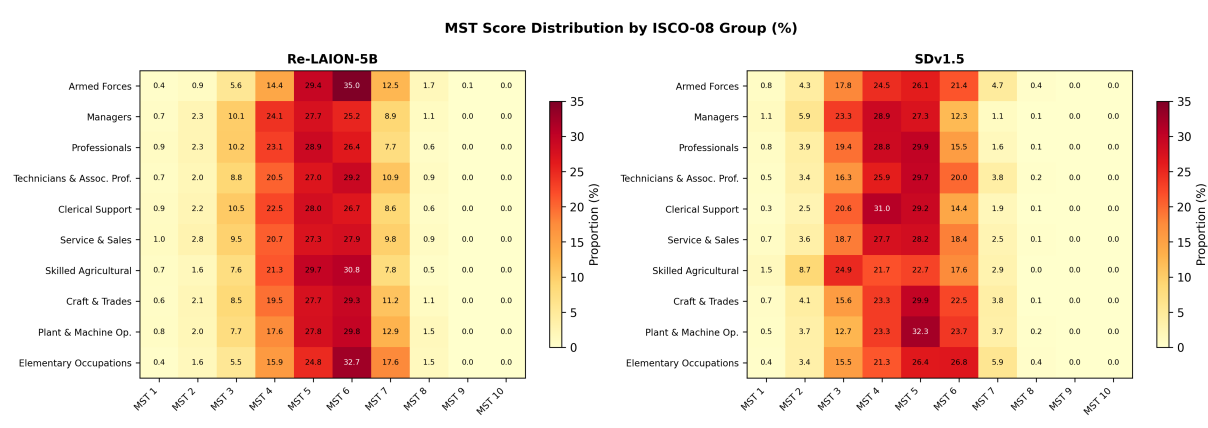


Figure 4.6 MST 1–10 score distribution (%) by ISCO-08 major occupational group for Re-LAION-5B and Stable Diffusion v1.5. Cell values indicate the proportion of images in each group assigned to each MST score.

Table 4.1 Ten professions with the strongest gender skew in Re-LAION-5B and Stable Diffusion v1.5. Gender skew is the absolute deviation of the female proportion from 0.5; higher values indicate stronger skew regardless of direction. F% = female percentage; M% = male percentage.

Dataset	Profession	F%	M%	Skew
Re-LAION-5B	Administrative Assistant	94.7	5.3	0.447
Re-LAION-5B	Makeup Artist	92.7	7.3	0.427
Re-LAION-5B	Receptionist	92.0	8.0	0.420
Re-LAION-5B	Executive Assistant	91.5	8.5	0.415
Re-LAION-5B	Office Assistant	90.2	9.8	0.402
Re-LAION-5B	Dietitian	88.6	11.4	0.386
Re-LAION-5B	Paralegal	87.8	12.2	0.378
Re-LAION-5B	Bodyguard	13.0	87.0	0.370
Re-LAION-5B	Priest	13.3	86.7	0.367
Re-LAION-5B	Blacksmith	14.5	85.5	0.355
SDv1.5	Logger	2.5	97.5	0.475
SDv1.5	Nurse	97.5	2.5	0.475
SDv1.5	Receptionist	97.3	2.7	0.473
SDv1.5	Farmer	3.0	97.0	0.470
SDv1.5	Dietitian	96.9	3.1	0.469
SDv1.5	Makeup Artist	96.9	3.1	0.469
SDv1.5	Flight Attendant	96.6	3.4	0.466
SDv1.5	Administrative Assistant	96.4	3.6	0.464
SDv1.5	Construction Worker	3.6	96.4	0.464
SDv1.5	Miner	3.9	96.1	0.461

Table 4.2 Ten professions with the highest Light-bin (MST 1–3) skin tone proportion in Re-LAION-5B and Stable Diffusion v1.5.

Dataset	Profession	Light%	Mid%	Dark%
Re-LAION-5B	Makeup Artist	27.6	72.3	0.1
Re-LAION-5B	Hair Dresser	21.8	78.0	0.2
Re-LAION-5B	Dentist	20.4	79.3	0.3
Re-LAION-5B	Carer	19.9	79.8	0.3
Re-LAION-5B	Florist	18.7	81.1	0.2
Re-LAION-5B	Singer	18.6	81.1	0.3
Re-LAION-5B	Barber	18.5	80.2	1.3
Re-LAION-5B	Customer Support Specialist	18.5	81.2	0.3
Re-LAION-5B	Comedian	18.1	81.0	0.9
Re-LAION-5B	Actor	17.2	80.7	2.1
SDv1.5	Librarian	56.8	43.1	0.1
SDv1.5	Carer	55.5	44.4	0.1
SDv1.5	Politician	50.2	49.8	0.0
SDv1.5	Meteorologist	45.3	54.7	0.0
SDv1.5	Mayor	45.2	53.9	0.9
SDv1.5	News Reader	45.2	54.7	0.1
SDv1.5	Farmer	43.7	56.3	0.0
SDv1.5	Comedian	41.3	58.7	0.0
SDv1.5	Florist	40.5	59.5	0.0
SDv1.5	Conservation Officer	40.4	59.1	0.5

Stable Diffusion v1.5, the direction of skew is broadly preserved — Clerical Support remains the only group at or near parity at 51.93% female — but the magnitude increases substantially across every group without exception. The most severe amplifications occur in Skilled Agricultural (from 43.3% to 7.3% female, a drop of 36 percentage points), Armed Forces (from 23.5% to 7.7%, a drop of 15.8 percentage points), and Plant & Machine Operators (from 22.15% to 13.55%). This systematic reduction in female representation across all occupational categories confirms that the amplification effect identified at the aggregate level is not driven by a small number of outlier professions but reflects a broad structural tendency of the generative model.

Comparing these distributions against the US BLS 2024 and Global ILOSTAT reference markers in Figure 4.4 reveals a consistent pattern of under-representation of women relative to real-world labour force composition. Both reference baselines show broad agreement in their directional signal: groups identified as female-majority in US BLS data are also female-majority in the global statistics, while groups such as Skilled Agricultural, Craft & Trades, and Plant & Machine Operators are male-dominated in both. This convergence between the two independent baselines strengthens the validity of the comparison and indicates that the observed gaps are not artefacts of geographic or methodological differences between the two reference sources.

Against these baselines, Re-LAION-5B falls below the real-world female proportion in eight of ten ISCO groups, most notably in Professionals (approximately 43% vs. 50% US BLS), Service & Sales (approximately 46% vs. 51% US BLS), and Technicians & Associate Professionals (approximately 44% vs. 49% US BLS). The training corpus therefore does not accurately reflect the gender composition of the workforce it implicitly represents, already exhibiting a systematic male-skew relative to observed labour force statistics. Stable Diffusion v1.5 substantially widens these gaps in nearly every group, with the most pronounced deviations occurring in Managers (approximately 19% vs. 38% US BLS, a gap of 19 percentage points), Professionals (approximately 29% vs. 50% US BLS), and Service & Sales (approximately 37% vs. 51% US BLS). The single exception is Clerical Support, where both Re-LAION-5B and Stable Diffusion v1.5 broadly align with the US BLS reference at approximately 60% and 52% female respectively — a case where the model's learned stereotype coincides with real-world labour force composition, rather than reflecting an accurate learned prior.

Taken together, these comparisons establish a two-stage bias pattern. Re-LAION-5B under-represents women relative to real-world labour force statistics, reflecting the gender imbalances inherent in the web-scraped imagery from which it is constructed. Stable Diffusion v1.5 then amplifies this under-representation further, producing occupational gender distributions that are more male-skewed than even the already biased training corpus. The generative model therefore does not merely inherit the distributional properties of its training data but actively exacerbates them, producing representations that diverge substantially from both demographic parity and empirically observed occupational gender composition.

At the individual profession level, 167 of 200 professions (83.5%) show gender skew in the same direction in both datasets, confirming that Stable Diffusion v1.5 largely inherits the directional gender associations of its training corpus. However, in 178 of 200 professions (89.0%) the magnitude of skew is greater in Stable Diffusion v1.5 than in Re-LAION-5B. The most skewed professions in Table 4.1 illustrate this pattern clearly: Administrative Assistant, which already shows a strong female skew in Re-LAION-5B at 94.7%, reaches 96.4% in SDv1.5; Dietitian rises from 88.6% to 96.9%; and Receptionist rises from 92.0% to 97.3%. At the male-skewed end, the stereotyping is equally pronounced, with Logger (97.5% male), Farmer (97.0%), and Construction Worker (96.4%) among the most extreme SDv1.5 outputs. The 33 professions where the datasets disagree in skew direction represent cases where the model's learned priors override the distributional signal in its training data. These cases warrant further investigation but are beyond the scope of the current analysis.

The skin tone analysis at the ISCO group level, shown in Figures 4.5 and 4.6, presents a consistent story across all occupational categories. In Re-LAION-5B the Light-bin proportion varies narrowly between 6.9% (Armed Forces) and 13.59%

(Clerical Support), indicating that no occupational group is strongly associated with light skin tones in the training corpus. In Stable Diffusion v1.5, all ISCO groups show a substantially elevated Light-bin proportion, ranging from 16.85% (Plant & Machine Operators) to 35.1% (Skilled Agricultural). The MST heatmap confirms that this shift is not caused by a redistribution within the Light bin but by a leftward shift in the modal MST score: while Re-LAION-5B concentrates mass at MST 5–6 across all groups, Stable Diffusion v1.5 systematically concentrates mass at MST 4–5, with correspondingly elevated MST 3 proportions. The Dark-bin collapse observed at the aggregate level is equally consistent across ISCO groups: Re-LAION-5B dark proportions range from 0.5% to 1.8%, while in SDv1.5 they approach zero in most groups, with Skilled Agricultural recording 0.0% dark-bin images entirely.

At the individual profession level, the scale of the skin tone amplification in SDv1.5 is particularly stark. The most light-skewed profession in Re-LAION-5B is Makeup Artist at 27.6% Light, yet in SDv1.5 this figure rises to 55.5% for Carer and 56.8% for Librarian — professions that in Re-LAION-5B reach only 19.9% and are absent from the top 10 entirely. The SDv1.5 top 10 light-skewed professions all exceed 40% Light, a threshold not reached by any profession in Re-LAION-5B. In 175 of 200 professions (87.5%), SDv1.5 generates a higher proportion of Light-bin images than Re-LAION-5B does for the same profession, establishing a near-universal tendency toward lighter skin tone generation that is independent of occupational category.

Unlike the gender analysis, a direct comparison of skin tone distributions against real-world reference statistics was not conducted. The BLS CPS data reports labour force composition in terms of self-identified racial categories rather than skin tone measurements, and converting race proportions into expected MST distributions requires an intermediate race-to-skin-tone mapping. As described in Section 3.4.5, this mapping was derived empirically from the FairFace dataset using the VGG-16 MST regression model; however, inspection of the resulting conditional distributions revealed that all racial groups map predominantly to the Mid bin (MST 4–7), reflecting the tonal range of the FairFace corpus rather than the full MST scale. The derived per-profession reference distributions therefore lack sufficient discriminative resolution to serve as meaningful baselines for the skin tone comparison, collapsing to near-identical Mid-dominant profiles regardless of occupational racial composition. As a result, skin tone analysis in this work is evaluated solely against the demographic parity baseline. The construction of a more granular and reliable race-to-skin-tone reference distribution, whether through a larger and more tonally diverse calibration dataset or through direct MST annotations on labour force samples, remains an open challenge and constitutes a limitation of the present analysis.

4.2 Dataset Classification

Dataset classification experiments assess whether a ConvNeXt-Tiny classifier trained on transformed image representations can distinguish Re-LAION-5B, Stable Diffusion v1.5, and COCO images as a three-class problem. Each transformation is treated as an independent experiment: images are transformed, split into training, validation, and test sets, and the classifier is trained on the transformed training set with early stopping (patience = 2) and evaluated on the held-out test set. Chance level for all experiments is $\frac{1}{3} \approx 33.3\%$ under a balanced three-class prior. Above-chance accuracy on a given transformation indicates that the transformation retains dataset-discriminating signal; the magnitude of accuracy quantifies how strongly that signal is preserved. Classification accuracy is used as the primary metric throughout this section, consistent with the three-class evaluation protocol of Zeng et al. [9] from which the transformation framework is adopted.

COCO is included solely to extend the classification task beyond a binary comparison between Re-LAION-5B and Stable Diffusion v1.5, enabling per-class accuracy attribution that identifies which dataset drives separability under each transformation. As COCO carries no profession-conditioned annotation for gender or skin tone within this pipeline, it does not appear in the demographic analyses that follow; the results in this section therefore reflect general distributional separability across datasets and should not be interpreted as direct evidence of demographic bias, which is examined independently in Section 4.3.

Table 4.3 summarises the results for all transformations evaluated.

1. The three datasets are separable across every level of representation tested

Every transformation achieves well above the chance level of 33.3%, with even the most aggressive information compression — mean RGB at 64.19% — retaining substantial discriminating signal. This demonstrates that dataset-level differences between Re-LAION-5B, Stable Diffusion v1.5, and COCO persist even under extreme reduction of image content. Even when spatial structure, semantics, and most visual detail are removed, the classifier can still attribute images to their source dataset significantly better than chance.

The persistence of separability across such compressed representations suggests that dataset identity is encoded in multiple correlated visual cues rather than a single isolated artefact. These cues may include colour statistics, spatial patterns, semantic composition, or other structural regularities that differ systematically between datasets. Because these signals remain detectable even after large portions of the visual information are removed, they likely reflect broad statistical regularities rather than isolated features.

Table 4.3 Dataset classification results per image transformation (three-class: Re-LAION-5B, Stable Diffusion v1.5, COCO). All models trained with ConvNeXt-Tiny, early stopping (patience = 2), seed = 0.

Transformation	Val Acc (%)	Val AUC	Test Acc (%)	Test AUC
Base (unmodified)	99.56	0.9996	99.42	0.9996
VAE reconstruction	99.15	0.9992	99.15	0.9992
High-pass filter	98.65	0.9991	98.64	0.9988
Patch shuffle 8×8	98.41	0.9989	97.91	0.9983
Patch shuffle 16×16	98.60	0.9986	97.69	0.9983
Low-pass filter	97.30	0.9977	96.99	0.9969
Depth	96.41	0.9968	96.78	0.9965
Patch shuffle 4×4	97.20	0.9968	94.38	0.9949
Edge	95.17	0.9934	93.62	0.9905
Semantic segmentation	90.47	0.9814	91.36	0.9837
Captions [†]	—	—	90.99	0.9820
Patch shuffle 2×2	88.97	0.9839	89.91	0.9766
Object detection	85.88	0.9617	88.62	0.9728
Pixel shuffle	86.01	0.9574	80.67	0.9430
Mean RGB	70.80	0.8443	64.19	0.8207

[†] Logistic regression probe on BLIP Large captions encoded via SentenceT5-Base; all other rows use ConvNeXt-Tiny

Importantly, these experiments demonstrate separability rather than its causal origins. The observed differences may arise from a combination of dataset curation practices, object distribution differences, scene composition biases, or visual statistics introduced by generative models. Consequently, the results indicate that the datasets possess distinct statistical signatures without directly identifying the specific factors responsible for those differences.

2. Dataset separability cannot be explained by acquisition artefacts alone The VAE result (99.15%) is virtually identical to the unmodified base (99.42%), a reduction of only 0.27 percentage points. Variational autoencoder reconstruction compresses each image through a latent bottleneck before reconstructing it, which typically suppresses certain low-level acquisition artefacts such as compression noise or encoding irregularities while preserving perceptually meaningful image structure.

Because VAE reconstruction largely preserves colour distributions, texture statistics, and spatial structure while reducing some superficial artefacts associated with file encoding or compression, the near-identical classification performance suggests that simple acquisition artefacts are unlikely to be the dominant source of discriminating signal. If dataset separability were primarily driven by compression differences, file-format characteristics, or other encoding artefacts, the VAE reconstruction process would be expected to reduce classification accuracy more substantially.

Instead, the results suggest that dataset differences arise primarily from content and structural properties of the images themselves. These may include differences in scene composition, object arrangement, photographic style, or statistical regularities introduced during dataset creation. However, it should be noted that VAE reconstruction does not eliminate all low-level image statistics, and therefore the results indicate that acquisition artefacts are unlikely to dominate rather than definitively ruling out all low-level contributions.

3. Local spatial statistics are highly dataset-discriminative Even 2×2 patch shuffling – which destroys global layout while preserving local spatial structure including texture patterns, gradient relationships, and colour correlations within each patch – achieves 89.91% test accuracy. Increasing patch size substantially improves performance, with 8×8 patch shuffling reaching 97.91% and 16×16 achieving 97.69%.

Patch shuffling disrupts global object arrangement and scene layout while preserving local spatial statistics within each patch. These statistics include short-range correlations between neighbouring pixels, micro-texture patterns, and small-scale colour gradients that often characterise the visual appearance of images. Because these local structures remain intact even when global spatial coherence is removed, classifiers trained on shuffled images must rely primarily on these fine-scale statistical cues.

The strong performance under patch shuffling therefore indicates that dataset-specific signatures are embedded in local spatial statistics rather than solely in high-level scene composition. Such statistics may arise from differences in photographic style, image processing pipelines, or generative model priors that produce consistent patterns at fine spatial scales. However, the experiment does not isolate the origin of these statistics, and the observed separability may reflect a combination of generative artefacts, dataset curation biases, or inherent distributional differences between the source corpora.

4. Both frequency bands carry strong discriminating signal, with a modest advantage for high-frequency components High-pass filtering achieves 98.64% test accuracy, outperforming low-pass filtering at 96.99% by 1.65 percentage points. High-frequency components correspond primarily to edges, fine textures, and small-scale spatial variations, whereas low-frequency components capture coarse illumination gradients and large-scale structural information.

The strong performance under both filtering operations indicates that dataset-level differences are distributed across multiple spatial scales. High-frequency signals such as edge patterns and texture statistics appear to carry slightly more discriminating information than low-frequency signals, suggesting that fine-grained

visual details contribute strongly to dataset identity. At the same time, the high performance under low-pass filtering indicates that broader structural cues such as global shape patterns or scene layouts also play an important role.

The relatively small difference between the two results suggests that dataset bias is not concentrated exclusively in either frequency band but instead permeates the entire frequency spectrum.

5. A hierarchy of dataset-discriminating signal exists across representational levels

Table 4.3 reveals a consistent ordering of discriminating power from strongest to weakest across the full transformation suite. Unmodified and VAE-reconstructed images achieve near-perfect accuracy ($\geq 99\%$), confirming that the raw statistical properties of each dataset are highly distinctive. Frequency and fine spatial statistics — high-pass filter, 8×8 and 16×16 patch shuffle — follow closely at 97–99%, indicating that dataset-specific signatures are preserved in local spatial structure even when global layout is destroyed. Structural geometry transformations — depth and Canny edge — occupy the next tier at 93–97%, demonstrating that scene geometry alone is sufficient for reliable classification without any colour or texture information. Dense semantic representations — segmentation at 91.36% — and semantic content probes — captions at 90.99% and object detection at 88.62% — follow, showing that object identity and linguistic content carry substantial but somewhat weaker discriminating power than structural cues. Global colour statistics form the lowest tier: 2×2 patch shuffle at 89.91%, pixel shuffle at 80.67%, and mean RGB at 64.19%, the last of which retains only the three-dimensional average colour vector yet still achieves 31 percentage points above chance. This hierarchy indicates that dataset bias is distributed across multiple correlated visual attributes rather than concentrated in any single representation, and that structural and low-level statistical differences between the datasets are more pronounced than semantic content differences.

6. Semantic dataset differences survive compression to natural language

Caption-based classification using SentenceT5 embeddings achieves 90.99% test accuracy. Because the classifier receives only textual descriptions rather than visual input, this result indicates that systematic differences between the datasets are sufficiently consistent to be reflected in natural language descriptions of the images.

Differences in object co-occurrence patterns, scene composition, and contextual relationships between objects may therefore manifest in textual representations as well as in visual statistics. The persistence of dataset separability under this modality shift suggests that dataset identity is not solely encoded in low-level visual patterns but also in semantic content that can be described linguistically.

7. Absolute accuracies exceed those reported in the reference framework Across comparable transformations, the obtained accuracies exceed those reported by Zeng et al. by approximately 15–25 percentage points. This discrepancy is most likely attributable to dataset composition rather than methodological differences.

The reference framework analyses separability between multiple web-scraped image datasets, which aggregate images from a wide range of photographic sources and therefore exhibit considerable internal diversity. In contrast, the dataset combination studied here includes Stable Diffusion v1.5 generated images, which are produced through a consistent generative pipeline that imposes shared priors over visual structure, texture statistics, and colour distributions.

The presence of a generative dataset therefore introduces a class whose visual statistics are influenced by the generative process, potentially producing more pronounced dataset-level signatures than those observed in purely web-scraped corpora. Consequently, the results reported here should be interpreted in the context of this specific three-dataset configuration rather than as a direct replication of the reference-paper findings.

4.3 Demographic Attribute Separability

To investigate whether demographic attributes are encoded in non-appearance visual features, a series of probe experiments were conducted using transformations designed to isolate distinct information channels: spatial structure, semantic content, body geometry, and global colour statistics. These probes follow the same methodological framework applied in the dataset classification experiments, substituting demographic labels for dataset identity as the classification target. Crucially, the occlusion family of probes directly tests whether demographic signal persists in the scene context once the person region is removed, providing evidence about spurious correlations between demographic attributes and non-person visual features. Results are reported as AUC, with chance level at 0.500 for binary gender classification and 0.500 macro-averaged AUC for eight-class skin tone classification.

To assess result stability, each probe was trained independently across four random seeds (0, 1, 2, 3); the best-performing seed result is reported for each probe and dataset combination. Variation across seeds was negligible for near-chance conditions on Re-LAION-5B, confirming the absence of recoverable signal rather than seed sensitivity.

MST categories 9 and 10 are absent from both datasets: no images in either corpus received these labels under the VGG-16 MST10 Regression annotator. This outcome is consistent with the distributional analysis in Section 4.1, which shows

near-zero Dark-bin proportions in both corpora. The annotator is capable of assigning these categories as confirmed during annotator evaluation, making it likely that MST 9 and 10 skin tones are extremely rare or entirely absent in the profession-conditioned image sets rather than systematically missed by the classifier. All probing experiments therefore operate over MST categories 1–8.

4.3.1 Gender Artifact Analysis

Table 4.4 Gender separability results per image transformation and dataset. All probes trained with seed 0 unless otherwise noted. AUC is the primary metric; chance level is 0.500 for binary classification.

Transformation	Probe	Re-LAION-5B AUC	SD v1.5 AUC
Occlusion (full, no background)	ResNet-50	0.499 [†]	0.948
Occlusion (masked rectangle)	ResNet-50	0.501 [†]	0.747
Occlusion (masked rectangle, no background)	ResNet-50	0.499 [†]	0.587 [‡]
Occlusion (masked segmentation)	ResNet-50	0.501 [†]	0.852
Occlusion (masked segmentation, no bg.)	ResNet-50	0.501 [†]	0.834
Depth at pose keypoints	Logistic Regression	0.559	0.592
Mean RGB	Logistic Regression	0.600	0.579
Object detection (restricted)	Logistic Regression	0.615	0.608
Object detection	Logistic Regression	0.692	0.608
Pose keypoints	MLP	0.697	0.680
Base (unmodified)	ResNet-50	0.986	0.946
Captions	SentenceT5	0.901	0.771

[†] AUC at chance level; model collapsed to majority-class prediction.

[‡] Near-chance AUC; recall ≈ 1.0 indicates near-complete majority-class prediction.

Unmodified images establish a strong separability upper bound The base ResNet-50 probe achieves AUC 0.986 on Re-LAION-5B and 0.946 on SD v1.5, confirming that gender-correlated visual signal is encoded at high levels in both datasets before any transformation is applied. The slightly higher value on Re-LAION-5B likely reflects the greater photographic diversity of a web-scraped corpus, which produces a wider variety of appearance cues correlated with gender labels, such as facial structure, body morphology, hairstyle, and clothing. These results establish an upper bound against which all reduced representations are compared and confirm that the annotation pipeline produces labels sufficiently consistent to support downstream probing.

Occlusion probes reveal a fundamental asymmetry between datasets The occlusion family of probes constitutes the most consequential set of results in this analysis, as it directly tests whether gender information is localised within the person or distributed across the broader scene context.

On Re-LAION-5B, every occlusion variant collapses performance to chance ($AUC \approx 0.500$), regardless of whether masking uses a bounding rectangle or a tight segmentation contour, and regardless of whether the background is retained or removed. This uniform collapse establishes that gender-correlated signal in Re-LAION-5B resides exclusively within the visible appearance of the depicted person. No residual signal is recoverable from background objects, spatial layout, or any other contextual feature once the person region is occluded.

The pattern for SD v1.5 is qualitatively different and constitutes the central finding of this analysis. The primary carrier of gender signal is the scene background: when the person is masked with a rectangle and the background is retained, the probe achieves AUC 0.747 from background context alone. However, the person-shaped silhouette is also independently informative. When the segmentation mask is applied without any background — leaving the classifier with only the person-shaped occluded region and no surrounding scene — performance rises to AUC 0.834, a 8.7 percentage-point increase over the rectangle-with-background condition. The silhouette outline itself therefore encodes gender signal beyond what background context alone provides, likely through the body shape and spatial proportions preserved by the tight contour. When both silhouette and background are simultaneously available under segmentation masking, performance reaches 0.852, confirming that both sources contribute additively.

The strongest evidence for the background as the dominant signal source comes from the sharpest drop in the table: rectangle masking with background removal, which eliminates both the person region and the immediately surrounding context while retaining only the distal image periphery, reduces performance to AUC 0.587. Removing the background is therefore far more damaging to classification performance than any other single intervention, directly demonstrating that the scene context is the primary carrier of gender information in SD v1.5 images. This finding directly replicates the contextual gender artifact identified by Meister et al. and demonstrates that the effect is substantially more pronounced in generatively produced images than in web-scraped corpora.

Global colour statistics carry modest but consistent signal in both datasets Mean RGB classification achieves AUC 0.600 on Re-LAION-5B and 0.579 on SD v1.5. Both values are in close agreement with the AUC of 0.580 reported by Meister et al. for mean RGB gender probing on COCO, providing a useful external reference point for the magnitude of this signal. Both datasets therefore exhibit gender-correlated photometric statistics at a level consistent with a general-purpose image benchmark, indicating that systematic differences in clothing colour or other photometric properties associated with gender labels produce detectable signal even under

compression to three global scalars. The near-identical performance across datasets further suggests that this source of signal reflects how occupational images are composed generally, rather than a characteristic specific to either corpus.

Geometric probes reveal body configuration as an independent signal source Pose keypoints achieve AUC 0.697 on Re-LAION-5B and 0.680 on SD v1.5, indicating that normalised skeletal geometry correlates measurably with gender labels in both datasets. Because pose keypoints encode body proportions, posture, and limb arrangement without conveying colour or texture information, this result demonstrates that systematic differences in depicted body configurations contribute to gender separability independently of photometric properties.

Depth values sampled at pose keypoint locations yield AUC 0.559 on Re-LAION-5B and 0.592 on SD v1.5 – the lowest values recorded for any probe that produces above-chance predictions. Although both exceed chance, the magnitude of separation is small, indicating that 3D geometric structure at skeletal locations contributes only weak gender-correlated signal. This positions depth at keypoints as a substantially less informative probe than skeletal geometry alone, and suggests that the dominant spurious features in these datasets are photometric and contextual in nature rather than geometric.

Semantic probes reveal scene composition and language as additional signal carriers

Object detection probes show that scene composition carries gender-correlated information beyond what body geometry alone explains. On Re-LAION-5B, the unrestricted object detection probe achieves AUC 0.692, compared to 0.615 for the person-category-restricted variant. This gap of 7.7 percentage points demonstrates that the objects appearing alongside people in Re-LAION-5B reflect gender-stereotyped scene compositions independently of the person’s appearance. On SD v1.5, both variants converge at AUC 0.608, indicating that the generative model does not amplify this form of object-level contextual stereotyping to the same degree as is observed for spatial background signal.

Caption-based classification using SentenceT5 embeddings achieves AUC 0.771 on SD v1.5, demonstrating that gender-correlated signal survives compression to natural language. Because the probe receives only textual input, this result indicates that the generative pipeline systematically produces scene descriptions whose semantic content co-varies with the gender of the depicted person: occupational contexts, object references, and activity settings described in captions differ systematically across gender categories. The persistence of this signal through a complete modality shift suggests that the stereotyping identified through visual probes is coherent and cross-modal rather than confined to low-level image statistics.

The two datasets exhibit qualitatively distinct spurious gender feature profiles

Considered jointly, the results reveal a structural divergence between the two datasets. In Re-LAION-5B, gender separability is driven almost entirely by the person’s own appearance: occlusion collapses performance to chance, and the residual contributions of colour statistics, object co-occurrence, and skeletal geometry are modest. In SD v1.5, gender information is distributed across the full image: background context alone reproduces near-complete classification performance, every region of the image contributes demographic signal, and stereotyping is recoverable even from natural language descriptions. This profile is consistent with the hypothesis that SD v1.5 amplifies the demographic priors of its Re-LAION-based training corpus, projecting them not only onto the generated person but onto the surrounding scene, the spatial layout, and the textual register of the generated content.

4.3.2 Skin Tone Artifact Analysis

Table 4.5 Skin tone separability results per image transformation and dataset (MST 1–8). All probes trained with seed 0 unless otherwise noted. Macro-averaged AUC is the primary metric; chance level is 0.500.

Transformation	Probe	Re-LAION-5B AUC	SD v1.5 AUC
Occlusion (full, no background)	ResNet-50	0.514 [†]	0.899
Occlusion (masked rectangle)	ResNet-50	0.514 [†]	0.728
Occlusion (masked rectangle, no background)	ResNet-50	0.515 [†]	0.660
Occlusion (masked segmentation)	ResNet-50	0.514 [†]	0.774
Occlusion (masked segmentation, no bg.)	ResNet-50	0.515 [†]	0.704
Mean RGB	Logistic Regression	0.563	0.543
Object detection (restricted)	Logistic Regression	0.591	0.626
Pose keypoints	MLP (seed 4)	0.596	0.666
Depth at pose keypoints	Logistic Regression	0.598	0.655
Object detection	Logistic Regression	0.611	0.663
Captions [‡]	SentenceT5	0.749	0.673
Base (unmodified)	ResNet-50	0.968	0.900

[†] Near-chance macro AUC; per-class AUCs at chance except MST8 (small-sample artefact).

[‡] Probe operates on BLIP Large captions encoded via SentenceT5-Base.

Unmodified images establish a strong separability upper bound The base ResNet-50 probe achieves macro-AUC 0.968 on Re-LAION-5B and 0.900 on SD v1.5, confirming that skin tone information is strongly encoded in person appearance within both datasets. The per-class results are notably consistent across MST categories for Re-LAION-5B (range 0.948–0.980), whereas SD v1.5 exhibits considerably wider variation (0.779–0.996), with the lowest values in the mid-range tones MST4 and

Table 4.6 Per-class skin tone AUC per image transformation and dataset (MST 1–8). MST 9 and MST 10 are absent from both datasets and left blank. All probes trained with seed 0 unless otherwise noted.

Transf.	Dataset	MST1	MST2	MST3	MST4	MST5	MST6	MST7	MST8
Occ. (full., no bg)	Re-LAION-5B	0.499	0.501	0.501	0.499	0.501	0.499	0.499	0.614
Occ. (full., no bg)	SD v1.5	0.969	0.936	0.881	0.778	0.783	0.888	0.963	0.996
Occ. (rect.)	Re-LAION-5B	0.499	0.499	0.499	0.499	0.501	0.501	0.499	0.614
Occ. (rect.)	SD v1.5	0.837	0.777	0.715	0.594	0.625	0.685	0.741	0.853
Occ. (rect., no bg.)	Re-LAION-5B	0.499	0.501	0.501	0.501	0.501	0.499	0.501	0.614
Occ. (rect., no bg.)	SD v1.5	0.783	0.679	0.631	0.538	0.586	0.595	0.618	0.846
Occ. (seg.)	Re-LAION-5B	0.499	0.499	0.499	0.501	0.499	0.501	0.499	0.614
Occ. (seg.)	SD v1.5	0.876	0.821	0.751	0.625	0.645	0.738	0.815	0.921
Occ. (seg., no bg.)	Re-LAION-5B	0.501	0.501	0.499	0.501	0.501	0.499	0.501	0.614
Occ. (seg., no bg.)	SD v1.5	0.817	0.719	0.669	0.581	0.596	0.655	0.712	0.883
Mean RGB	Re-LAION-5B	0.546	0.586	0.570	0.576	0.518	0.563	0.583	0.558
Mean RGB	SD v1.5	0.585	0.594	0.525	0.520	0.529	0.538	0.551	0.499
Obj. det. (restr.)	Re-LAION-5B	0.682	0.596	0.578	0.550	0.540	0.548	0.577	0.653
Obj. det. (restr.)	SD v1.5	0.691	0.648	0.630	0.538	0.579	0.599	0.627	0.692
Pose keypoints	Re-LAION-5B	0.691	0.633	0.576	0.565	0.554	0.543	0.573	0.630
Pose keypoints	SD v1.5	0.770	0.668	0.640	0.539	0.588	0.610	0.638	0.875
Depth at pose	Re-LAION-5B	0.730	0.638	0.581	0.560	0.541	0.541	0.565	0.624
Depth at pose	SD v1.5	0.756	0.657	0.630	0.536	0.583	0.602	0.615	0.859
Obj. det.	Re-LAION-5B	0.705	0.639	0.586	0.564	0.550	0.564	0.588	0.691
Obj. det.	SD v1.5	0.735	0.683	0.649	0.566	0.589	0.637	0.672	0.775
Captions [‡]	Re-LAION-5B	0.784	0.732	0.719	0.760	0.734	0.751	0.745	0.769
Captions [‡]	SD v1.5	0.772	0.688	0.653	0.582	0.594	0.653	0.692	0.747
Base	Re-LAION-5B	0.976	0.976	0.969	0.961	0.948	0.959	0.973	0.980
Base	SD v1.5	0.966	0.935	0.883	0.779	0.783	0.895	0.964	0.996

[‡] Probe operates on BLIP Large captions encoded via SentenceT5-Base.

MST5. This disparity indicates that the generative model renders skin tone signal less uniformly across the tonal spectrum than is observed in real imagery, a tendency that becomes more pronounced under reduced representations.

Occlusion probes reveal the same dataset asymmetry observed for gender, with per-class variation All occlusion variants on Re-LAION-5B produce macro-AUC values at chance (0.514–0.515). The per-class breakdown confirms this is not an artefact of cancellation: individual AUCs for MST1 through MST7 are uniformly at chance (0.499–0.501) under every masking strategy. The exception is MST8, which registers AUC 0.614 consistently across all variants including those where the background is fully removed; given that this value is stable across masking strategies, it most plausibly reflects a small-sample artefact in this category rather than genuine contextual signal. The results establish that skin tone information in Re-LAION-5B is localised entirely within the visible person region, replicating the finding observed for gender in this dataset.

On SD v1.5, occlusion probes retain substantial above-chance performance across all masking strategies, with macro-AUC ranging from 0.660 to 0.774. The per-class results reveal that this contextual signal is concentrated at the extremes of the tonal distribution: MST1 and MST8 achieve the highest per-class AUCs across all variants (e.g., 0.876 and 0.921 under segmentation masking), while mid-range tones MST4 and MST5 remain substantially lower. This indicates that the generative model encodes the strongest contextual skin tone priors at the tonal extremes, where stereotypical scene associations may be most pronounced in the training data.

The same occlusion gradient observed for gender is present here: segmentation masking consistently outperforms rectangle masking, and retaining the background alongside person occlusion consistently outperforms background removal. Both patterns confirm that the immediately surrounding scene is the primary carrier of contextual skin tone signal in SD v1.5.

Global colour statistics carry weaker photometric signal than for gender Mean RGB achieves macro-AUC 0.563 on Re-LAION-5B and 0.543 on SD v1.5, noticeably lower than the corresponding gender values of 0.600 and 0.579. This reduction is consistent with the expectation that global colour statistics carry stronger signal for an attribute whose visual correlates include clothing and hairstyle than for one whose primary cue is a specific tissue-level photometric property: averaging colour across the entire image dilutes the localised skin tone signal more severely than it dilutes the broader gender-correlated colour variation. Per-class mean RGB AUCs are broadly distributed across all MST categories for Re-LAION-5B (range 0.518–0.586), with no clear concentration at tonal extremes, suggesting that colour-based skin tone stereotyping

at the global level is diffuse rather than category-specific.

Geometric probes produce comparable signal for skin tone and gender, with SD v1.5 showing stronger encoding

Pose keypoints and depth at pose keypoints yield macro-AUC values of 0.596 and 0.598 respectively on Re-LAION-5B, and 0.666 and 0.655 on SD v1.5. These values are slightly higher than the corresponding gender results and, notably, close to one another within each dataset, indicating that skeletal geometry and 3D keypoint depth carry roughly equivalent amounts of skin tone information. The consistently higher SD v1.5 figures across both probes reinforce the pattern that the generative model encodes demographic attributes more pervasively into structural representations than the web-scraped corpus.

The per-class breakdown for pose keypoints reveals that, as with the occlusion probes, the strongest signal in SD v1.5 is concentrated in MST1 and MST8 (AUC 0.770 and 0.875 respectively), while MST4 remains close to 0.539. This convergent finding across multiple probe types suggests that the systematic scene stereotyping identified under occlusion propagates through body configuration priors as well as background context.

Object detection probes show that scene composition encodes skin tone in both datasets

Both object detection variants achieve moderate macro-AUC values across datasets: the restricted probe yields 0.591 on Re-LAION-5B and 0.626 on SD v1.5, while the unrestricted probe achieves 0.611 and 0.663 respectively. Unlike the gender case, where the unrestricted probe on Re-LAION-5B substantially outperformed the restricted variant (0.692 vs. 0.615), the gap here is modest on both datasets (approximately 2 percentage points for Re-LAION-5B, 3.7 for SD v1.5), indicating that beyond-person object co-occurrence contributes less additional skin tone signal than it does for gender. The consistently higher SD v1.5 values, however, confirm that the generative model encodes skin tone more strongly in scene-level object distributions than the web-scraped corpus, consistent with the pattern across all non-appearance probes.

Caption probing reveals a cross-dataset asymmetry absent from the visual probes

Caption-based classification produces a striking reversal of the pattern observed for all visual probes: Re-LAION-5B achieves macro-AUC 0.749, substantially higher than SD v1.5 at 0.673. This asymmetry — where the real dataset exhibits stronger linguistic skin tone signal than the generated dataset — directly contrasts with the visual occlusion results, where SD v1.5 substantially outperformed Re-LAION-5B.

The Re-LAION-5B per-class caption AUCs are notably consistent across tones (range 0.719–0.784), indicating that skin tone stereotyping at the linguistic level is

distributed broadly across the full tonal spectrum rather than concentrated at the extremes. For SD v1.5, per-class caption AUCs show the familiar concentration at MST1 and MST8, while mid-range tones produce lower values, mirroring the pattern seen in the visual probes. Together, these results suggest that while SD v1.5 embeds skin tone priors more pervasively into visual scene statistics than Re-LAION-5B, the corresponding linguistic stereotyping in the captions of real images is no weaker — and for mid-range tones, stronger — than in the generated dataset. Skin tone stereotyping in web-scraped data may therefore be more evenly distributed across the tonal spectrum at the semantic level than the visual probe results alone would suggest.

Per-class results reveal systematic difficulty for mid-range skin tones across all probes

Across all non-appearance probes and both datasets, a consistent ordering emerges: MST1 and MST8 achieve the highest per-class AUCs while MST4 and MST5 produce the lowest, with the disparity most pronounced in SD v1.5. This pattern reflects the greater visual similarity of mid-range tones to neighbouring categories, which reduces the distinctiveness of any indirect correlates. It also raises a methodological concern: audit methods relying solely on macro-averaged metrics may mask substantially weaker separability for the categories that are simultaneously most visually ambiguous and most likely to be underrepresented in training data, potentially understating bias for mid-range skin tones.

Skin tone and gender exhibit analogous but not identical spurious feature profiles

The skin tone results largely recapitulate the structural pattern observed for gender: Re-LAION-5B localises demographic signal within the person region, while SD v1.5 distributes it throughout the image. The principal differences are quantitative rather than qualitative — skin tone contextual AUCs under occlusion are somewhat lower than those for gender, mean RGB is a weaker probe for skin tone, and the caption asymmetry reverses direction — but the fundamental finding holds for both attributes: SD v1.5 systematically projects demographic priors into scene context, object composition, body geometry, and linguistic register in a manner that is not observed in the web-scraped corpus.

4.3.3 Connections to Dataset Classification Results

Several structural parallels link the demographic probe experiments to the earlier dataset classification findings, and together the two sets of results provide a mutually reinforcing account of how statistical bias is distributed within and between the datasets. Given the uniformly high overall accuracies observed across all transformations, near-perfect separability of each individual class can be inferred

without requiring explicit per-class reporting: at 95%+ overall accuracy under a balanced three-class prior, no single class can be pulling performance substantially below chance while the others achieve high accuracy.

The most direct parallel is the persistence of signal across representation levels. In the dataset classification experiments, separability remained above chance across every transformation tested, from global colour statistics at 64.19% through to natural language at 90.99%, indicating that dataset identity is encoded in multiple correlated visual cues simultaneously. The demographic probing results recapitulate this structure: neither gender nor skin tone is recoverable from a single isolated representation, but both remain above chance across photometric (mean RGB), geometric (pose keypoints, depth at keypoints), semantic (object detection, captions), and contextual (occlusion) probes. The implication in both cases is the same: the statistical regularities in question are not artefacts of a single image property but are embedded coherently across the full representational spectrum.

The sharpest connection between the two analyses concerns the role of Stable Diffusion v1.5. In the dataset classification task, the presence of SDv1.5 as one of the three classes was proposed as the primary explanation for why classification accuracies substantially exceed those reported by Zeng et al. for purely web-scraped datasets: the generative pipeline imposes consistent visual priors — over texture, colour, spatial layout, and scene composition — that make SDv1.5 images far more internally homogeneous than Re-LAION-5B or COCO, and therefore far easier to identify as a class. The occlusion results in the demographic probing experiments provide a mechanistic account of this homogeneity. SDv1.5 encodes gender at AUC 0.947 in the unoccluded person and AUC 0.747 from background context alone; the corresponding skin tone values are 0.900 and 0.774. These figures confirm that the generative model has learned coherent joint priors over person appearance and scene context that are stable enough across 200 professions to be exploited by a linear classifier. Re-LAION-5B shows no such contextual signal under any masking condition, indicating that its demographic information is confined to person pixels and that its scene context is essentially uninformative about the gender or skin tone of the depicted individual. The dataset classification experiments cannot reveal which class is responsible for the high separability; the demographic probing experiments establish that the answer is predominantly SDv1.5.

The comparison between the two caption-based results further sharpens this picture. Dataset identity survived compression to natural language at 90.99% classification accuracy, a figure that places caption-based classification at the same tier as semantic segmentation and above object detection in the discrimination hierarchy. Gender persists in captions at AUC 0.771 for SDv1.5 and 0.901 for Re-LAION-5B; skin tone at AUC 0.673 and 0.749 respectively. The reversal between datasets in the

caption domain — where Re-LAION-5B shows stronger demographic separability than SDv1.5, in contrast to every visual probe — is the one result that does not replicate the pattern seen in the classification task. It suggests that while SDv1.5’s generative priors produce stronger scene-level visual stereotyping, the real-image corpus contains richer linguistically recoverable demographic associations, likely because web-scraped images are drawn from a broader semantic space of human activity where occupational, environmental, and social contexts co-vary more systematically with demographic attributes at the description level. Taken together, the caption findings across both experiments confirm that stereotyping in these datasets is not confined to pixel statistics or geometric structure but extends into the semantic content of the imagery, recoverable from text alone.

Finally, the depth and edge classification results connect to the novel Depth-at-Pose-Keypoints representation introduced in the demographic probing pipeline. Depth achieves 96.78% dataset classification accuracy and Canny edge 93.62%, confirming that structural geometry carries strong dataset-discriminating signal. At the demographic level, depth at pose keypoints yields AUC 0.559 for gender and 0.598 for skin tone on Re-LAION-5B, and 0.592 and 0.655 on SDv1.5 — the lowest above-chance values recorded for any geometric probe. The contrast between the high dataset classification accuracy of full depth maps and the relatively weak demographic signal from sparse keypoint depth sampling suggests that the structural differences captured in full depth maps are primarily compositional — camera distance, scene depth layering, background structure — rather than body-geometric, and that this compositional signal does not substantially co-vary with demographic attributes at the individual level. The sparse depth-at-keypoints representation, by isolating only the 3D configuration of the skeletal joints, extracts the demographically relevant portion of the structural signal while discarding the scene-level compositional information that drives the high classification accuracy.

4.4 Debiasing Evaluation

The debiasing evaluation compares the demographic distribution and spurious feature profile of the 10,000-image debiased doctor corpus against the full Stable Diffusion v1.5 generation results reported in Sections 4.1 and 4.3. Rather than constructing a separate doctor-specific baseline — which would be expected to replicate or amplify the patterns already identified across the 200,000-image standard corpus — the evaluation uses the aggregate SDv1.5 findings as the reference point. This comparison directly assesses whether the combined ITI-GEN, ControlNet, and Chamfer colour alignment pipeline produces a measurable departure from the demographic and

spurious feature profile characteristic of standard Stable Diffusion v1.5 generation.

An important methodological distinction applies throughout this section. Demographic labels for the debiased corpus are not derived from automated classifiers but are assigned directly from the generation configuration: each image is labelled with the ITI-GEN gender token and MST skin tone token used to produce it. This eliminates annotation uncertainty as a confound in the evaluation and ensures that the probing classifiers measure the degree to which low-level image representations encode the demographic conditioning signal rather than classifier-predicted attributes. For gender, high probing accuracy on person-level representations is therefore an expected consequence of successful ITI-GEN conditioning rather than evidence of spurious signal. For skin tone, the 10-class problem is substantially harder than the binary gender case, and person-level probe values should be interpreted relative to the chance level of 0.500 for macro-averaged AUC rather than in direct comparison with the binary gender results.

4.4.1 Demographic Distribution Shift

The debiased generation pipeline was configured to produce images uniformly distributed across gender and skin tone categories. ITI-GEN embeddings were applied for male and female gender categories in equal proportions across the 10,000-image corpus, with MST skin tone conditioning applied uniformly across all ten MST levels (MST 1–10), yielding a target distribution of approximately 500 images per gender–skin tone combination. After structural validation using the same acceptance criteria as the standard generation pipeline, the effective composition of the debiased corpus reflects this intended uniform distribution, subject to profession-specific rejection rates.

By construction, the debiased corpus achieves a gender distribution of approximately 50% female and 50% male. This stands in direct contrast to the strongly male-skewed output of the standard Stable Diffusion v1.5 pipeline, which produced 29.02% female images across all 200 professions, a figure consistent with the amplification of training data gender imbalance documented in Section 4.1.

The skin tone distribution of the debiased corpus is equally balanced by construction across MST 1–10, in contrast to the pronounced light-bin skew of the standard pipeline, where 76.99% of generated images fell in the Mid bin and Dark-bin representation approached zero across most professions. The debiasing pipeline therefore produces a distribution that would require radical change to the model’s learned prior to arise naturally from standard generation.

Figure 4.7 shows a representative sample of debiased generated images at a fixed seed, with rows corresponding to female and male gender conditioning and columns corresponding to MST skin tone levels 1 through 10.



Figure 4.7 Representative debiased generation samples at a fixed seed. Rows correspond to female (top) and male (bottom) ITI-GEN gender conditioning; columns correspond to MST skin tone levels 1 through 10. Skin tone transitions progressively from lighter values at MST 1 to darker values at MST 10, demonstrating that the ten-point conditioning axis produces a visually coherent and continuous tonal gradient. Gender presentation is clearly differentiated between rows through facial features, body morphology, and clothing style. The ControlNet OpenPose conditioning enforces a consistent standing, full-body, centred pose across all twenty images, and the Chamfer colour alignment is reflected in the broadly consistent warm tonal palette maintained throughout the grid. The background exhibits some variation across columns, reflecting the residual scene-level variation that pose conditioning does not fully constrain, consistent with the background-level gender signal identified in Section 4.4.2.

4.4.2 Probing Accuracy on the Debiased Corpus

To assess whether the debiasing interventions reduced not only the aggregate demographic imbalance but also the learnability of demographic signal from low-level image representations, the same probing classifiers applied to the standard SDv1.5 corpus were applied to the debiased corpus under identical experimental conditions. Tables 4.7 and 4.8 report gender and skin tone probing AUC for the standard SDv1.5 corpus and the debiased corpus across all evaluated representations.

Gender Probing Results

Person-level representations confirm successful demographic conditioning. The unmodified image probe achieves $AUC \approx 1.000$ on the debiased corpus, and person-centric occlusion variants are uniformly high: 0.998 for full person appearance without background (FullNoBg), 0.999 for the segmentation-masked person region with background retained (MaskSegm), and 0.985 for the silhouette alone without background (MaskSegmNoBg). These values exceed the corresponding standard SDv1.5 figures by substantial margins. As noted above, this is expected: ITI-GEN directly conditions the person’s visual appearance and labels are assigned from the conditioning token rather than a classifier. High person-level separability confirms that the conditioning pipeline successfully produced perceptually gendered images and that

Table 4.7 Gender probing AUC on the standard Stable Diffusion v1.5 corpus and the debiased corpus. Chance level is 0.500 for binary classification. Labels for the debiased corpus are derived from the generation configuration rather than automated annotation. † Standard SDv1.5 value reflects near-complete majority-class collapse; see Section 4.3.1 for detail.

Representation	Probe	SDv1.5	Debiased
Base (unmodified)	ResNet-50	0.946	≈ 1.000
Occlusion (full, no background)	ResNet-50	0.948	0.998
Occlusion (masked segmentation)	ResNet-50	0.852	0.999
Occlusion (masked segmentation, no bg.)	ResNet-50	0.834	0.985
Occlusion (masked rectangle)	ResNet-50	0.747	0.995
Occlusion (masked rectangle, no bg.)	ResNet-50	0.587 [†]	0.701
Mean RGB	Logistic Regression	0.579	0.507
Object detection	Logistic Regression	0.608	0.449
Object detection (restricted)	Logistic Regression	0.608	0.457
Depth at pose keypoints	Logistic Regression	0.592	0.454
Pose keypoints	MLP	0.680	0.475
Captions	SentenceT5	0.771	0.530

Table 4.8 Skin tone probing macro-averaged AUC on the standard Stable Diffusion v1.5 corpus and the debiased corpus. The SDv1.5 column reports 8-class macro-averaged AUC (MST 1–8, as MST 9 and MST 10 are absent from the standard corpus); the debiased column reports 10-class macro-averaged AUC (MST 1–10, from the generation configuration). Chance level is 0.500 in both cases. Labels for the debiased corpus are derived from the generation configuration rather than automated annotation.

Representation	Probe	SDv1.5	Debiased
Base (unmodified)	ResNet-50	0.900	0.861
Occlusion (full, no background)	ResNet-50	0.899	0.823
Occlusion (masked segmentation)	ResNet-50	0.774	0.729
Occlusion (masked segmentation, no bg.)	ResNet-50	0.704	0.663
Occlusion (masked rectangle)	ResNet-50	0.728	0.632
Occlusion (masked rectangle, no bg.)	ResNet-50	0.660	0.543
Mean RGB	Logistic Regression	0.543	0.482
Object detection	Logistic Regression	0.663	0.439
Object detection (restricted)	Logistic Regression	0.626	0.461
Depth at pose keypoints	Logistic Regression	0.655	0.443
Pose keypoints	MLP	0.666	0.483
Captions	SentenceT5	0.673	0.490

the demographic signal is encoded at the level of the depicted person, precisely as intended.

Low-level spurious signal is eliminated across all non-person representations. Mean RGB classification falls from 0.579 on the standard SDv1.5 corpus to 0.507 on the debiased corpus, effectively at the binary chance level of 0.500. Object detection probes fall to 0.449 and 0.457 for the unrestricted and restricted vocabularies respectively, both at or near chance, indicating that the debiased corpus exhibits no systematic gender-correlated object co-occurrence patterns. Depth at pose keypoints collapses to 0.454 and pose keypoints to 0.475 (mean across five seeds), both within the range expected under a null hypothesis of no demographic signal in geometric representations. These reductions stand in direct contrast to the standard SDv1.5 results, where all of these probes achieved AUC values between 0.579 and 0.680.

Caption-based probing falls from 0.771 on the standard SDv1.5 corpus to 0.530 on the debiased corpus, a reduction of 24.1 percentage points. The near-halving of this signal indicates that the debiasing pipeline substantially disrupts the generation of gender-stereotyped semantic content, even though caption generation is not directly conditioned by any of the debiasing interventions. This provides evidence that the debiasing effect propagates into the semantic register of generated content.

The background-only probe (MaskRectNoBg) achieves AUC 0.701 on the debiased corpus. Direct comparison with the standard SDv1.5 value of 0.587 is complicated by the different label distributions: the 0.587 value reflected near-complete majority-class collapse in a strongly male-skewed corpus, whereas the debiased corpus with balanced 50/50 labels requires genuine discrimination. The 0.701 value indicates that some background-level signal persists, likely because ITI-GEN’s conditioning of scene content and the ControlNet pose constraint together influence global composition in ways that partly correlate with the gender conditioning token. Nonetheless, the background-only AUC of 0.701 is substantially lower than the person-level probes (0.985–0.999), confirming that the primary carrier of gender signal in the debiased corpus is the person’s appearance as intended.

Skin Tone Probing Results

The skin tone probing results follow a consistent pattern of reduction across all representations, with every probe falling relative to the standard SDv1.5 corpus. Unlike the gender case, person-level probes decrease rather than increase in the debiased corpus: the unmodified base probe falls from 0.900 to 0.861, and the full person appearance probe (FullNoBg) falls from 0.899 to 0.823. This reduction reflects the substantially greater difficulty of 10-class skin tone discrimination under a balanced uniform distribution across MST 1–10 compared to the standard corpus,

where the model’s strong light-skewed prior produces a concentrated distribution that is easier to learn. The person-level values of 0.823–0.861 remain substantially above the 10-class chance level of 0.500, confirming that the ITI-GEN conditioning successfully encodes skin tone signal at the person level across the full ten-point scale.

The most direct evidence for the effectiveness of the Chamfer colour alignment is the mean RGB probe, which falls from 0.543 on the standard SDv1.5 corpus to 0.482 on the debiased corpus – the skin tone probe closest to the 10-class chance level of 0.500. Since mean RGB compresses each image to its global average colour per channel, a value at or near chance confirms that the debiased corpus exhibits no systematic skin-tone-correlated colour statistics at the global level, directly demonstrating that the Chamfer colour alignment disrupts the photometric correlations that enable colour statistics to carry demographic signal.

All remaining non-person probes collapse to at or near chance. Object detection falls from 0.663 to 0.439 (unrestricted) and from 0.626 to 0.461 (restricted), both at or near chance, indicating that the debiased corpus exhibits no systematic skin-tone-correlated object co-occurrence patterns. Depth at pose keypoints falls from 0.655 to 0.443 and pose keypoints from 0.666 to 0.483, both at or near chance, confirming that geometric representations retain no skin tone signal. Caption probing falls from 0.673 to 0.490, approaching chance, indicating that the semantic content of generated descriptions is no longer systematically correlated with the skin tone conditioning token.

The occlusion probes show consistent reductions across all variants. MaskRectNoBg falls from 0.660 to 0.543, close to chance, while MaskRect falls from 0.728 to 0.632, indicating a partial but incomplete elimination of background-level skin tone signal. The retention of above-chance signal in background-only probes suggests that the scene composition generated by the model retains some skin-tone-correlated properties even under pose conditioning, consistent with the background entanglement observed in the gender analysis.

Summary

Taken together, the gender and skin tone probing results establish a coherent and consistent pattern. The debiasing pipeline successfully conditions demographic attributes at the person level while substantially eliminating the low-level spurious correlations, geometric signal, and semantic stereotyping that characterise the standard Stable Diffusion v1.5 generation pipeline. The near-chance mean RGB skin tone probe provides the clearest mechanistic confirmation that the Chamfer colour alignment addresses its intended target. Residual background-level signal persists in both the gender and skin tone analyses, indicating that background-level demographic

conditioning remains partially entangled with the generation process even under explicit pose and appearance control, and represents the primary avenue for further improvement.

4.5 Limitations

Several limitations qualify the interpretation of the results presented in this chapter.

Scope of the Re-LAION-5B sample. The analysis is conducted on four of 128 shards of Re-LAION-5B, representing approximately 3.1% of the full dataset. Findings are therefore estimates of corpus-level properties rather than exact population parameters, and the presence of unknown crawl-time biases cannot be excluded.

Annotation as measurement instrument. Demographic labels are produced by automated classifiers rather than ground-truth human annotation. Gender predictions are binary and do not capture non-binary gender expression. Skin tone estimates are sensitive to imaging conditions and lighting variability inherent in web-scraped data. Observed demographic distributions should therefore be interpreted as classifier outputs rather than definitive ground truth labels.

Dataset classification and causation. Above-chance classification accuracy confirms that a transformation retains dataset-discriminating signal, but does not establish which specific properties of each dataset produce that signal, nor does it constitute a causal claim about how those properties arose. The per-class accuracy attribution localises signal to a dataset but cannot decompose it further without additional controlled experiments.

Debiasing evaluation scope. The debiasing evaluation is restricted to a single profession and a single combination of inference-time interventions. It does not evaluate weight editing or fine-tuning approaches, and results may not generalise to other professions, prompts, or model architectures. The probing results demonstrate that low-level spurious correlations are eliminated at the representation level for the doctor profession, but whether this elimination generalises across the full 200-profession set or persists under different prompt formulations remains untested. The residual background-level signal identified in both the gender and skin tone analyses indicates that background-level demographic conditioning remains partially entangled with the generation process, and constitutes the primary open question for future work on this pipeline.

Synthetic image annotation reliability. Demographic classifiers and face detectors are trained on real photographs. Their reliability on synthetic images produced by latent diffusion models has not been formally validated in this work, and the presence of generative artefacts may affect annotation accuracy in ways that differ systematically between the real and synthetic corpora.

4.6 Synthesis of Evaluation Findings

The dataset classification experiments establish that Re-LAION-5B, Stable Diffusion v1.5 generated images, and COCO are distinguishable with high accuracy across every level of visual representation tested, from unmodified images at 99.42% down to mean RGB compression at 64.19%. The persistence of above-chance separability even under the most aggressive information reduction demonstrates that dataset identity is encoded in multiple correlated visual cues simultaneously rather than in any single isolated artefact. The VAE experiment confirms that this separability is not a consequence of low-level acquisition artefacts, as reconstruction through the Stable Diffusion latent bottleneck reduces accuracy by only 0.27 percentage points.

The demographic distribution analysis reveals that Stable Diffusion v1.5 does not merely reflect the imbalances present in its training corpus but actively amplifies them. Gender amplification is confirmed in 178 of 200 professions, with the female proportion falling from 41.5% in Re-LAION-5B to 29.02% in Stable Diffusion v1.5. The skin tone analysis reveals a near-complete collapse of dark-toned representation in generative outputs: the Dark-bin proportion falls from 0.81% in Re-LAION-5B to 0.14% in Stable Diffusion v1.5, an 83% reduction, while the Light-bin proportion almost doubles from 12.45% to 22.87%. Skin tone amplification toward lighter tones is confirmed in 175 of 200 professions.

The demographic probing experiments reveal qualitatively distinct spurious feature profiles between the two datasets. In Re-LAION-5B, gender and skin tone information is localised almost entirely within the visible appearance of the depicted person: every occlusion variant collapses performance to chance, and residual signal from colour statistics, object co-occurrence, and skeletal geometry is modest. In Stable Diffusion v1.5, demographic information is distributed across the full image and is recoverable from background context alone, from the person silhouette independently of background, and from natural language descriptions of generated content. This cross-modal persistence demonstrates that the stereotyping embedded in the generative model is coherent and pervasive rather than confined to low-level pixel statistics.

The debiasing evaluation demonstrates that the combined ITI-GEN, ControlNet,

and Chamfer colour alignment pipeline produces a measurable and substantive departure from the demographic and spurious feature profile of the standard generation pipeline. Every non-person probe collapses to at or near chance for both gender and skin tone, confirming that the debiasing interventions substantially reduce the low-level spurious correlations, object-level stereotyping, and semantic stereotyping that characterise standard Stable Diffusion v1.5 generation. Residual background-level signal persists in both analyses, indicating that background-level demographic conditioning remains partially entangled with the generation process even under explicit pose and appearance control.

5 Conclusions and Future Work

5.1 Conclusions

This thesis applied and extended established spurious feature detection and demographic probing methodologies to Re-LAION-5B and Stable Diffusion v1.5, producing a systematic empirical characterisation of how demographic bias is encoded, distributed, and amplified across a web-scale training corpus and the generative model trained on it. The findings address each of the five research objectives and collectively advance understanding of bias mechanisms in large-scale vision systems.

Objective 1: Dataset acquisition, preparation, and annotation. A

profession-conditioned subset of 200,000 images was retrieved from Re-LAION-5B across 200 occupations aligned with ISCO-08 categories, and a corresponding set of 200,000 images was generated using Stable Diffusion v1.5 with LCM-LoRA acceleration for the standard generation pipeline. CLIP embeddings were recomputed from scratch using `openai/clip-vit-large-patch14` in the absence of pre-computed embeddings, and a two-stage structural validation pipeline was developed to enforce human-centric image suitability across both corpora. Demographic annotation was performed using the Realistic Gender Classifier and a VGG-16 MST10 Regression model selected through systematic evaluation against CCv2, MSTE, and FACET benchmarks, with skin tone extended to the full ten-point Monk Skin Tone scale beyond the binary and six-point scales used in prior work.

Objective 2: Spurious feature detection. The transformation-based framework of Zeng et al. was applied and extended to a three-dataset configuration — Re-LAION-5B, Stable Diffusion v1.5, and COCO — enabling per-class accuracy attribution that a binary design cannot provide. Fifteen transformations spanning frequency, geometry, semantics, and colour were evaluated. Every transformation exceeded the 33.3% chance level, from unmodified images at 99.42% down to mean RGB at 64.19%, establishing that dataset-specific spurious signatures persist even under extreme information compression and are distributed across multiple correlated visual attributes rather than concentrated in any single representation.

Objective 3: Spurious feature correlations with demographics. Demographic probing following the occlusion framework of Meister et al. revealed a fundamental asymmetry between the two datasets. In Re-LAION-5B, gender and skin tone information is localised entirely within the visible appearance of the depicted person: every occlusion

variant collapses to chance, and no residual signal is recoverable from background context. In Stable Diffusion v1.5, demographic information is distributed across the full image, recoverable from background context alone (gender AUC 0.747; skin tone AUC 0.728), from person silhouette geometry (gender AUC 0.834), from object co-occurrence, skeletal geometry, and natural language descriptions (gender AUC 0.771). This cross-modal, cross-representational persistence demonstrates that the generative model does not merely inherit the demographic priors of its training corpus but projects them coherently onto the surrounding scene, spatial layout, and semantic content of generated images. Gender amplification was confirmed in 178 of 200 professions and skin tone amplification toward lighter tones in 175 of 200 professions, with the Dark-bin proportion collapsing by 83% from Re-LAION-5B to Stable Diffusion v1.5.

Objective 4: Inference-time debiasing. A combined pipeline of ITI-GEN demographic conditioning, ControlNet OpenPose structural conditioning, and Chamfer latent-space colour alignment was implemented and evaluated on a 10,000-image debiased doctor corpus. The pipeline successfully overrides the model’s strongly skewed learned prior, achieving a uniform gender and skin tone distribution by construction. At the representation level, every non-person probe collapsed to at or near chance for both gender and skin tone, including mean RGB (0.507 for gender; 0.482 for skin tone), object detection, pose keypoints, depth at keypoints, and caption-based probing (gender 0.530; skin tone 0.490). The near-chance mean RGB skin tone result provides the most direct mechanistic evidence that the Chamfer colour alignment successfully disrupts the photometric correlations that enable demographic signal to propagate through colour statistics. Residual background-level signal persists in both analyses, indicating that background-level demographic conditioning remains partially entangled with the generation process under the current pipeline.

Objective 5: Conclusions on bias mechanisms and mitigation. The combined evidence characterises a two-stage bias propagation process. Re-LAION-5B under-represents women relative to real-world labour force statistics and exhibits modest demographic signal distributed across colour, geometry, and semantic representations, but confines this signal to person-level appearance. Stable Diffusion v1.5 then amplifies these imbalances substantially and transforms their distribution: the generative process projects demographic priors beyond the person onto background, scene composition, object layout, and textual description, producing a substantially more entrenched and pervasive form of bias than is present in the training corpus. Inference-time debiasing can measurably reduce spurious correlations at the representation level without model retraining, but background-level entanglement

persists and represents the primary remaining challenge for this class of interventions.

Impact on the field. These findings have several direct implications for practitioners and researchers working with large-scale visual datasets and generative systems. First, auditing a generative model’s demographic outputs is insufficient without also auditing the spurious feature distributions of the training corpus, as the generative process amplifies and transforms biases rather than simply reproducing them. Second, occlusion-based probing provides a practical and interpretable diagnostic for distinguishing person-localised from scene-distributed demographic signal, a distinction that aggregate distribution metrics cannot detect. Third, the persistence of demographic signal in natural language descriptions of generated images demonstrates that stereotyping in generative systems is not confined to pixel statistics but extends into the semantic register, with implications for any downstream application that consumes generated text-image pairs. Fourth, the debiasing evaluation demonstrates that inference-time interventions can eliminate low-level spurious correlations effectively, but background-level demographic conditioning requires interventions that target scene composition directly rather than person appearance alone.

5.2 Future Work

Several directions present themselves as natural extensions of the work presented in this thesis, spanning improvements to the debiasing methodology, broader evaluation scope, and deeper investigation of the mechanisms identified here.

The most immediate extension concerns the debiasing pipeline. Residual background-level demographic signal persists at AUC 0.701 for gender and 0.632 for skin tone under background-only probing, indicating that scene composition remains partially entangled with the demographic conditioning signal even under pose and appearance control. Future work could investigate targeted background conditioning mechanisms that explicitly decorrelate scene composition from demographic attributes, either through adversarial background debiasing approaches or through extension of the ITI-GEN framework to condition on background properties independently of person appearance. The weight editing approach of Unified Concept Editing [85] offers a complementary avenue, as its direct modification of cross-attention projections may suppress background-level demographic associations that inference-time conditioning does not address. A systematic comparison of conditioning-based approaches against weight editing methods on the same probing protocol would establish whether background entanglement is specific to the conditioning paradigm or is a general property of all inference-time interventions.

The debiasing evaluation in this thesis is restricted to a single profession. Extending the evaluation to a stratified sample spanning all ten ISCO-08 major groups would determine whether the elimination of spurious correlations observed for the doctor profession generalises across occupational contexts with different baseline demographic distributions and different magnitudes of stereotyping. Professions at the extremes of the observed skew distributions represent particularly informative test cases, as the debiasing pipeline would be required to override the strongest learned priors in the model.

The skin tone analysis in this thesis uses the full ten-point MST scale for the debiased corpus, with labels derived from the generation configuration rather than from automated classifiers. A complementary evaluation using validated automated skin tone estimation on the debiased corpus would quantify the photometric fidelity of the conditioning, assessing whether images conditioned on darker MST tokens exhibit measurably different skin pigmentation statistics and whether the Chamfer colour alignment produces coherent skin tone variation rather than global colour shifts. This would also enable the debiasing evaluation to be compared directly with the eight-class standard corpus results.

The transformation-based probing framework applied in this work uses ConvNeXt-Tiny as the classification backbone. Future work could evaluate whether the separability ceiling identified here reflects genuine representational differences between datasets or the capacity limitations of the chosen probe, by applying larger classification architectures or self-supervised foundation model representations. The original Zeng et al. ablation demonstrates limited impact of model size on dataset classification accuracy for structural transformations, but this was conducted in a different dataset context and may not generalise to the Re-LAION-5B versus Stable Diffusion v1.5 comparison.

Finally, the analysis is grounded in a subset of four Re-LAION-5B shards, covering approximately 3.1% of the full corpus. Extending the analysis to a larger shard sample would determine whether the spurious feature profiles and demographic distributions identified here are stable properties of the corpus as a whole or are influenced by domain concentration effects specific to the retrieved subset. The link attrition and domain concentration analyses conducted in this work establish that the retrieved subset is non-uniform in its domain composition, and access to LAION crawl metadata would enable a more principled characterisation of the representativeness of the surviving URL pool.

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Appendix A: Occupation List

The following 200 occupations, listed alphabetically, constitute the full profession set used for profession-conditioned image retrieval from Re-LAION-5B and for image generation using Stable Diffusion v1.5. The set was constructed by sampling job titles systematically across all ten ISCO-08 major groups, ensuring representative coverage of skill levels, professional domains, and social prestige ranges.

#	Occupation	#	Occupation	#	Occupation
1	Accountant	51	Cleaner	101	Hair Dresser
2	Actor	52	Clerk	102	Handyperson
3	Actuary	53	Coach	103	Health Inspector
4	Administrative Assistant	54	Comedian	104	Historian
5	Administrator	55	Compliance Officer	105	Hotel Concierge
6	Air Traffic Controller	56	Composer	106	Hotel Manager
7	Animal Trainer	57	Conservation Officer	107	Human Resources
8	Anthropologist	58	Construction Worker	108	IT Support Specialist
9	Appraiser	59	Copywriter	109	Illustrator
10	Archaeologist	60	Court Reporter	110	Industrial Designer
11	Architect	61	Crime Scene Investigator	111	Insurance Underwriter
12	Archivist	62	Customer Support Specialist	112	Janitor
13	Art Director	63	DJ	113	Jeweller
14	Artist	64	Dancer	114	Journalist
15	Astronaut	65	Data Scientist	115	Judge
16	Astronomer	66	Database Administrator	116	Lawyer
17	Athlete	67	Debt Counselor	117	Librarian
18	Audio Technician	68	Dentist	118	Lifeguard
19	Auditor	69	Detective	119	Loan Officer
20	Automotive Designer	70	Development Officer	120	Logger
21	Baker	71	Dietitian	121	Logistics Manager
22	Banker	72	Director	122	Magician
23	Bankruptcy Specialist	73	Doctor	123	Makeup Artist
24	Barber	74	Dog Walker	124	Marine Biologist
25	Barista	75	Draughtsperson	125	Marketing Manager
26	Bartender	76	Driver	126	Masseur
27	Basketball Player	77	Economist	127	Mathematician
28	Biologist	78	Editor	128	Mayor

Table 1 – continued

#	Occupation	#	Occupation	#	Occupation
29	Biomedical Engineer	79	Electrician	129	Mechanic
30	Blacksmith	80	Emergency Management Specialist	130	Meteorologist
31	Bodyguard	81	Entrepreneur	131	Midwife
32	Bounty Hunter	82	Environmental Engineer	132	Miner
33	Boxer	83	Ergonomist	133	Model
34	Brand Manager	84	Estate Planner	134	Musician
35	Brewer	85	Event Coordinator	135	News Reader
36	Bricklayer	86	Executive Assistant	136	Nurse
37	Broker	87	Exterminator	137	Nutritionist
38	Builder	88	Facilities Manager	138	Oceanographer
39	Butcher	89	Farmer	139	Office Assistant
40	CEO	90	Fashion Designer	140	Operations Manager
41	Carer	91	Firefighter	141	Optician
42	Carpenter	92	Fishmonger	142	Painter
43	Cartographer	93	Flight Attendant	143	Paralegal
44	Cashier	94	Florist	144	Paramedic
45	Chef	95	Football Player	145	Park Ranger
46	Chemical Engineer	96	Forklift Operator	146	Payroll Specialist
47	Chemist	97	Gardener	147	Personal Trainer
48	Chiropractor	98	Geologist	148	Pharmacist
49	Civil Engineer	99	Graphic Designer	149	Photographer
50	Claims Adjuster	100	Grocer	150	Physicist